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Laboratory of Computer Engineering, Data Science and Artificial Intelligence

MASTER'S THESIS

Design of a Telemonitoring System for Patients with Chronic Diseases (Hypertension, Diabetes) Using Consultation Data, Home Measurements and Possibly Mobile Sensors

Presented by
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Master of Science in Data Science and Artificial Intelligence

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Declaration

I, **Nsame Ayuni Rodrick**, declare that this Master's thesis entitled *"Design of a Telemonitoring System for Patients with Chronic Diseases (Hypertension, Diabetes) Using Consultation Data, Home Measurements and Possibly Mobile Sensors"* is the result of my own research, carried out under the supervision of Dr. Malong Yannick and Dr. Noulapeu Ngaffo at the National Higher Polytechnic School of Douala (NHPSD), University of Douala, Cameroon.

I declare that, to the best of my knowledge, this work has not been submitted, in whole or in part, for any other degree, diploma or certification at any other institution.

I further declare that all sources of information, ideas and data used in this thesis have been acknowledged and referenced.

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Abstract

Hypertension and type 2 diabetes are common in Cameroon, and the follow-up of patients between consultations is fragmented. This thesis designs a telemonitoring system for patients with either or both conditions and evaluates its predictive component. The system comprises a patient application and a clinician view built with Flutter on Firebase, a guideline rule that grades every reading on four levels offline on the phone, a two-stage home triage in which a single elevated reading first prompts repeated measurements, and a server worker for escalation, reminders and notifications; it passes 220 automated tests. The predictive component forecasts the risk level of the next reading, so that the label, derived from official thresholds, never enters the inputs, and the guideline rule applied to the last reading (persistence) is the baseline. Eight model families were compared in a nested cross-validation grouped by hospital on the eICU demonstration database (4 182 haemodynamic and 1 356 glycaemic samples, 174 hospitals), and validated externally on 366 readings of 32 inpatients of a public hospital in Douala. On blood pressure, five of the eight families forecast significantly better than persistence; the deployed CatBoost model lowers the ranked probability score from 0.1384 to 0.1341 ($p = 0.002$), a gain of about 3%. On glucose, no family does so significantly (GRU-D 0.1275 against 0.1329, $p = 0.14$). Deep temporal models showed no significant advantage over tree ensembles. On the Douala patients, every model lost accuracy and under-predicted the alert; updating the intercept corrected the offset but not the slopes, and no model improved the alert decisions over the rule. Measures of the change in the patient's state discriminated weakly (AUROC at most 0.60). These results support a design in which the guideline rule is the floor and a model may only raise the alert level, and they call for revalidation on home data before any clinical use.

Keywords: telemonitoring; hypertension; type 2 diabetes; risk prediction; external validation; mobile health; Cameroon.

Résumé

L'hypertension artérielle et le diabète de type 2 sont fréquents au Cameroun, et le suivi des patients entre deux consultations est fragmenté. Ce mémoire conçoit un système de télésurveillance pour les patients atteints de l'une ou des deux affections et évalue sa composante prédictive. Le système comprend une application patient et une vue clinicien développées avec Flutter sur Firebase, une règle issue des recommandations qui classe chaque mesure en quatre niveaux, hors connexion, sur le téléphone, un triage à domicile en deux étapes dans lequel une mesure élevée isolée déclenche d'abord des mesures répétées, et un service serveur pour l'escalade, les rappels et les notifications ; il passe 220 tests automatisés. La composante prédictive prévoit le niveau de risque de la mesure suivante, de sorte que l'étiquette, dérivée de seuils officiels, n'entre jamais dans les variables d'entrée, et la règle appliquée à la dernière mesure (persistance) sert de référence. Huit familles de modèles ont été comparées par validation croisée imbriquée, groupée par hôpital, sur la base de démonstration eICU (4 182 échantillons hémodynamiques et 1 356 glycémiques, 174 hôpitaux), puis validées sur 366 mesures de 32 patients hospitalisés dans un hôpital public de Douala. Pour la pression artérielle, cinq familles sur huit prévoient significativement mieux que la persistance ; le modèle CatBoost retenu abaisse le score de probabilité ordonné (RPS) de 0,1384 à 0,1341 ($p = 0{,}002$), soit un gain d'environ 3 %. Pour la glycémie, aucune famille n'y parvient de façon significative (GRU-D 0,1275 contre 0,1329, $p = 0{,}14$). Les modèles temporels profonds n'ont montré aucun avantage significatif sur les ensembles d'arbres. Sur les patients de Douala, tous les modèles ont perdu en exactitude et sous-estimé l'alerte ; la mise à jour de l'intercept a corrigé ce décalage mais pas les pentes, et aucun modèle n'a amélioré les décisions d'alerte par rapport à la règle. Les mesures de l'évolution de l'état du patient discriminent faiblement (AUROC au plus 0,60). Ces résultats confortent une conception dans laquelle la règle fixe le niveau minimal et un modèle ne peut que relever l'alerte, et ils imposent une revalidation sur des données recueillies à domicile avant tout usage clinique.

Mots-clés : télésurveillance ; hypertension artérielle ; diabète de type 2 ; prédiction du risque ; validation externe ; santé mobile ; Cameroun.

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List of Acronyms

ADA American Diabetes Association
AI Artificial intelligence
AUC Area under the curve
AUROC Area under the ROC curve
BP Blood pressure
CI Confidence interval
CMS Centers for Medicare & Medicaid Services
CPU, GPU Central, graphics processing unit
DBP Diastolic blood pressure
ECG Electrocardiogram
eICU eICU Collaborative Research Database
ENSPD École Nationale Supérieure Polytechnique de Douala (NHPSD)
ESH European Society of Hypertension
GRU-D Gated recurrent unit with decay
HbA1c Glycated haemoglobin
HEARTS WHO technical package for cardiovascular care
IHSG International Hypoglycaemia Study Group
IoT Internet of Things
ISO International Organization for Standardization
LASSO Least absolute shrinkage and selection operator
LSTM Long short-term memory
MAD Median absolute deviation
MAR, MCAR, MNAR Missing at random, completely at random, not at random
mHealth Mobile health
MQTT Message Queuing Telemetry Transport
NEWS2 National Early Warning Score 2
NHPSD National Higher Polytechnic School of Douala
PCA Principal component analysis
PPG Photoplethysmography
PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROBAST Prediction model Risk Of Bias Assessment Tool
RCT Randomised controlled trial
ROC Receiver operating characteristic
RPS Ranked probability score
RQ Research question
SBP Systolic blood pressure
SD Standard deviation
SDK Software development kit
SHAP Shapley additive explanations
SMS Short message service
STEPS WHO STEPwise approach to surveillance
TLS Transport Layer Security
TPE Tree-structured Parzen estimator
TRIPOD Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis
WHO World Health Organization

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Introduction

Hypertension is the leading modifiable risk factor for premature death worldwide. The number of adults aged 30–79 years living with hypertension doubled between 1990 and 2019 to reach 1.3 billion, about 78% of whom live in low- and middle-income countries, and elevated systolic blood pressure causes an estimated 10.8 million deaths each year [1]. Sub-Saharan Africa combines a high and rising prevalence with poor detection, treatment and control [2]: in the WHO African Region only 12% of hypertensive adults have their blood pressure controlled. Diabetes compounds the burden. In Africa, 58% of people living with diabetes are also hypertensive [3]; the two conditions share their risk factors, aggravate each other, and call for the same care: repeated measurement and treatment adjusted over decades.

The Follow-up Challenge

Care for hypertension and diabetes in Cameroon is episodic. Blood pressure and glucose are measured at consultations a few times a year, recorded on paper, and rarely assembled into a history that a clinician can read as a trend. Yet diagnosis and treatment adjustment are longitudinal acts: the WHO protocol requires measurements repeated over several visits and monthly follow-up until the target is reached [1, 4]. Two failures follow. Loss to follow-up affects half of patients or more in resource-limited settings, and therapeutic inertia leaves treatment unchanged while blood pressure stays above target [1]. In the African Region, 72% of the patients already diagnosed are not controlled: they are known to the health system and lost during follow-up.

The Potential of Telemonitoring and Prediction

Mobile telephony now reaches most of the population [5], and a controlled study in Cameroonian district hospitals has shown that remote clinical support is feasible in primary care [6]. Telemonitoring, the transmission of home measurements to a care team, lowers systolic pressure by 3 to 7 mmHg and glycated haemoglobin by 0.4 to 0.5 points, provided a clinician acts on the data [7, 8]. Where clinicians are scarce, the system must also decide which readings deserve their time. The only validated protocol for patients with both conditions sorts readings into three levels with fixed thresholds [8]: it reacts to the level of a reading, not to its trend, and uses no learned model. Machine learning could forecast deterioration from a patient's own series, but the reported results of the field are fragile. The same algorithm is reported as the best on one dataset [9] and as no better than chance on another [10], because

the first study predicts a label already encoded in its inputs.

Research Problem and Motivation

This thesis addresses two problems.

1. **A practical problem.** No telemonitoring system evaluated in Central Africa combines consultation records, home measurements and a predictive model for patients with hypertension and diabetes, and orders the clinician's attention under intermittent connectivity and scarce specialist time [11, 12].
2. **A methodological problem.** The predictive results of the field rest on labels that leak into the features, improper metrics and validation on the training population. Whether a learned model forecasts deterioration better than the guideline rule, and whether a model developed elsewhere transfers to Cameroonian patients, has not been measured.

The data available to this work were not collected at home. The public eICU Collaborative Research Database, which records intensive-care stays in United States hospitals, is used to develop and compare the models. The observation charts of 32 patients hospitalised in Douala, with blood pressure, pulse and glucose recorded morning and evening, are used only to validate them externally. No home measurement or sensor data were collected: the system is built for home follow-up, but its predictive component is evaluated on hospital data, and its transfer to home data remains to be established.

Objectives and Research Questions

The general objective is to design a telemonitoring system for patients with hypertension and type 2 diabetes that supports the clinician through risk assessment and alert prioritisation, and to evaluate its predictive component without the biases of previous work. The specific objectives are:

- to derive the requirements of the system and the gaps of the field from the literature (Chapter 1);
- to formalise the risk-tiering problem: a sourced labelling rule, a forecasting task free of label leakage, and the combination of rule and model (Chapter 2);
- to build a documented pipeline of data collection and treatment, and a benchmark protocol that selects the model instead of presupposing it (Chapter 3);
- to compare the candidate models on public multi-centre data, validate them externally on Cameroonian patients, and build the system around the result (Chapter 4).

Four research questions follow:

RQ1. Can a telemonitoring system combining consultation records, home readings and risk prediction be built under the connectivity, workload and cost constraints of Cameroonian care?

RQ2. Do learned models forecast the risk level of the next reading better than the guideline rule, and does a deep temporal model outperform classical models?

RQ3. Does the advantage persist when the forecast is made 24 and 48 hours ahead?

RQ4. Do models developed on another health system transfer to Cameroonian patients?

Hypotheses

RQ1 is a design question, answered by the system and by the traceability of each of its elements to a requirement of the literature (Table 3.12). RQ2 to RQ4 lead to four testable hypotheses, stated with the measures of Section 3.10:

H1. At least one learned model forecasts the level of the next reading with a lower ranked probability score than the guideline rule applied to the last reading (persistence).

H2. This advantage persists at lead times of 24 and 48 hours.

H3. The deep temporal models do not outperform the tabular models.

H4. Performance is lower on the Douala patients than on the development data, and updating the intercept on the local data recovers part of the calibration.

Contributions and Thesis Outline

This work contributes:

- **A data collection and treatment pipeline**: a labelling rule in which every threshold comes from an official source, quality controls that tell artefacts from events, an analysis of the instruments from the readings themselves, and the harmonisation of intensive-care records to the rhythm of ward charts.
- **A benchmark protocol**: candidates taken from published benchmarks rather than chosen in advance, nested validation grouped by hospital, proper scoring rules combined in a declared selection cost, and external validation on Cameroonian patients. The same discipline selects the measure that tracks the evolution of a patient's state.
- **A telemonitoring system**: a patient application and a clinician dashboard implementing the three-level home protocol, a local risk service sharing the code of the benchmark, and server functions for escalation, reminders and verdicts.

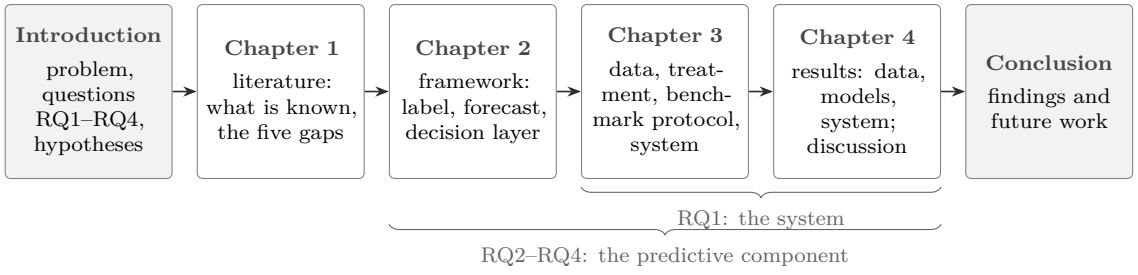


Figure 1: Organisation of the thesis and where each research question is answered.

Figure 1 summarises the organisation. Chapter 1 reviews the clinical and technical literature and identifies the gaps. Chapter 2 formalises the risk-tiering problem. Chapter 3 describes the data, their treatment, the benchmark protocol and the system. Chapter 4 reports the results and discusses them, and the Conclusion summarises the findings and the directions for future work. Every number of the results is produced by a script of the accompanying code, which can be rerun.

Chapter 1

Theoretical Foundations and Literature Review

1.1 Introduction to the Review

A literature review situates a piece of research in its scientific field: it establishes what is already known, how it was established and with what limits, so that the contribution of the work can be stated precisely and defended. For this thesis it is all the more necessary as the subject lies at the crossing of three fields that rarely meet: the epidemiology of chronic diseases in sub-Saharan Africa, the clinical evaluation of telemonitoring, and the engineering of connected and predictive systems. The general context is that of the Introduction: hypertension and type 2 diabetes are frequent and often associated in Cameroon, their care fails mainly at follow-up, specialists are few, records are fragmented and connectivity is intermittent, but mobile telephony is widespread.

The review is built on nineteen core sources, eighteen peer-reviewed studies and the global report of the World Health Organization (WHO) on hypertension, each checked against PubMed, Crossref or the publisher's record; Appendix A summarises each of them. Two further studies, the TELEMED-CAM evaluation [13] and a home monitoring system [14], complete them, together with the guidelines of the WHO [4, 15] and of the European Society of Hypertension (ESH) [16], from which the blood pressure thresholds are taken. Two architecture references [17, 18] are used for the system design.

The chapter is organised as the school plan requires. Section 1.2 defines the key concepts and the models that structure the reasoning. Sections 1.3 to 1.8 analyse previous work in six themes, from the problem to the response (Table 1.1); within each theme the studies are presented in chronological order, each study once, with its method, its main results and its limits. Section 1.9 compares the approaches critically, Section 1.10 identifies the limits of the existing literature, Section 1.11 links them to the present research, and Section 1.12 summarises the review and prepares the transition to the methodology.

Table 1.1: Sources of the review, by theme and in chronological order within each theme.

Theme	Source	Design and setting
1. Burden of disease	Hinneh et al. [3]	systematic review and meta-analysis, Africa
	World Health Organization [1]	institutional global report
2. Care and digital health in sub-Saharan Africa	Kingue et al. [13]	interventional pilot, rural Cameroon
	Bediang et al. [6]	controlled multicentre study, Cameroon
	Dodoo, Al-Samarraie, and Alsswey [11]	review of 66 empirical studies, sub-Saharan Africa
	Gafane-Matemane et al. [2]	narrative review, sub-Saharan Africa
	Agbeyangi and Lukose [12]	systematic review, South Africa, Kenya, Nigeria
	Segun-Omosehin et al. [5]	perspective, sub-Saharan Africa
3. Mobile health and clinical evidence	Alessa et al. [19]	review and content analysis of applications, United Kingdom
	Mao et al. [7]	meta-analysis of 51 randomised trials
	Zhou et al. [20]	meta-analysis of randomised trials
	Bandeira et al. [21]	updated meta-analysis of randomised trials
	Mihevc et al. [8]	multicentre randomised trial, Slovenia
4. Sensors and IoT	Alhaddad et al. [22]	review of non-invasive glucose sensing
	Tai et al. [23]	IoT architecture and deep learning technique
	Efendi et al. [9]	IoT and cloud implementation with device validation
	Liu and Wang [24]	review of IoT sensing and machine learning
	Liu et al. [25]	smartphone photoplethysmography study
5. Machine learning	Chatrati et al. [14]	home monitoring system with machine learning
	Mendoza-Cano et al. [10]	cross-sectional predictive study, Mexico
6. Privacy and ethics	Alhammad et al. [26]	systematic review of patient perspectives

1.2 Conceptual Framework

1.2.1 Key concepts

The terms of Table 1.2 are used throughout the thesis with the meaning given here. Several of them are used differently in the literature; fixing them avoids ambiguity.

1.2.2 Models structuring the work

Four models structure the reasoning of the thesis (Figure 1.1).

The care cascade. The WHO describes hypertension care as a cascade in which patients are lost at each stage [1]. The model locates where an intervention acts: screening acts on diagnosis, whereas longitudinal follow-up acts on treatment and control, where most diagnosed patients in Africa are lost (Section 1.3.2).

The measurement model. A reading is the patient's usual pressure plus a variable error, and repeated readings reduce the error in proportion to their number. This simple model justifies home self-measurement and the use of series rather than isolated values (Section 1.3.5).

Table 1.2: Key concepts and the definitions adopted in this thesis.

Concept	Definition adopted
Hypertension	systolic pressure $\geq$ 140 mmHg, diastolic pressure $\geq$ 90 mmHg, or antihypertensive treatment [1]; the ESH grades (grade 1 from 140/90, grade 2 from 160/100, grade 3 from 180/110 mmHg) are used for the risk levels [16]; the home threshold is 135/85 mmHg
Type 2 diabetes	chronic hyperglycaemia due to insulin resistance and relative insulin deficiency; its diagnosis and follow-up require a biological measurement (capillary or venous glucose, glycated haemoglobin)
Comorbidity	coexistence of hypertension and type 2 diabetes in the same patient
Care cascade	nested stages of care—hypertensive, diagnosed, treated, controlled—measured on the same population [1]
Loss to follow-up	interruption of the planned follow-up of a patient already in care
Therapeutic inertia	failure to intensify treatment while blood pressure remains above target
Telemedicine	delivery of care at a distance through information and communication technologies; it includes teleconsultation, tele-expertise between professionals and telemonitoring
Telemonitoring	transmission of physiological data from the patient's home to a care team, which provides clinical feedback
Mobile health (mHealth)	health practices supported by mobile devices: applications, text messages, connected devices
Self-measurement	measurement of blood pressure or capillary glucose by the patient at home, with a validated device and a standard protocol
Internet of Things (IoT)	connected devices that acquire and transmit data without human intervention; in health, sensors and instruments linked to a platform
Edge and cloud	processing on a device close to the patient, such as the phone (edge), and storage and processing on remote servers (cloud)
Photoplethysmography (PPG)	optical measurement of blood volume variations in the skin, from which heart rate is derived
Clinical decision support	system that turns patient data into alerts or recommendations for a clinician, who keeps the decision
Triage	ordering of readings or patients by priority for clinical review
Supervised learning	estimation of a function mapping input features to a known label; here, a forecast of a future clinical state from a past window of readings
Label leakage	situation in which the label can be recovered from the inputs, so that measured performance reflects a rule rather than a prediction
Proper scoring rule	measure of probabilistic forecasts that a model cannot improve by reporting probabilities other than its own [27]
External validation	evaluation of a model, without refitting, on patients from another setting than those it was developed on

The layered telemonitoring architecture. Remote monitoring systems are described as a chain from acquisition to processing and visualisation [23], distributed over body, environment, edge and cloud layers [17]. The model structures the system of Section 3.11.

Supervised learning for clinical prediction. A model maps features computed from a past window to a future clinical state. The framework fixes what a valid evaluation requires: a label outside the input window, proper measures suited to rare events, and a comparison with the guideline rule (Section 2.3). Behaviour change techniques, coded with the taxonomy used by Zhou et al. [20], complete it on the patient side.

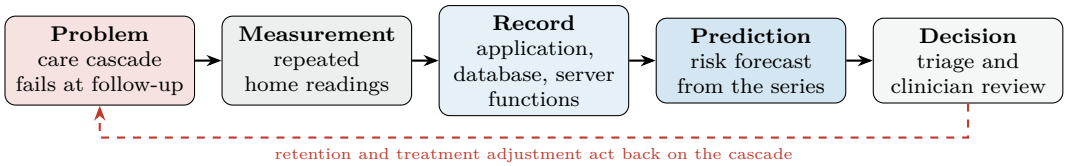


Figure 1.1: Conceptual framework of the thesis.

1.3 Burden of Hypertension and Diabetes

1.3.1 The global burden of hypertension

The WHO global report on hypertension [1] is the reference framing of the problem. For population surveillance it defines hypertension as a systolic blood pressure (SBP) $\geq$ 140 mmHg, a diastolic blood pressure (DBP) $\geq$ 90 mmHg, or the use of antihypertensive medication, in adults aged 30–79 years; this threshold is used throughout this thesis for epidemiological figures. The ESH uses the same threshold and divides hypertension into three grades [16], which are used later to define the risk levels; some publications use the lower 130/80 mmHg threshold of the American guidelines, which explains part of the discrepancies between them.

The report establishes that the number of hypertensive adults rose from 650 million in 1990 to 1.3 billion in 2019, while the age-standardised prevalence barely changed, from 32% to 33%: population growth and ageing, not a rise in individual risk, drive the increase. Elevated systolic pressure is the leading risk factor for premature death among the 87 factors compared, ahead of smoking and hyperglycaemia, with 10.8 million deaths and 235 million disability-adjusted life years each year. Progress towards the global target is insufficient: the prevalence of uncontrolled hypertension fell only from 29% in 2010 to 26% in 2019, and no WHO region is on track. The report notes, however, that the target would be met if about half of the adults already diagnosed were effectively treated. Among the recommended responses, it endorses team-based care, community and home blood pressure measurement, and telemonitoring. As an institutional synthesis, the report relies on modelled estimates and does not address patient-level monitoring technology; it is used here for the global framing and for its normative positions, not as primary evidence.

1.3.2 The failure of the care cascade

The WHO structures care as a cascade of nested stages measured on the same denominator, all hypertensive adults (Table 1.3). Two observations follow. The European Region has a higher prevalence than the African Region but a control rate more than twice as high: the African deficit is organisational rather than epidemiological. And the conditional losses are spread along the whole chain: in the African Region, 57% of hypertensive adults are undiagnosed, 37% of the diagnosed are untreated and 56% of the treated are not controlled. Among patients already diagnosed, 72% (31 of 43 points) are not controlled: they are known to the health system and lost during follow-up. Screening acts only on the first link of the chain; longitudinal follow-up acts on the second and third.

Table 1.3: Hypertension care cascade by WHO region, adults aged 30–79 years, 2019 [1].

Indicator	African Region	World	Americas	Europe
Age-standardised prevalence	36%	33%	35%	37%
Diagnosis coverage	43%	54%	70%	66%
Treatment coverage	27%	42%	60%	53%
Effective treatment (control)	12%	21%	36%	26%

1.3.3 The hypertension–diabetes comorbidity

The association between hypertension and type 2 diabetes rests on shared risk factors—physical inactivity, unbalanced diet and obesity—and on a direct mechanism: diabetes stiffens small vessels, which raises peripheral resistance and blood pressure, and hypertension accelerates the vascular damage of diabetes.

Hinneh et al. [3] produced the first systematic quantification of this comorbidity in Africa. Their meta-analysis, conducted according to PRISMA on PubMed, EMBASE and HINARI up to March 2023, pooled 41 cross-sectional studies totalling 144 433 participants: 58.1% of people living with diabetes in Africa are also hypertensive (95% CI 52.0–63.2), and 60.8% of those with type 2 diabetes. The risk rises with age, excess weight, the number of years lived with diabetes, urban residence and male sex, and patients with both conditions paradoxically have lower chances of receiving appropriate care before complications occur. The sub-regional estimates place Central Africa highest (Table 1.4 and Figure 1.2).

Table 1.4: Sub-regional pooled prevalence of hypertension among people with diabetes in Africa [3].

Sub-region	Pooled prevalence	95% CI	Studies
Central Africa	77.6%	53.0–91.4	2
Southern Africa	69.1%	59.8–77.1	10
Eastern Africa	53.0%	45.8–59.1	15
Western Africa	51.5%	41.8–61.1	10

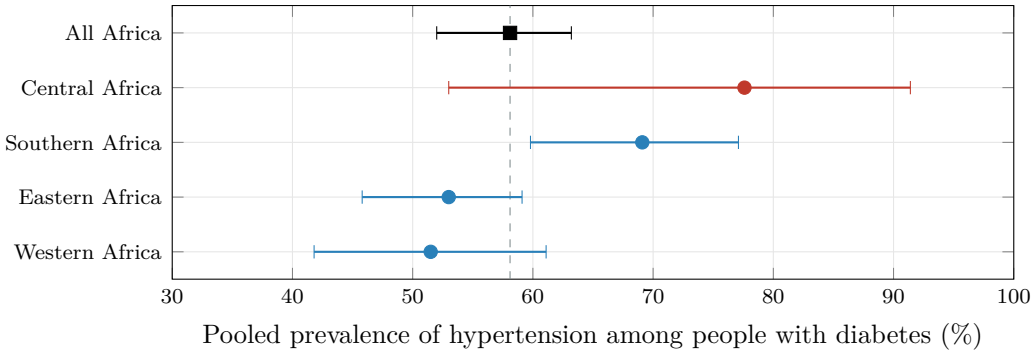


Figure 1.2: Estimates and 95% intervals of Table 1.4; the dashed line marks the pooled African estimate. The Central African interval spans 38 points.

Reliability of the Central African estimate. The value of 77.6% must be read with its interval. It rests on two small Cameroonian hospital series (109 and 210 patients), while three other Central African studies listed in the article (1 832 patients, prevalences of 35.0–50.4%) do

not enter the subgroup. Heterogeneity is extreme ($I^2 \approx 98\%$), the included studies are mostly hospital-based with varying diagnostic thresholds, and the restriction to English-language publications under-represents francophone Central Africa, as the authors acknowledge. The figure also measures hypertension among people with diabetes, not the reverse. It cannot carry the case for this work on its own. That case rests on the continental estimate of 58.1%, on the mechanism, and on the position of the WHO [1]: uncontrolled hypertension is among the risk factors most strongly associated with premature death in people with diabetes; Member States have adopted a target of 80% of people with diagnosed diabetes having good blood pressure control; and the HEARTS package developed for hypertension is explicitly applicable to diabetes and other long-term conditions.

1.3.4 Comparability of the estimates

The prevalences quoted in this chapter are not directly comparable. They differ in reference population (general population, people with diabetes or hospital attendees), in diagnostic threshold (140/90 or 130/80 mmHg), in age structure, in the number of measurement visits—a single-visit measurement overestimates prevalence—and in recruitment, since opportunistic screening and self-selection produce samples that differ from probability sampling. None of the available sources is nationally representative for Cameroon. The nature of each estimate is therefore stated where it is used.

1.3.5 Why one reading is not enough

The WHO diagnostic protocol is longitudinal by construction [1]. Blood pressure is measured twice at the first encounter, under standard conditions (seated, back supported, appropriate cuff), and the decision depends on the level observed (Table 1.5); hypertension is diagnosed only if the threshold is exceeded at two visits. A care model that cannot guarantee the second visit does not produce imperfect diagnoses: it produces none.

Table 1.5: Recommended action by confirmed pressure at the first encounter [1].

Pressure on both readings of the day	**Recommended action**
SBP $\geq$ 160 or DBP $\geq$ 100 mmHg	start treatment the same day
SBP 140–159 or DBP 90–99 mmHg	confirm at another visit, one to four weeks later

The reason is the variability of blood pressure. A reading of patient i at time t can be written $b_{i,t} = \mu_i + \varepsilon_{i,t}$, where μ_i is the patient's usual pressure, the clinically relevant quantity, and $\varepsilon_{i,t}$ gathers circadian variation, activity, stress, the measurement setting and instrument error. With n readings far enough apart for their errors to be independent,

$$\mathrm{Var}(\hat{\mu}_i) = \sigma_\varepsilon^2 / n. \tag{1.1}$$

A consultation provides two to four readings a year; home follow-up provides several dozen (Figure 1.3), which estimate μ_i far more precisely and also estimate the patient's own variability σ_ε. This is the central argument for home measurement.

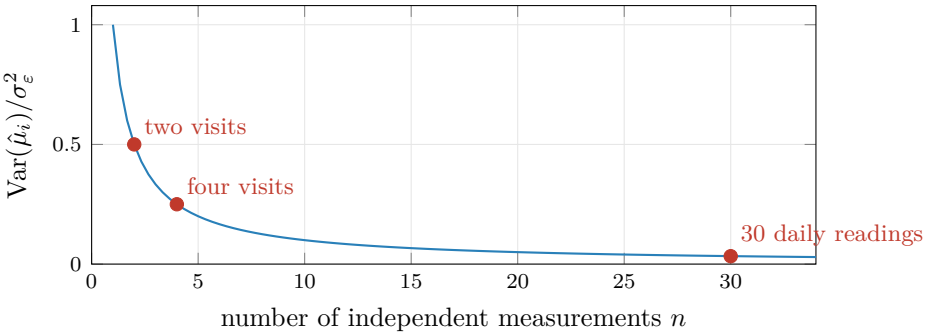


Figure 1.3: Relative variance of the estimate of the usual pressure against the number of independent measurements.

Follow-up must moreover be frequent. Recommendation 7 of the WHO guideline on the pharmacological treatment of hypertension sets monthly follow-up after any treatment change until the target is reached, then every three to six months [4], and the HEARTS treatment-protocol module adds that the return interval should usually be four weeks or less during titration [15]. With 56% of treated patients uncontrolled in the African Region, the number of contacts required per patient and per year is

$$\bar{n} = 0.56 \times 12 + 0.44 \times n_c \approx 7.6 \text{ to } 8.5, \qquad n_c \in [2, 4], \tag{1.2}$$

where n_c is the number of yearly contacts of a controlled patient. An episodic model that delivers one or two contacts a year delivers between an eighth and a quarter of what the guideline requires. This cumulative gap, more than the length of each consultation, is the structural deficit that telemonitoring must fill. It also explains therapeutic inertia: intensifying treatment requires a history showing that the target is persistently missed, and that history does not exist.

1.4 Hypertension Care and Digital Health in Sub-Saharan Africa

1.4.1 Burden, determinants and care gaps

Gafane-Matemane et al. [2] synthesise the hypertension literature of sub-Saharan Africa published between January 2017 and June 2023, drawn from MEDLINE, PubMed, CINAHL, African Journals Online and grey literature. Most countries of the region reported prevalence data over the period, confirming a high burden with poor detection and control. The authors relate the rise of hypertension to the shift towards a westernised lifestyle—salt, sugar and fat, physical inactivity, obesity, tobacco and alcohol—together with socio-economic and psychosocial factors, and they identify infection with the human immunodeficiency virus as a regional specificity; among the mechanisms still under investigation they list sodium handling, endothelial function and circadian profiles such as nocturnal dipping. They distinguish individual obstacles to care (health literacy, cost, distance) from system obstacles (medicine availability, staffing and training, uneven political will). Community-based task shifting

improves linkage to care and medication adherence, but its effect on blood pressure over time is inconsistent.

Their inventory of national surveys is the most recent available for Central Africa (Table 1.6). Dispersion is wide, from 13.5% to 42.1%, and reflects the diversity of instruments as much as real differences; three countries of the sub-region have no data for the period. In Cameroon, the May Measurement Month campaign of 2019 found a prevalence of 20.8%, with only 11.2% of hypertensive participants controlled. The review is narrative rather than pooled, the surveys it gathers are heterogeneous, and some figures are inconsistent: the same pair of values (48% and 34%) is presented once as prevalence and once as diagnosis rate. It is used here for the regional profile and the care cascade, not for pooled prevalence.

Table 1.6: Most recent hypertension surveys in Central Africa (adapted from [2], Table 1).

Country	Instrument	Year	n	Prev.	Aware.	Treat.	Control
Burundi	hospital screening	2018	353	16.7%	–	–	–
Cameroon	May Measurement Month	2019	30 167	20.8%	29.9%	24.0%	11.2%
Central African Rep.	WHO STEPS	2017	3 265	42.1%	30.2%	–	–
Comoros	national survey	2019	2 602	38.4%	47.9%	45.4%	30.2%
DR Congo	May Measurement Month	2019	29 857	25.5%	33.1%	23.2%	11.9%
Congo (Rep.)	May Measurement Month	2019	3 157	33.5%	42.6%	37.3%	23.3%
Gabon	retrospective study	2013–15	2 104	13.5%	–	–	–
Chad, Eq. Guinea, São Tomé	no data	–	–	–	–	–	–

1.4.2 Why the episodic model of care fails

The WHO documents the mechanisms behind the failure of the cascade [1]. *Loss to follow-up* often affects half of patients or more in resource-limited settings; the report therefore holds that screening is useful only if linked to effective follow-up, and it recommends opportunistic detection in health facilities together with the retention of identified patients rather than population screening. *Therapeutic inertia*, the tendency not to intensify treatment despite uncontrolled blood pressure, explains why the last link of the cascade is the weakest; a simple standardised protocol helps reduce it. Lifestyle change alone lowers systolic pressure by only about 1.3 mmHg in primary-care trials, so almost every diagnosed patient needs medication, and its timely adjustment. *Consultation time* is often below five minutes in primary care, which requires digital tools to be fast and to replace paper rather than add to it. Finally, *paper records* are duplicated, lost and unavailable during the visit, whereas a secure longitudinal record is described as essential for hypertension, diabetes and other long-term conditions.

The core indicators of the HEARTS programme define what such a record must produce (Table 1.7). Two of the four are temporal and can only be computed from a dated history of each patient, which a register of consultations does not keep.

Table 1.7: Core indicators of the HEARTS programme [1].

Indicator	Definition	Temporal
Programme coverage	hypertensive patients registered in the catchment area	no
Blood pressure control	registered patients with BP $<$ 140/90 mmHg over the last three months	yes
Retention in care	registered patients with a visit in the last three months, among those registered at least twelve months ago	yes
Availability	essential medicines and measurement devices available	no

The programmes cited by the WHO point in the same direction. In Sri Lanka, 86% of enrolled patients return every three months and 57% are controlled; in the Philippines, India and Cuba, part of the follow-up was moved out of the facility, towards the home or the community. These are associations observed in programmes that changed several components at once, not causal estimates. The Indian programme also shows that retention depends on medicine supply: stock-outs soon after launch discouraged patients from returning, so that a telemonitoring system deployed during stock-outs would probably not improve control, whatever its technical performance.

1.4.3 Telemedicine evidence from Cameroon

Two controlled evaluations of cardiovascular telemedicine have been conducted in Cameroonian primary care (Table 1.8).

TELEMED-CAM (2008). Kingue et al. [13] evaluated a hypertension management package in thirty rural health centres supported by a telemedicine centre at Yaoundé General Hospital. The package combined a standardised protocol restricted to generic drugs, lifestyle counselling, training of general practitioners and nurses, and a telecare channel through which providers reported patient data daily by text or voice message and received telephone feedback. Among 268 participants followed for 24 weeks, attendance at scheduled visits was higher in the intervention arm at every grade of hypertension ($p \leq 0.04$), and systolic pressure fell from the second visit. The benefit on control was confined to stage III hypertension (no improvement in 16.7% against 33.3%, $p = 0.04$); for stages I and II the proportions at target were similar (65.2% against 70%, $p = 0.20$). Allocation was not strictly random, clustering was ignored, the arms differed at baseline (systolic pressure 169 against 161 mmHg, diabetes 9.1% against 43.6%), and telecare was only one component of the package, so its own effect cannot be isolated. The study also quantifies the cost of systematic remote review: about six calls per provider per day, of about five minutes each.

Tele-expertise in district hospitals (2016–2017). Bediang et al. [6] compared two district hospitals equipped with a twelve-lead electrocardiograph and an internet tele-expertise platform (Mbouda and Akonolinga) with two matched control hospitals (Foumbot and Sa'a). Among 171 patients, access to an electrocardiogram with a cardiologist's opinion reached 98.9% against 33.3% (relative risk 2.97), the expert modified the local provider's diagnosis in 46.2% of cases against 17.9%, and patient satisfaction was higher. The study was not randomised and the groups differed at baseline; its primary outcome measures the availability

of the service rather than its clinical effect. The reported blood pressure benefit (40.9% against 17.9% of patients with a favourable evolution) counts the 52 control patients who never returned as not improved; restricted to the patients actually reviewed at three months, the proportions become 41.3% against 53.8%. The specialist's opinion also took 10.3 days on average against a protocol maximum of 72 hours, no faster than in the control centres (11.2 days, $p = 0.82$): the bottleneck was expert availability, not transmission.

Table 1.8: The two Cameroonian evaluations of cardiovascular telemedicine.

	Kingue et al. [13]	**Bediang et al. [6]**
Period	2008	2016–2017
Function evaluated	telemanagement of treatment	tele-expertise for diagnosis
Design	prospective, partly randomised by centre	controlled, matched centres, not randomised
Participants	268	171
Technology	mobile handset, text and voice messages	portable 12-lead ECG, web platform
Principal result	control improved in stage III only	access to ECG with expertise, 98.9% vs 33.3%
Retention	visit adherence higher at every grade	52 control patients not seen at 3 months
Main weakness	clustering ignored; effect of a package	non-comparable groups; differential attrition

Read together, the two studies support three conclusions. Retention is the most robust result, observed by different mechanisms eight years apart. An effect on blood pressure control is not established in Cameroon. And the value added lies in the clinical decision—a revised diagnosis, a supported prescriber—rather than in a tool placed in the patient's hands. Both studies also show that systematic expert review of every case does not scale where specialists are scarce.

1.4.4 Digital health landscape and barriers to adoption

Dodoo, Al-Samarraie, and Alsswey [11] reviewed 66 empirical studies and government reports on telemedicine in sub-Saharan Africa, drawn from eight databases under adapted PRISMA guidelines. Progress is markedly uneven between countries, and the barriers fall into six dimensions (Table 1.9). Two findings bear directly on design: most projects lost momentum once external funding ended, and health professionals facing a heavy workload perceive telemedicine as an additional task. The authors also report that store-and-forward services suffered from low user satisfaction for lack of real-time interaction. Their inventory lists for Cameroon telecardiology, wound-care and tele-education services, including the TELEMED-CAM project, and records 41 countries with national digital health strategies. Francophone countries, Cameroon included, are under-represented in the literature they review.

Agbeyangi and Lukose [12] extend this picture with a PRISMA-guided systematic review of 53 studies published between 2014 and 2024, comparing South Africa, Kenya and Nigeria. South Africa leads adoption, including specialist teleconsultation and chronic disease management; Kenya shows strong integration of mobile health; Nigeria is advancing remote diagnosis under infrastructural constraints. Regulatory gaps, digital literacy and infrastructure remain

Table 1.9: Six dimensions of barriers to telemedicine in sub-Saharan Africa [11].

Dimension	Definition given by the authors
Technological	information systems and connectivity
Organisational	stakeholder engagement, sustainability of systems
Individual	users' experience, knowledge and awareness
Financial	financial resources and operating cost
Cultural	social constraints and traditional beliefs
Legal and regulatory	policies, legislation, confidentiality and privacy protection

the persistent barriers. Central Africa is absent from the comparison.

Segun-Omosehin et al. [5] assess the digital health landscape of the region from the perspective of cardiovascular care. More than 80% of the population holds a mobile telephone subscription, mobile internet penetration was a quarter in 2023, and the adoption rate of smartphones is expected to reach 88% by 2030; yet about 550 million people lack reliable access to electricity and the internet, and up to 18% of countries had no registered cardiologist in 2020. The collapse of the Rwandan platform Babyl in 2023, after about 2.5 million users had registered, illustrates that adoption is not sustainability. Artificial intelligence remains largely untested in the region, with few validation studies in African populations and a documented risk that externally trained algorithms perform poorly there. As a perspective article without a declared search strategy, this source is used for orders of magnitude, not as pooled evidence.

Taken together, these sources describe a population that is reachable by mobile networks and a specialist who is not. A system that forwards every reading to a clinician reproduces the scarcity it aims to relieve; a system that sorts readings and orders the clinician's attention addresses it. They also impose offline operation, a simple interface for both patients and providers, and a documented operating cost.

1.5 Mobile Health and Clinical Evidence

1.5.1 What self-management applications offer

Alessa et al. [19] analysed the 186 hypertension self-management applications available in English on the two main application stores in January 2018, coding thirty of them in depth. Most offered a narrow set of functions: 58.1% offered a single function, only 17.7% three or more, and none all eight identified. Self-monitoring was present in 53.2% of applications, feedback in 28.0% and reminders in 21.0%; most of the applications that tracked blood pressure required the patient to type in the value displayed by the monitor. None of the thirty analysed in depth had been scientifically evaluated or certified as a medical device, and none reported the involvement of patients in its design. Three applications claimed to measure blood pressure with the smartphone camera without any evidence of validity. Of the thirty, only twenty had a privacy policy, and fourteen of these declared sharing data with third parties. Coded with the taxonomy of behaviour change techniques, the applications contained a median of nine techniques, with self-monitoring, prompts and cues, and action planning present in all thirty. As a content analysis of a market snapshot, the study describes

what applications contain, not whether they work.

1.5.2 Pooled clinical effects

Mao et al. [7] pooled 51 randomised trials published between 2008 and 2019, with 13 054 participants with diabetes or hypertension. Mobile health interventions reduced glycated haemoglobin (HbA1c) by 0.39 points (−0.50 to −0.29), fasting glucose by 0.52 mmol/L, systolic pressure by 2.99 mmHg (−4.19 to −1.80) and diastolic pressure by 1.14 mmHg. Two subgroup results are decisive for this work. Interventions combining mobile health with professional management reduced systolic pressure by 6.17 mmHg (−8.83 to −3.50), whereas mobile health alone produced a non-significant 2.16 mmHg. And the effect on systolic pressure was present in developed countries (−5.72 mmHg) but absent in developing countries (+0.25 mmHg, −3.10 to 3.59), whose trials used mobile health without professional management; the glycaemic effect, by contrast, was larger in developing countries (−0.52 against −0.35 points). Only one of the 51 trials was conducted in sub-Saharan Africa. Heterogeneity was moderate to high (I^2 of 57–67%) and most trials were short.

Zhou et al. [20] focused on self-management through mobile applications for adults with hypertension: twenty randomised trials published from 2017 onwards, sixteen of them meta-analysed, with 4 168 participants from nine countries, none in sub-Saharan Africa. Systolic pressure fell by 5.78 mmHg (−7.97 to −3.59; low certainty, $I^2 = 82\%$) and diastolic pressure by 3.28 mmHg (moderate certainty). The subgroup analyses link the effect to the design of the intervention (Table 1.10): interventions with at least eleven behaviour change techniques and those addressing several risk factors were markedly more effective, and three-month interventions tended to outperform longer ones, a difference that is not statistically significant ($P = .09$) and that the authors attribute to the weakening of habit after about twelve weeks. Fifteen of the twenty trials were at high risk of bias, and interventions based mainly on text messages or calls—the type most often evaluated in Africa—were excluded.

Table 1.10: Subgroup analyses on systolic pressure [20]. BCT: behaviour change technique.

Factor	Modality	Effect on SBP (95% CI)	P
Risk factors addressed	multiple	−6.50 mmHg (−9.00 to −3.99)	.007
	single	−1.54 mmHg (−4.15 to 1.06)	
Number of techniques	≥ 11 BCT	−9.68 mmHg (−13.49 to −5.87)	<.001
	< 11 BCT	−2.88 mmHg (−3.90 to −1.86)	
Duration	3 months	−8.87 mmHg (−10.90 to −6.83)	.09
	≥ 6 months	−5.76 mmHg (−8.74 to −2.77)	

Bandeira et al. [21] updated the synthesis of randomised trials of mobile health interventions on blood pressure. They confirm statistically and clinically significant reductions across diverse populations and settings, and identify telemonitoring with clinician feedback as the most effective modality. Heterogeneity between interventions, short follow-up in several trials and the scarcity of data from low- and middle-income countries limit the generalisation of these results.

1.5.3 A randomised trial in the target population

Mihevc et al. [8] conducted the trial most directly relevant to this work: a multicentre, open-label randomised trial in three Slovenian primary-care centres between 2021 and 2023, enrolling 128 patients aged 65 or more with both hypertension and type 2 diabetes. In addition to usual care, the intervention group measured blood pressure and glucose at home with Bluetooth devices that transmitted automatically through a smartphone to a cloud platform, and a telemedicine centre coordinated by a general practitioner reviewed the values according to a three-level protocol (green, yellow, red), adopted in this work (Section 2.4.2). After twelve months, systolic pressure was 6.9 mmHg lower than in the control group (-11 to -2.7), after a difference of 10.5 mmHg at six months, and HbA1c was 0.5 points lower.

Three further results matter. Telemonitoring had no effect on other cardiovascular risk factors, on diabetes-related quality of life or on lifestyle: the authors credit the clinical effect to better contact with the care team, better adherence, earlier recognition of worsening and treatment adapted on reliable measurements, not to behaviour change. Seven of the 64 patients of the intervention arm (10.9%) withdrew in the first month because they could not use the equipment, despite one hour of individual training. And about a third of the systolic effect was lost between six and twelve months. The trial was conducted in a high-income setting, among older adults, without blinding; its application to low-resource settings requires adaptation.

1.5.4 Synthesis of the clinical evidence

Table 1.11 and Figure 1.4 gather the available estimates in chronological order.

Table 1.11: Estimates of the effect of mobile health and telemonitoring on systolic pressure and glycaemic control.

Source	Design	Effect on SBP (95% CI)	HbA1c
Mao et al. [7], overall	30 RCTs on SBP	-2.99 mmHg (-4.19 to -1.80)	-0.39
with professional management	subgroup	-6.17 mmHg (-8.83 to -3.50)	–
mobile health alone	subgroup	-2.16 mmHg (-5.07 to 0.75), n.s.	–
developing countries	subgroup	$+0.25$ mmHg (-3.10 to 3.59), n.s.	-0.52
Zhou et al. [20]	16 RCTs	-5.78 mmHg (-7.97 to -3.59), low certainty	–
Mihevc et al. [8]	RCT, 128 patients	-6.9 mmHg (-11 to -2.7)	-0.5

Four conclusions emerge. First, mobile health and telemonitoring produce a modest but clinically relevant effect, of the order of -3 to -7 mmHg on systolic pressure and -0.4 to -0.5 points on HbA1c; these are intermediate outcomes, and none of the sources measures morbidity or mortality. Second, the effect depends on professional oversight: Mao et al. [7], Bandeira et al. [21] and Mihevc et al. [8] converge on this point by independent routes, and so do the Cameroonian studies of Section 1.4.3. Third, the evidence from sub-Saharan Africa is almost absent, and the absence of effect observed in developing countries is plausibly a design gap—no clinician at the end of the chain—rather than a context effect. Fourth, two problems remain open: the erosion of the effect over time, and the inability of a fraction of patients to

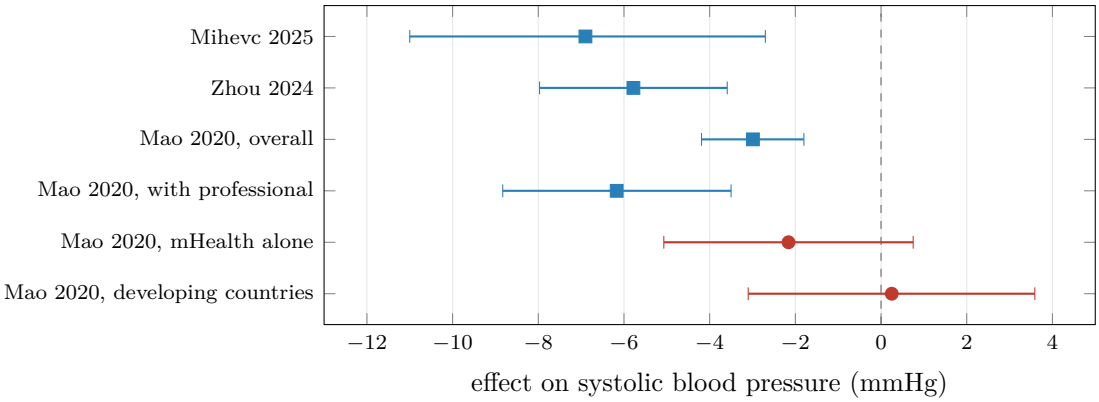


Figure 1.4: Estimates of Table 1.11 with their 95% intervals. Squares: intervals that exclude no effect; circles: intervals that include it. The two estimates without a significant effect describe interventions without professional management or conducted in developing countries.

use the equipment. Telemonitoring must therefore be built as decision support linked to a clinician, not as a self-tracking tool handed to the patient.

1.6 Devices, Sensors and Architectures

1.6.1 The normative position on measuring devices

The WHO sets the frame within which any sensor must be assessed [1]. Manual devices should be replaced by independently validated automated electronic devices, and automated upper-arm cuffs are recommended. Cuffless devices, including popular connected wristbands, are not recommended for the management of hypertension, because of unresolved questions about their accuracy and their ability to track blood pressure change; as of 2023, no alternative to the validated upper-arm cuff is accurate enough for diagnosis and management. Only about one automated device in five on the market has evidence of validation.

1.6.2 Non-invasive sensing of glucose and blood pressure

Alhaddad et al. [22] reviewed the non-invasive sensing modalities proposed for blood glucose, namely electrocardiography, electromagnetic sensing, bioimpedance and photoplethysmography (PPG), and the machine learning models applied to them. PPG- and ECG-based sensing combined with deep learning shows promise for detecting glucose trends and hypoglycaemia, but not for clinical-grade absolute measurement: many studies have small samples, inter-individual variation limits generalisation, and accuracy remains below the standards recognised by the Food and Drug Administration.

Liu et al. [25] estimated blood pressure from the PPG signal recorded by a smartphone camera alone, without additional hardware. Combining waveform feature engineering with machine learning, and using SHAP values to trace each prediction back to interpretable physiological features, they report a strong correlation with reference devices and assess agreement by Bland–Altman analysis on 127 participants. The sample is modest and single-site, accuracy remains below that of clinical cuff devices, and generalisation across skin tones, devices and lighting conditions is not established.

These two studies, set against the WHO position and the unvalidated camera applications found by Alessa et al. [19], define a consistent position. Glucose must come from a glucometer and blood pressure from a validated upper-arm cuff, because these are the measurements on which clinical decisions rest. Smartphone PPG is a plausible low-cost source of heart rate and an exploratory source of blood pressure trends, to be evaluated rather than relied upon.

1.6.3 IoT architectures for remote monitoring

Tai et al. [23] address the difficulty of applying conventional machine learning to noisy, heterogeneous and high-dimensional IoT health data. They propose a human-centric deep learning technique for remote chronic disease monitoring, embedded in a standardised three-phase architecture: data acquisition, data processing and data visualisation. The architecture supports accurate and interpretable predictions in their evaluation, which was conducted in controlled conditions and does not address the connectivity constraints of low-resource settings.

Efendi et al. [9] describe and validate a complete IoT system for monitoring elderly people. Its three layers are a physical layer (an optical sensor for heart rate and oxygen saturation and a temperature sensor, read by an ESP32 microcontroller), a network layer (MQTT transmission to a Firebase real-time database) and an application layer (mobile application and website). Firebase was chosen for real-time synchronisation, built-in authentication, and local persistence with automatic merging when connectivity returns. Against a digital oximeter and thermometer, six participants measured for ten minutes at rest gave a mean absolute percentage error of 0.90%, and XGBoost was the best of five classifiers of the user's state as stable or not stable. The system measures heart rate, oxygen saturation and temperature, not blood pressure or glucose; its validation sample is small, the reference instruments are not identified as validated, and its classifier is examined in Section 1.7.

Liu and Wang [24] survey the use of IoT mobile sensing and machine learning for cardiovascular disease, cancer, chronic respiratory disease and diabetes. They classify devices into portable, wearable, intelligent clinical, remote, implantable and ingestible categories, describe a seven-layer platform from device access to decision support, and identify the weaknesses of traditional monitoring: manual processes, information silos, lack of real-time feedback and of early-warning mechanisms. They also note that electronic medical records require cleaning of duplicates, unit errors and values carried over identically between visits. Although presented as a systematic review, the article reports no protocol, flow diagram or number of included studies, so it is treated here as a narrative survey. Controlled settings dominate the studies it covers, and deployment in low- and middle-income countries is poorly represented.

Two architecture references complete the picture. Mucchi et al. [17] decompose remote monitoring into a body layer of worn sensors, an environment layer of ambient sensors, an edge layer, typically a smartphone, that checks the readings and raises a first warning, and a cloud layer that stores the data, performs long-term analysis and manages alerts; they tested their system with ten nurses on simulated patients for twenty days. Usability was rated well, but half of the nurses rated the system's efficiency for chronic disease as sufficient or below, two thirds rated their own digital skills as only fair, and they asked for faster alert notification

and immediate feedback from the doctor. Yazdani et al. [18] add a clinician view that lists patients by priority rather than chronologically and a checkup period configurable for each patient, without implementing them.

Table 1.12: The reviewed architectures: layers, edge processing, implementation and validation.

Source	Layers	Edge	Implementation	Validation
Yazdani et al. [18]	user, function, database	absent	design only	none
Mucchi et al. [17]	body, environment, edge, cloud	smartphone	mobile application and cloud repository	ten nurses, twenty days, simulated patients
Tai et al. [23]	acquisition, processing, visualisation	–	deep learning technique	controlled conditions
Efendi et al. [9]	physical, network, application	absent	hardware and cloud	six volunteers, ten minutes, seated
Liu and Wang [24]	seven layers, from device access to decision support	data transfer only	survey diagram	none

The architectures converge on a common structure: acquisition, transmission and storage, processing, and presentation (Table 1.12). They disagree on the edge, and the disagreement matters: without an edge, every reading travels to the cloud and every decision is taken there; with an edge, readings can be checked and given a first warning locally, without a round trip. Where some 550 million people lack reliable electricity and internet access [5], the edge is a precondition rather than an optimisation. None of these architectures has been validated on patients in routine care in a low-resource setting, and none acquires blood pressure and glucose through its own sensors.

1.7 Machine Learning for Clinical Risk Prediction

Chatrati et al. [14] proposed a smart home monitoring system that analyses home blood pressure and glucose readings and notifies the healthcare provider when an abnormality is detected, combining conditional rules for hypertension with a learned classifier for diabetes. It is the closest published precedent of a system combining both measurements with machine learning. As described, however, it classifies single readings rather than series, and it has not been evaluated on patients in routine care.

Liu and Wang [24] report that artificial neural networks, support vector machines, random forests, gradient boosting and deep learning are the dominant algorithm families for chronic disease monitoring. The accuracies they collect cluster between 95% and 100%, whereas the only recurrent network applied to a genuine clinical prediction task, the readmission of lupus patients, reaches an area under the curve (AUC) of 0.70. They name the black-box nature of these models and their dependence on large labelled datasets as their main limitations.

The classifier comparison of Efendi et al. [9] shows why such figures must be read with care. XGBoost reached an accuracy of 0.973 and the four other models exceeded 0.92. However, the label *stable* is defined as all three parameters lying in their normal ranges, and each feature is recoded as low, normal or high using the same ranges, so the label is a deterministic function

of the inputs:

$$y = 1 \iff (x_1 = 1) \wedge (x_2 = 1) \wedge (x_3 = 1), \tag{1.3}$$

and the perfectly balanced classes (2 250 against 2 250) show that the dataset was constructed rather than observed. The models recover a rule instead of predicting a state. Some reported metrics are also mutually inconsistent, and the training configuration described (epochs, sigmoid activation, Adam optimiser) does not apply to tree-based models. The result does not show that XGBoost outperforms other algorithms.

Mendoza-Cano et al. [10] provide the counter-example of a genuine prediction task. Using 1 308 hypertensive adults from the nationally representative Mexican survey ENSANUT 2022, they predicted uncontrolled blood pressure (130/80 mmHg threshold, prevalence 30.3%) from seven predictors selected by LASSO: sex, age, socio-economic status, years since diagnosis, history of diabetes, source of medical care and adherence to a nutritional plan. Random forest reached an AUC of 0.75 (0.72–0.77), logistic regression 0.61 and XGBoost 0.54 (0.49–0.60), the last being indistinguishable from chance. Variable importance and SHAP values ranked age and years since diagnosis first, and the data and code are public. The best model nevertheless detects only 51% of uncontrolled cases (specificity 0.98); the logistic regression row is computed with a different positive class from the other two, and the discussion attributes to random forest a sensitivity that is in fact its specificity. The design is cross-sectional, and the predictors are demographic and administrative rather than physiological.

The same algorithm is thus the best in one study and the worst in the other. Reported performance is governed by how the task is constructed—the integrity of the label and the informativeness of the features—far more than by the algorithm; across clinical prediction studies, machine learning shows no systematic benefit over logistic regression [28]. Accuracy is moreover an improper measure when the positive class is rare: with prevalence π, a model that never alerts reaches $1 - \pi$. Proper scoring rules [27] and calibration [29] are the measures this literature lacks. Table 1.13 sets the studies side by side with the design adopted here.

Three lessons follow. Reported performance depends on the construction of the task before the algorithm. Ensemble tree models are plausible candidates, and interpretability through feature importance and SHAP is now expected practice, as Mendoza-Cano et al. [10] and Liu et al. [25] both show. And no source forecasts a future physiological state from a patient's own series of blood pressure and glucose readings, nor evaluates such a model in a low-resource setting.

Tracking how a patient changes is a related problem. Patient-similarity research uses many distances without a common benchmark [30, 31]; the largest independent comparison of distance measures for series, nine measures on 38 data sets, separates lock-step measures, which compare aligned elements, from elastic ones [32]; and statistical control charts, which compare a patient's readings with their own baseline, are promising in hypertension and diabetes but little evaluated [33].

Table 1.13: How the reviewed machine learning studies were evaluated, against the design of this work.

Study	Data	Predicted	Reported	Limit for this problem
Chatrati et al. [14]	home BP and glucose readings	state of the current reading	rules and a classifier	single readings, not a series
Tai et al. [23]	monitoring architecture	chronic disease status	controlled conditions	no patient validation
Efendi et al. [9]	constructed set, 2 250 against 2 250	"stable", defined from the input ranges	XGBoost accuracy 0.973	label is a function of the inputs
Mendoza-Cano et al. [10]	1 308 hypertensive adults, national survey	uncontrolled pressure	AUC 0.75 (forest), 0.54 (XGBoost)	cross-sectional; sensitivity 0.51
Liu and Wang [24]	review of IoT studies	various	accuracy 95–100%; AUC 0.70 for readmission	accuracy on unclear tasks
This work	eICU, 174 hospitals; Douala, 32 inpatients	level of the next reading	proper scores against persistence	evaluated on hospital data

1.8 Data Privacy and Ethics

Alhammad et al. [26] conducted the first systematic review of how patients view the protection of the data that mobile health applications collect about them—confidentiality, privacy and security: 33 studies published between 2014 and 2022, identified in PubMed, Ovid and ScienceDirect under PRISMA guidance. Patients' concerns about their data are a consistent and significant barrier to adoption. Encryption, secure authentication and regular risk assessment are recommended as baseline safeguards, and transparency about how data are handled improves trust. Most included studies come from high-income countries and measure self-reported perceptions rather than actual incidents.

These findings converge with earlier observations of this review: most commercial applications either lack a privacy policy or declare sharing data with third parties [19], the legal and regulatory dimension is one of the six barriers to telemedicine in the region [11], and the reference implementation of Efendi et al. [9] stores its data in a cloud region outside the country of use, which raises the question of data residency. Since the present work processes hospital-sourced patient data through a mobile and cloud pipeline, data protection is a design constraint from the start: anonymisation before modelling, encrypted transmission, role-based authentication, and identification of the applicable Cameroonian legal framework (Section 3.13).

1.9 Critical Comparison of the Approaches

The studies reviewed follow five approaches (Table 1.14). None combines what the problem requires: clinical validity, professional oversight, longitudinal data, prediction and suitability for low-resource settings.

Table 1.14: Comparison of the approaches found in the literature.

Approach	Studies	Strengths	Weaknesses
Self-management applications	Alessa et al. [19] and Zhou et al. [20]	low cost, behaviour change techniques	little clinical validation; weak effect without a professional
Rule-based telemonitoring with a clinician	Mihevc et al. [8] and Kingue et al. [13]	validated effect on BP and HbA1c; simple, transparent	fixed thresholds, reacts to level not trend, no learning; costly expert time
Tele-expertise between professionals	Bediang et al. [6]	changes clinical decisions; feasible in Cameroon	no home data; specialist delay unchanged
IoT platforms with machine learning	Efendi et al. [9], Tai et al. [23], and Liu and Wang [24]	complete technical chain; offline storage	no BP or glucose sensors; tiny or controlled validation; label leakage
Predictive models on clinical data	Mendoza-Cano et al. [10] and Chatrati et al. [14]	interpretable risk predictors	cross-sectional or single readings; modest sensitivity; no deployment

Three trends emerge from the chronology. Telemonitoring moves from self-management towards systems linked to a clinician, which the most recent evidence identifies as the condition of effectiveness [7, 8, 21]. Prediction moves from rules towards machine learning, with interpretability tools such as SHAP becoming standard [10, 25]. And sensing moves towards cuffless and smartphone-based measurement [22, 25], with cloud platforms designed to work offline [9].

Four points remain controversial. Cuffless blood pressure measurement is promising in research [25] but not accepted for clinical decisions [1]. The very high comorbidity reported for Central Africa rests on two small hospital series (Section 1.3.3). Whether mobile health works in Africa is open, since the effect disappears in developing countries when no professional is involved [7]. And the same algorithm, XGBoost, is reported as the best [9] and as the worst [10], a contradiction explained by how the tasks were constructed rather than by the algorithm (Section 2.3).

1.10 Limits of the Existing Literature

The analysis reveals five limits that matter for the follow-up of patients with hypertension and diabetes in Central Africa.

Blind spots. Central Africa is almost absent from the evidence: one of the 51 trials of Mao et al. [7] was conducted in sub-Saharan Africa, the comparative reviews cover South Africa, Kenya and Nigeria [12], and francophone countries are under-represented [11]. No evaluated system combines consultation records, home measurements and sensor data for patients with both hypertension and diabetes.

Methodological weaknesses. The Cameroonian studies are not randomised and have unbalanced or incomplete groups [6, 13]; most trials are short and measure intermediate outcomes only; the technical validations rely on few volunteers, favourable conditions and reference instruments of unknown validation [9]; and the machine learning studies suffer from

label leakage, improper metrics or reporting errors, and are validated on the population they were trained on (Section 1.7).

Absence of local data and of measured transfer. No nationally representative estimate exists for Cameroon, and no artificial intelligence tool in circulation has been validated on an African population [5]. Early-warning scores derived elsewhere lose discrimination in African wards and require local validation and recalibration [34, 35]; no learned model trained elsewhere has been validated on Cameroonian patients, so models trained elsewhere cannot be assumed valid here.

A rule without memory. The only validated home protocol for both conditions [8] uses fixed thresholds identical for all patients, reacts to the level of a reading rather than to its trend, and uses no learned model; no source forecasts a future state from a patient's own series of blood pressure and glucose readings.

Clinician time. Systematic expert review does not scale [6, 13], yet no design sizes the alert volume or orders the clinician's work, and none reports its operating cost.

Four questions therefore remain to be explored: forecasting deterioration from home series rather than reacting to single readings; measuring whether such a forecast transfers to Cameroonian patients; detecting a change in the patient's state before a threshold is crossed; and ordering scarce clinician time, with the operating constraints of the system documented.

1.11 Link with the Present Research

Continuity. The present work builds directly on the existing studies. It adopts the validated three-level home protocol of Mihevc et al. [8], the layered architecture with an edge tier of Mucchi et al. [17], offline-first storage as in Efendi et al. [9], the priority-ordered clinician view of Yazdani et al. [18], and the modelling and interpretability practice of Mendoza-Cano et al. [10]. It also follows the conclusion of the clinical literature that telemonitoring must be linked to a clinician.

Originality. It differs from them on five points: a system designed to combine consultation records, home measurements and, possibly, sensor data for both conditions; a learned model applied to each patient's series, to forecast the level of the next reading rather than classify the current one; a hybrid design in which the guideline rule is a floor that the model can raise but never lower; models selected by a benchmark rather than chosen in advance, evaluated with proper scores against the guideline rule; and models developed on public multi-centre data and validated externally on Cameroonian patients, a transfer that no study identified in the region has measured.

Research questions and hypotheses. The research questions and hypotheses stated in the Introduction follow from these limits; Table 1.15 links each of them to the limit it addresses and to the chapter where it is answered.

Table 1.15: From the limits of the literature to the research questions and hypotheses.

Limit of the literature	Research question	Hypothesis	Answered in
Blind spots; clinician time	RQ1: can the system be built under the constraints of Cameroonian care?	design question	Sections 3.11 and 4.7
A rule without memory; methodological weaknesses	RQ2: do learned models forecast the next level better than the guideline rule, and do deep temporal models do better than classical ones?	H1, H3	Section 4.3
A rule without memory	RQ3: does the advantage persist 24 and 48 hours ahead?	H2	Section 4.3
Absence of local data and of measured transfer	RQ4: do models developed elsewhere transfer to Cameroonian patients?	H4	Section 4.4

1.12 Synthesis of the Review

Table 1.16 summarises, for each theme, what the literature establishes, what it leaves open, and the consequence for this work.

Table 1.16: Synthesis of the literature review.

Theme	Established	Not established	Consequence for this work
Burden	cascade fails at follow-up; 72% of diagnosed not controlled; 58% of people with diabetes hypertensive	reliable Cameroonian estimates	longitudinal follow-up; joint blood pressure and glucose
Context	loss to follow-up, inertia, paper records; six barrier dimensions; specialist scarcity	effect of telemedicine on control in Cameroon	offline-first; triage of clinician attention
Clinical evidence	modest effect, conditional on a clinician	effect in sub-Saharan Africa; long-term effect	clinician-in-the-loop decision support
Devices	validated cuff and glucometer required; layered architectures with offline storage	validated cuffless measurement; validation in routine care	manual or Bluetooth entry from validated devices
Machine learning	task construction drives performance; interpretability expected	forecasting from individual series; transfer to African patients	leakage-free task; proper scores; benchmark; external validation
Privacy	data concerns hinder adoption; baseline safeguards known	evidence from African patients	anonymisation; declared data residency

The scientific achievements are clear. Care for both conditions fails at follow-up: in the African Region, 72% of the hypertensive adults already diagnosed are not controlled. Telemonitoring lowers systolic pressure by 3 to 7 mmHg and HbA1c by about half a point when a clinician acts on the data; without one, no effect was shown in developing countries. Validated instruments, layered architectures and interpretable models are available. The gap is equally clear: the only validated protocol for both conditions triages single readings with fixed thresholds, reported machine learning performance depends on how the task is built—the same algorithm is the best in one study and no better than chance in another—and no learned model has been validated on Cameroonian patients. No system combining multi-source data

with a forecast from each patient's series has been designed and evaluated for patients with hypertension and diabetes in Central Africa.

Filling this gap requires two things before any data are analysed: a formal statement of the problem, in which the label is sourced and cannot leak into the inputs (Chapter 2), and a methodology able to treat longitudinal data under local constraints and to evaluate predictive models without the biases identified in this chapter (Chapter 3).

Chapter 2

Risk-Tiering Framework

This chapter formalises the problem the system solves: turning a patient's irregular series of readings into a level of attention for the clinician. It defines the series and the samples, a label whose every threshold comes from an official source, a forecasting task in which the label cannot leak into the inputs, the decision layer that combines the guideline rule with a model, and a measure of how a patient's state evolves. Chapter 3 instantiates each element on the data.

2.1 Patient Series and Samples

A patient i is observed through four channels $c \in \{\mathrm{SBP}, \mathrm{DBP}, \mathrm{pulse}, \mathrm{glucose}\}$, each sampled on its own schedule: a reading is a pair $(t, x_{i,t}^{(c)})$, and a channel may be missing at any time. The series is not resampled onto a grid: interpolating sparse home readings would invent values and understate variability, one of the quantities of interest.

A sample anchored at time t pairs a window of past readings with a label taken outside it:

$$\mathcal{S}_{i,t} = \Big(\{x_{i,s} : t - L < s \le t\},\ y_{i,t'}\Big), \qquad t' = \min\{s > t + \varepsilon\}, \tag{2.1}$$

where L is the window length and $\varepsilon \ge 0$ an embargo that sets the lead time: the target is the first reading after $t + \varepsilon$ (Figure 2.1). Windows are defined in time rather than in number of readings, because every clinical reference is expressed in time—the seven-day profile, the monthly follow-up—and five readings may span four days for one patient and seven months for another. All the samples of a patient are correlated: the patient, not the sample, is the independent unit.

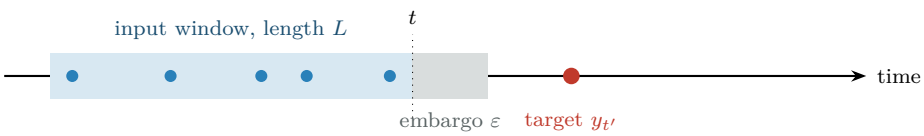


Figure 2.1: Construction of one sample: the readings of the window, the embargo and the first reading after it.

2.2 A Label from Official Thresholds

The label states how much clinical attention a reading calls for, on four ordered levels: 0 stable, 1 moderate, 2 high, 3 critical. It must satisfy six requirements: every threshold comes from an official source; low values count as well as high ones; it applies to a single reading; a missing channel does not bias it; thresholds are versioned; and its agreement with clinical judgement can be measured. The thresholds of Table 2.1 are those of in-hospital monitoring, the setting of the data actually collected; a home deployment substitutes the out-of-office thresholds of Section 2.4.2.

The ESH grades classify a patient's usual pressure from repeated measurements [16]; they are not defined for one reading. They are used here as severity bands of a single reading, and the repetition that diagnosis requires is carried by the input window. The other thresholds keep their own function: 135/85 mmHg governs the home triage, and 140/90 mmHg the epidemiological comparisons of Chapter 1. Four further choices need a justification.

- **Low pressure and pulse.** The National Early Warning Score 2 (NEWS2) adds the scores of six physiological parameters; only a score of 3 in a single parameter, the "red score", calls on its own for urgent review by a clinician [36]. Only that threshold is used for the two NEWS2 parameters recorded here, systolic pressure and pulse; the intermediate bands have meaning only inside the aggregate score. No official source sets a lower threshold for diastolic pressure, so none is used.
- **Pulse.** Tachycardia and hypotension accompany the hypovolaemia of hyperglycaemic crises [37], so the pulse belongs in the label of these patients, at the red-score threshold only.
- **Hypoglycaemia.** Level 1 (< 3.9 mmol/L) is an alert value and level 2 (< 3.0 mmol/L) is clinically significant [38]; they map to high and critical. Level 3 is defined by a clinical event, not a glucose value, and cannot be observed in the records.
- **Severe hyperglycaemia.** The diagnostic criteria of ketoacidosis and of the hyperosmolar state require ketones, pH or osmolality, which the charts do not record [37]. No accepted cut-off exists for severe hyperglycaemia; the 300 mg/dL of the official hospital-harm measure [39] is used as a convention, and a sensitivity analysis caps hyperglycaemia at the high level, so that a conclusion that changes between the two versions is reported as depending on it.

Boundaries are stored in the unit of their source and converted at comparison, so that no rounding moves a reading across a level; the table is versioned, and every result records the version it was produced with.

Within a condition the worst channel governs, and one label is computed per condition:

$$y_t^{\mathrm{H}} = \max\bigl(s_{\mathrm{SBP},t}, s_{\mathrm{DBP},t}, s_{\mathrm{pulse},t}\bigr), \qquad y_t^{\mathrm{G}} = s_{\mathrm{glu},t} \text{ if glucose is measured at } t,$$
$$y_t = \max\bigl(y_t^{\mathrm{H}}, y_t^{\mathrm{G}}\bigr), \tag{2.2}$$

where s is the level of a channel and the maximum runs over the labels that exist. Within a condition the worst channel governs because an additive score is validated only on the

Table 2.1: Severity of a single reading, per channel. Pressure in mmHg, pulse in beats per minute, capillary glucose in mmol/L.

Level	Haemodynamic channels	Glucose
3 critical	SBP $\geq$ 180 or DBP $\geq$ 110, grade 3 [16]; SBP $\leq$ 90; pulse $\leq$ 40 or $\geq$ 131 [36]	$<$ 3.0, level 2 hypoglycaemia [38]; $>$ 16.7 (300 mg/dL) [39]
2 high	SBP 160–179 or DBP 100–109, grade 2	3.0 to $<$ 3.9, level 1 hypoglycaemia; $\geq$ 10.0 (180 mg/dL) [40]
1 moderate	SBP 140–159 or DBP 90–99, grade 1	$>$ 7.8 (140 mg/dL) [40]
0 stable	otherwise	otherwise

complete NEWS2 parameter set, not on two or three of its parameters. A single label for both conditions would read a missing glucose as a normal one; since glucose is ordered for some patients and not for others, the error would be systematic, so the glycaemic label exists only where glucose is measured (Figure 2.2). The *alert* of a condition is a level of high or critical. Figure 2.3 draws the bands of the two channels that carry most of the alerts; both are U-shaped, with a critical level at each end.

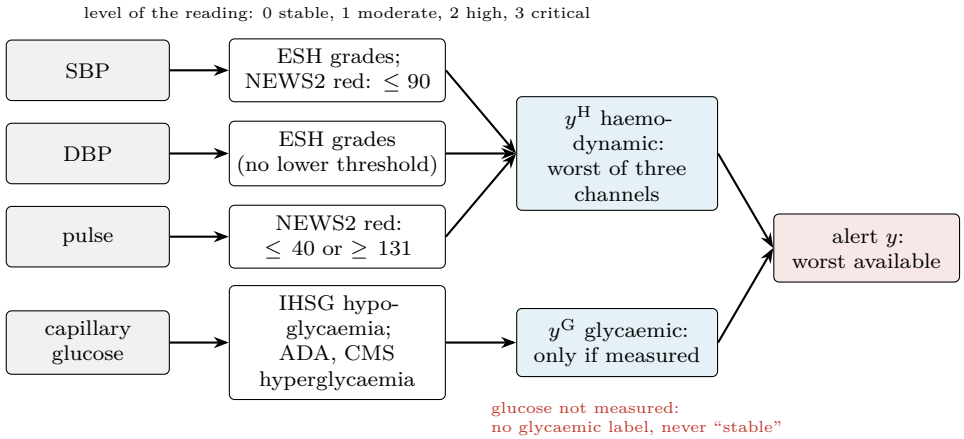


Figure 2.2: Construction of the label at a target reading. Each channel is graded by its sourced thresholds (Table 2.1); the worst channel governs within a condition; the alert is the worst of the condition labels that exist.

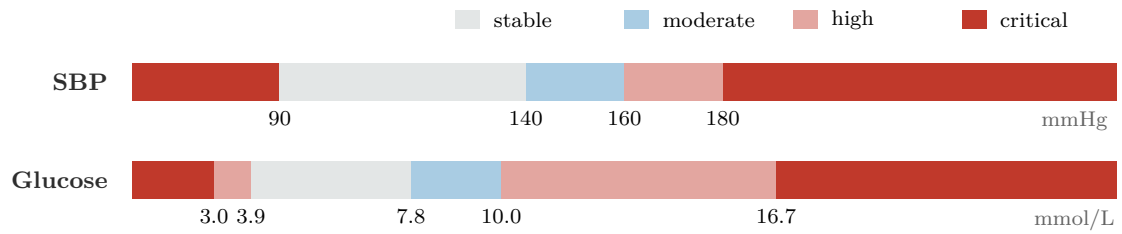


Figure 2.3: Level of a single reading for systolic pressure and capillary glucose (Table 2.1). Diastolic pressure and pulse follow the same pattern with their own thresholds.

2.3 The Forecasting Task

2.3.1 From label recovery to forecasting

The same algorithm, XGBoost, is reported as the best model by Efendi et al. [9] and as no better than chance by Mendoza-Cano et al. [10]. The difference does not come from the algorithm: the first study measures the recovery of a rule already encoded in the features, the second a genuine prediction from weak predictors (Section 1.7). For this work the consequence is direct. If the label is a rule applied to the current reading, $y_t = h(x_t)$, a model that sees x_t only recovers the rule. The legitimate task forecasts a future state from a past window,

$$\hat{y}_{t'} = g\big(\{x_{i,s}\}_{t-L<s\leq t}\big), \qquad t' > t + \varepsilon, \tag{2.3}$$

in which the label lies outside the input by construction (Figure 2.4). Its performance is lower, and that lower figure is the honest one: the forecasting formulation is not an extra ambition but the condition for the result to mean anything.

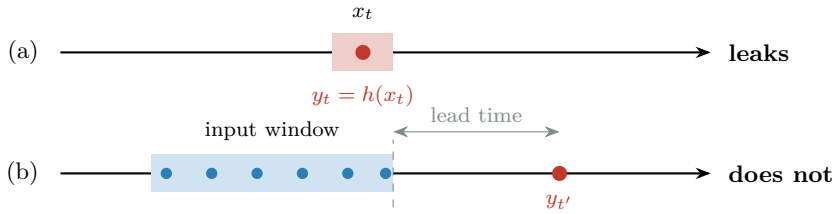


Figure 2.4: (a) Classifying the current state when the label is a rule applied to the same reading. (b) Forecasting a state outside the input window.

Four rules keep the label out of the inputs:

1. the label is computed on the target reading and the features on readings of the input window, with no overlap;
2. no feature measuring the distance to a threshold is computed at the label instant;
3. the guideline rule applied to the last observed value is a mandatory baseline, and the contribution of a model is its margin over that rule;
4. performance is also reported against labels that do not derive from the thresholds: the physician's relabelling of a sample of readings, and, once the system is used, the clinicians' verdicts.

The comparison that matters is with *persistence*, the guideline rule applied to the last reading of each channel, turned into probabilities by the matrix of transitions between levels observed in the training data. Physiological series are strongly autocorrelated, so carrying the current level forward is a strong baseline, and the contribution of a model is its margin over it. If that margin is small, the honest conclusion is that, at this sampling density and sample size, the models add little to the guideline rule—a result in itself, and more useful to a clinician than an inflated accuracy.

2.3.2 Two formulations of the forecast

Two formulations reach the same output, the probability $P(y_{t'} = k)$ of each level k, and neither is assumed better. The **direct ordinal classification** predicts the level of each condition, with a loss that respects the order of the levels. The **value forecast** predicts the value $\hat{x}^{(c)}$ of each channel at the target; with $\hat{\sigma}_c$ the root mean squared error of out-of-fold forecasts, the level probabilities follow by integrating the forecast distribution over the intervals of Table 2.1,

$$P(s_c = k) = \sum_{[a,b)\,:\,\text{level } k} \Big[\Phi\Big(\tfrac{b-\hat{x}^{(c)}}{\hat{\sigma}_c}\Big) - \Phi\Big(\tfrac{a-\hat{x}^{(c)}}{\hat{\sigma}_c}\Big)\Big], \qquad P(y \le k) = \prod_c P(s_c \le k), \tag{2.4}$$

the product assuming that the channel errors are independent given the inputs (Figure 2.5). The value forecast gives a training signal in clinical units on every sample and survives a revision of the guidelines without retraining; the classification needs no assumption on the channels. The alert probability is $p = P(y = 2) + P(y = 3)$.

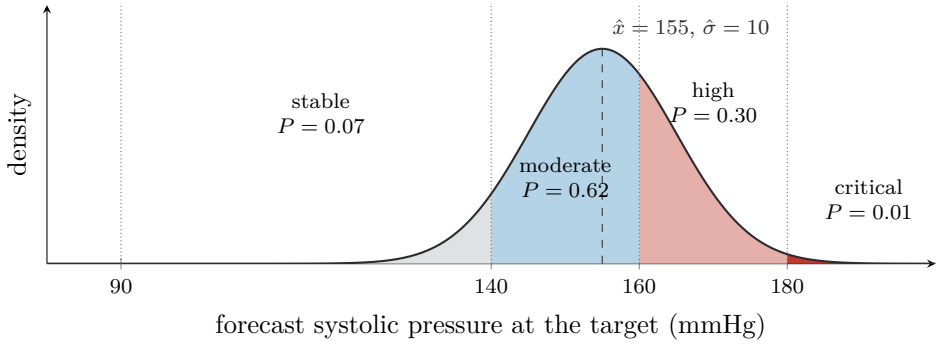


Figure 2.5: From a value forecast to level probabilities, for one channel. The forecast distribution is cut by the thresholds of Table 2.1; each area is the probability of a level. Within a condition, the channels are then combined by their worst level.

The four levels are ordered, so a standard multi-class loss, which treats predicting "critical" for a stable patient as no worse than predicting "moderate", is not appropriate: the value forecast carries the order naturally, and the classification uses an ordinal model.

2.3.3 Lead time and sampling frequency

How far ahead a forecast can look depends on how often the patient is measured. With routine readings at a rate λ, the expected number of readings inside a horizon of length Δ is

$$\mathbb{E}[N_\Delta] = \lambda\Delta. \tag{2.5}$$

Under the routine home protocol of Mihevc et al. [8], about two blood pressure sessions a week, $\lambda = 2/168$ per hour gives 0.29 readings within 24 hours and 0.57 within 48 hours (Figure 2.6): most forecasts of the next day would have nothing to forecast. The horizon must therefore match the regime of the patient. In the *routine* regime the useful target is the next scheduled reading, a few days ahead. In the *verification* regime, when the protocol prescribes two readings a day, a horizon of 24 to 48 hours contains about four readings, and it is also the moment when a clinical decision is pending, so a short-horizon forecast could change an

action.

The ward charts and the harmonised intensive-care data follow the verification regime, a reading every 12 hours. The lead time is then set by the embargo ε of the sample definition: with $\varepsilon = 0$ the target is the next reading, 12 hours ahead; with $\varepsilon = 12$ h, the reading 24 hours ahead; with $\varepsilon = 36$ h, the one 48 hours ahead. The primary task uses the next reading, and hypothesis H2 is tested by how the margin over persistence changes with the lead time, always against persistence at the same lead time: a forecast that only matches the rule applied to the last reading anticipates nothing.

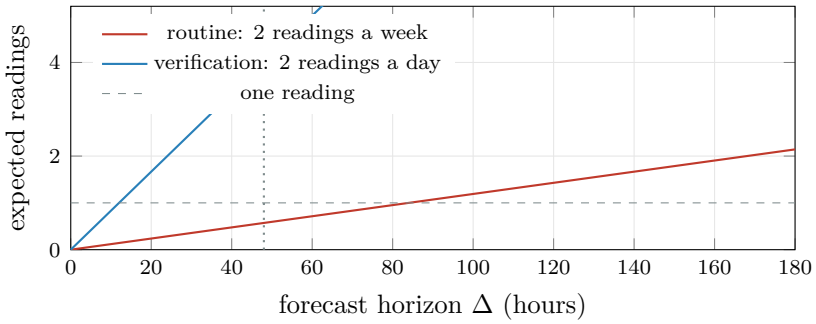


Figure 2.6: Expected number of readings inside the forecast horizon, by sampling regime. The dotted line marks 48 hours.

2.4 The Decision Layer

2.4.1 Three tiers and the rule floor

Three families of clinical decision support exist (Table 2.2). The system of this work is *hybrid*: the guideline rule provides a floor, and a learned model may refine it. If the model is uncertain or fails, the system falls back to guideline conformity rather than to nothing. Yazdani et al. [18] add a second argument for keeping the rules explicit: most decision-support systems are built for one disease and must be rebuilt for another, and separating the platform from a configurable rule set per condition avoids this, which a system covering two conditions with different guideline sources needs.

Table 2.2: Families of clinical decision support.

Family	Basis	Strength	Weakness
Rule-based	guideline thresholds	transparent, needs no training data	no personalisation, no trend
Model-based	learned from data	personalisation, trend detection	opaque, needs data, may not transfer
Hybrid	rules as floor, model as refinement	guideline conformity plus personalisation	two components to validate

The clinician needs a decision, not a probability: green (record), yellow (more measurements), red (review now). The model's alert probability is mapped to a tier by two cut-points $p_{\text{yellow}} < p_{\text{red}}$. Each cut-point is a decision threshold: alerting at p_t means accepting $(1-p_t)/p_t$ false alarms per true alert. The yellow tier costs the patient extra readings and the clinician

nothing, so its threshold is low; the red tier costs clinician time. Until the partner physician states the costs of errors, $p_{\text{yellow}} = 0.1$ and $p_{\text{red}} = 0.3$ are used, which accept nine and about two false alarms per true alert. The model may raise the tier set by the guideline rule, but never lower it:

$$\text{tier}_{\text{final}} = \max\Big(\text{tier}_{\text{rule}}(x_{\text{last}}),\ \text{tier}_{\text{model}}(p)\Big), \qquad \text{green} \prec \text{yellow} \prec \text{red}. \tag{2.6}$$

A patient at 185/110 mmHg is red whatever the model says; the best model of the reviewed literature misses half of the positive cases [10]; and if the model fails entirely, the system falls back to the standard of care rather than to nothing (Figure 2.7).

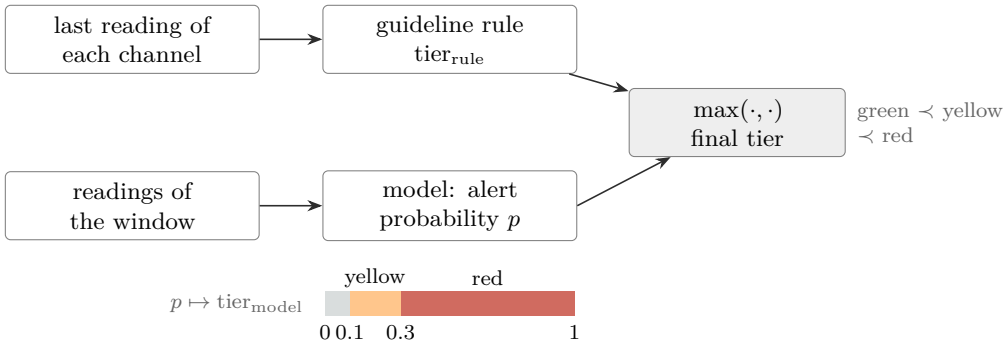


Figure 2.7: The decision layer. The model can raise the tier given by the guideline rule, never lower it; the cut-points 0.1 and 0.3 are provisional.

2.4.2 The home triage

The label of Table 2.1 uses in-hospital thresholds, those of the data actually collected. A home deployment substitutes the out-of-office thresholds of the only validated protocol for patients with both conditions [8] (Table 2.3), which this work adopts as its reference.

Measurement protocol. For blood pressure, two readings in the morning and two in the evening, after five minutes of seated rest; the device enforces a one-minute interval between the two readings and the second is retained. The reference value is the mean of the retained morning and evening readings over seven days: with fourteen readings, the standard error of the usual pressure is divided by $\sqrt{14} \approx 3.7$ compared with a single reading (Section 1.3.5). The home target is 135/85 mmHg, five points below the office threshold, because there is no white-coat effect at home. For glucose, routinely one fasting measurement and one 90 minutes after lunch, and, after an abnormal value, a one-day profile of six measurements (before and 90 minutes after each meal); the targets are a fasting glucose below 7 mmol/L (about 1.26 g/L) and a post-prandial glucose below 10 mmol/L (about 1.80 g/L). Routinely, blood pressure is measured twice a week and glucose once a week: a low routine frequency keeps the burden on the patient acceptable, and a high frequency is triggered only when a signal justifies it.

Two-stage triage. A yellow reading leads to a verification profile—seven days of blood pressure, or the one-day profile of six glucose readings—and only the mean of the profile reaches a clinical decision. The protocol has three properties that the system adopts. It is *bidirectional*: low values trigger as well as high ones (hypoglycaemia, hypotension in older

treated patients). It is *two-stage*: noise is separated from trend by increasing the number of readings rather than by lowering the threshold. And it is *economical in clinician time*: only the red level, and a profile off target, mobilise a clinician, which matters where professionals already perceive telemedicine as extra work [11]. It also has three limits, which define the room for a learned model: its thresholds are fixed and identical for all patients, it reacts to the level of a reading rather than to its trend, and it uses no learned model.

Decisions left open by the source. The source leaves five points open, which this work decides: the more severe level applies when systolic and diastolic pressures disagree; a glucose of unknown meal status is held to the after-meal targets; the pulse keeps the NEWS2 red score; a seven-day profile is concluded only with at least six retained readings on three days, the minimum of the ESH practice guidelines [41], and is otherwise reported as incomplete, never as on target; and a glucose profile is done within two days of the trigger. Figure 2.8 shows the resulting path of a reading.

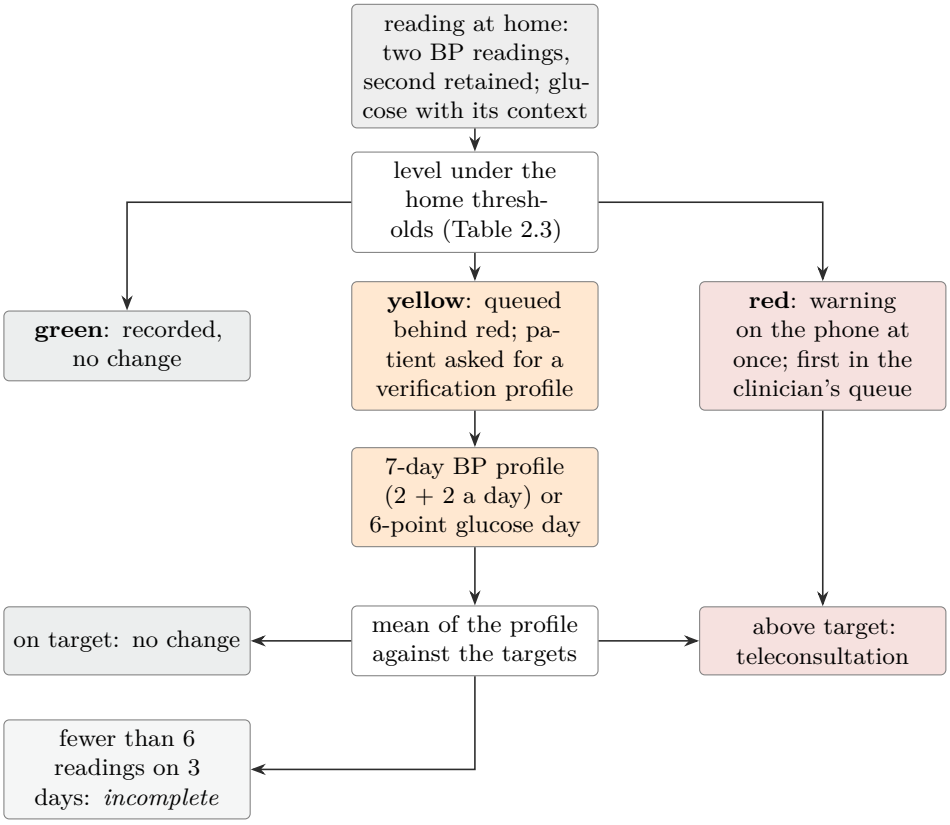


Figure 2.8: Path of a home reading through the two-stage triage. Only the mean of a verification profile, or a red reading, reaches a clinical decision.

2.5 Evolution of the Patient's State

The level answers whether a reading calls for attention; the clinician also needs to know whether the patient has changed. The state of a patient over a window is the vector of the mean of each channel, $\mathbf{s} = (\bar{x}_{\mathrm{SBP}}, \bar{x}_{\mathrm{DBP}}, \bar{x}_{\mathrm{pulse}}, \bar{x}_{\mathrm{glucose}})$. The current state $\mathbf{c}$ covers $(t - 24\,\mathrm{h}, t]$ and the baseline $\mathbf{b}$ the three days before, $(t - 96\,\mathrm{h}, t - 24\,\mathrm{h}]$; evolution is a distance $d(\mathbf{c}, \mathbf{b})$

Table 2.3: The three-level home triage of Mihevc et al. [8]. Pressure in mmHg, glucose in mmol/L.

Level	Blood pressure	Glucose	Action
Green	110–134 / 60–84	4–7 before a meal; 7–10 after	no change
Yellow	135–174 / 85–104	7.1–10 before; 10.1–14 after	verification profile
Red	≥ 175 / ≥ 105, or $<$ 110 / < 60	< 4; > 10 before; > 14 after	teleconsultation
Profile mean	$\geq 135/85$ or $< 110/60$	> 7 before or > 10 after	teleconsultation

computed on the channels present in both (Figure 2.9).

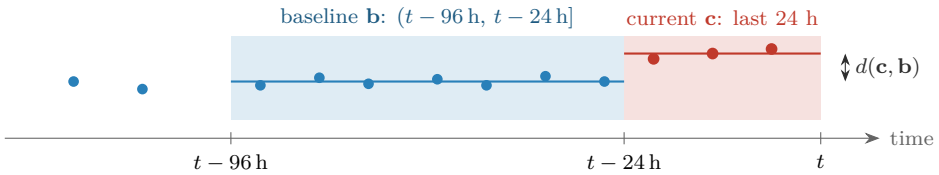


Figure 2.9: The two windows compared for one channel. Each window is summarised by its mean; the four channel means form the state vectors $\mathbf{c}$ and $\mathbf{b}$.

Cosine similarity compares directions, not sizes: if every channel rises by half, $\mathbf{c} = 1.5\,\mathbf{b}$ and the cosine distance is zero, so a patient whose pressures and pulse rise together appears unchanged. On raw values it is also dominated by the channel with the largest unit, and its behaviour depends on how the vectors are scaled [42]. No measure is therefore assumed: the candidates (Table 2.4) are the lock-step measures of the benchmark of Wang et al. [32], plus two families for monitoring an individual: scaling each change by the patient's own variability, the idea of the reference change value of laboratory medicine [43], and the Mahalanobis distance [44], the statistic of multivariate control charts for individual observations [45]. Elastic measures such as dynamic time warping align sequences; a state is a vector of four means, so they do not apply.

Table 2.4: Candidate measures of the change between current state and baseline, from the simplest. σ_j: between-patient SD of channel j; σ_{ij}: SD of patient i, shrunk towards the pooled within-patient SD; Σ: pooled within-patient covariance of changes.

Measure	Definition	Property
Cosine	$1 - \mathbf{c}\cdot\mathbf{b}/(\lVert\mathbf{c}\rVert\lVert\mathbf{b}\rVert)$	blind to a common change of size
Euclidean, raw	$\lVert\mathbf{c}-\mathbf{b}\rVert$	dominated by the largest unit
Euclidean, cohort scale	$\lVert(\mathbf{c}-\mathbf{b})/\boldsymbol{\sigma}\rVert$	one scale for all patients
Cosine, centred	cosine of the deviations from the cohort mean	direction of the deviation
Euclidean, patient scale	$\lVert(\mathbf{c}-\mathbf{b})/\boldsymbol{\sigma}_i\rVert$	change against own variability
Manhattan, patient scale	$\sum_j \lvert c_j - b_j\rvert/\sigma_{ij}$	less weight to one large change
Chebyshev, patient scale	$\max_j \lvert c_j - b_j\rvert/\sigma_{ij}$	the largest single change
Mahalanobis	$\sqrt{(\mathbf{c}-\mathbf{b})^\top\Sigma^{-1}(\mathbf{c}-\mathbf{b})}$	correlated channels counted once

2.6 Synthesis

The framework turns the gaps of Chapter 1 into five formal choices:

- one label per condition, on four ordered levels; every threshold comes from an official source, and a missing channel never counts as normal;
- a task that forecasts the level of the next reading from a past window, so that the label cannot leak into the inputs, with persistence as the baseline to beat;
- two formulations, ordinal classification and value forecasting, which give the same output and are compared rather than chosen in advance;
- a decision layer in which the model may raise the tier of the guideline rule but never lower it, and in which, at home, a yellow reading first leads to a verification profile;
- eight candidate measures of the evolution of the patient's state, among which a benchmark selects one.

None of these choices depends on the data. Chapter 3 describes the data to which they are applied, how the candidates are compared, and the system built around them.

Chapter 3

Methodology and System Implementation

3.1 Introduction

This chapter describes how the research was conducted and justifies each methodological choice. It instantiates the framework of Chapter 2 to answer the four research questions: whether a telemonitoring system combining consultation records, home readings and risk prediction can be built under the constraints of Cameroonian care (RQ1), whether learned models forecast the level of the next reading better than the guideline rule (RQ2), at longer lead times (RQ3), and on Cameroonian patients (RQ4).

The literature imposes three requirements on the method (Chapter 1): use longitudinal data whose instruments and quality can be assessed; evaluate predictive models without the biases found in previous work—label leakage, improper metrics, validation on the training population, overfitting of small samples; and respect an ethical and legal framework for real patient data. The chapter presents in turn the type of research, its setting and population, the data and how they were collected, the sampling plan, the treatment of the data, the features, the candidate models, the experimental protocol and the metrics; then the telemonitoring system built around them, the implementation details that make every result reproducible, the ethical considerations, the limitations of the methodology and a synthesis.

3.2 Type and Approach of Research

The research is **applied**: it addresses a practical problem, the follow-up of patients with chronic diseases, by designing an artefact, a telemonitoring system. It follows a design-science logic in three stages: requirements are derived from the literature (Chapter 1), the system is designed and built, and its predictive component is evaluated on retrospective data. Within this frame, the evaluation of the predictive component is **experimental**, since models are compared under a protocol fixed in advance, and the analysis of the cohorts is **descriptive**. The approach is **quantitative** (Table 3.1), since the hypotheses are measurable comparisons; the labels, however, are clinical judgements, so the partner physician is to validate the labelling rule and relabel a sample of readings. The data are secondary: no patient was recruited and

no measurement was taken for this work. Figure 3.1 shows how the datasets enter the analysis: models are developed and compared on public multi-centre data only, and the Douala charts are used once, for external validation.

Table 3.1: Type of research by objective.

Objective	Type	Approach
Design the telemonitoring system (RQ1)	applied, design science	requirements analysis, prototyping, traceability
Compare predictive models (RQ2; H1, H3)	experimental	quantitative
Evaluate longer lead times (RQ3; H2)	experimental	quantitative
Validate the models on Cameroonian patients (RQ4; H4)	experimental	quantitative
Describe the cohorts and the quality of the data	descriptive	quantitative
Validate the labels	evaluative	relabelling by the partner physician (planned)

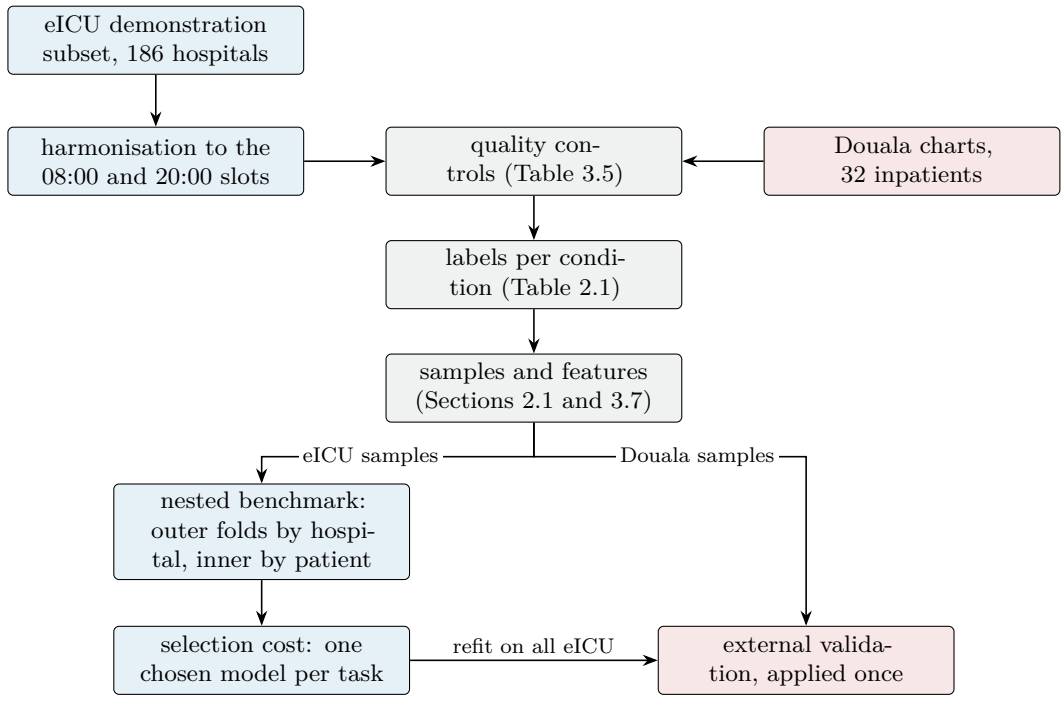


Figure 3.1: Study design. The same treatment, labels and samples apply to both datasets.

3.3 Research Setting and Population

The research was carried out at the Laboratory of Computer Engineering, Data Science and Artificial Intelligence of the NHPSD and, for the Douala records, at the Laquintinie Hospital of Douala, a public hospital, under the supervision of an endocrinologist of the hospital (Section 3.13). The deployment context is that described in Section 1.4: patients followed episodically, paper records, few specialists and intermittent connectivity.

3.3.1 Target population and eligibility

The target population of the system consists of adults with diagnosed hypertension, type 2 diabetes, or both, followed at a health facility and able to measure at home. The data used in this work come from two other populations, each with its own criteria. In eICU, a stay is eligible when the patient is an adult and hypertension or diabetes is recorded in the diagnoses or the past history; hypertensions of another nature (pulmonary, intracranial, portal, ocular, intra-abdominal, gestational) are excluded (Appendix B). In Douala, all the observation charts made available by the hospital for the period are used; the admission diagnosis was not transcribed, so eligibility cannot be checked patient by patient. The criteria of a prospective collection with the system (Table 3.2) are adapted from the trial of Mihevc et al. [8]; they would be finalised with the hospital and a research ethics committee before any recruitment.

Table 3.2: Eligibility criteria of a prospective collection with the system.

Inclusion	**Exclusion**
Age 18 years or more	type 1 or gestational diabetes; pregnancy
Hypertension and/or type 2 diabetes documented in the facility record	cognitive impairment preventing the use of the devices, without a caregiver
Followed at the partner facility	severe condition requiring hospitalisation at enrolment
Able to measure at home, alone or with a caregiver, with access to a smartphone	refusal or withdrawal of consent
Written informed consent	

3.3.2 Collection scenarios

Which data could be obtained determines most of what follows, so four scenarios were distinguished when the work was planned (Table 3.3). They differ in the interval between readings, which fixes the horizon a model can forecast (Section 2.3.3), and in their ethical requirements.

Table 3.3: Collection scenarios considered and their consequences.

	A: retrospective outpatient	**B: prospective at home**	**C: mixed**	**D: retrospective inpatient**
Source	consultation records	recruited cohort with home devices	records plus a prospective sub-cohort	ward observation charts
Interval between readings	about 90 days	3–4 days	both	about 12 hours
Ethics	secondary use	individual consent	both	secondary use
Feasible horizon	next consultation	a few days; 24–48 h during verification	per source	24–48 h
Obtained	yes, descriptive	no	no	yes

Two datasets were obtained, both transcribed from paper records of the Laquintinie Hospital. The consultation records (scenario A) give 81 readings of 58 outpatients, 49 of whom (84%) have a single reading: they document the fragmentation of follow-up but cannot

support longitudinal modelling. The observation charts of 32 inpatients (scenario D) give 366 readings taken morning and evening; they fix the rhythm of the data and are used for external validation. No home data (scenario B) could be collected, so the method is written for irregular home series and applied to the twice-daily charts, and the low-cost acquisition named in the title is not evaluated.

3.4 Data Sources and Collection

3.4.1 The three sources

Three sources are used, each for a single purpose (Table 3.4).

Table 3.4: The three data sources and their role.

Source	Content	Collection	Role
eICU Collaborative Research Database [46, 47]	intensive-care stays in United States hospitals, 2014–2015; open demonstration subset of 2 520 stays of 1 841 patients in 186 hospitals	electronic records of routine care, de-identified	development and comparison of the models
Douala inpatient charts	32 patients, 366 readings, 8 June to 30 August 2026; pressure, pulse and capillary glucose, morning and evening	paper observation charts, transcribed by hand without names	external validation only
Douala consultation records	58 outpatients, 81 readings, 49 with a single reading	paper consultation records, transcribed without names	description of the fragmentation of follow-up

The Douala charts were transcribed one row per reading. The transcription records whether a reading was taken in the morning or the evening, but not its hour nor the admission diagnosis. In the wards, pressure and pulse follow the nursing schedule, whereas glucose is ordered according to the clinical picture; the consequences of this difference for missing data are treated in Section 3.6.4. In eICU, vital signs come from nurse charting and periodic monitoring, and glucose from bedside measurements. Figure 3.2 traces the path from the paper record to the analysis data, and Figure 3.3 shows what the charts contain, patient by patient.

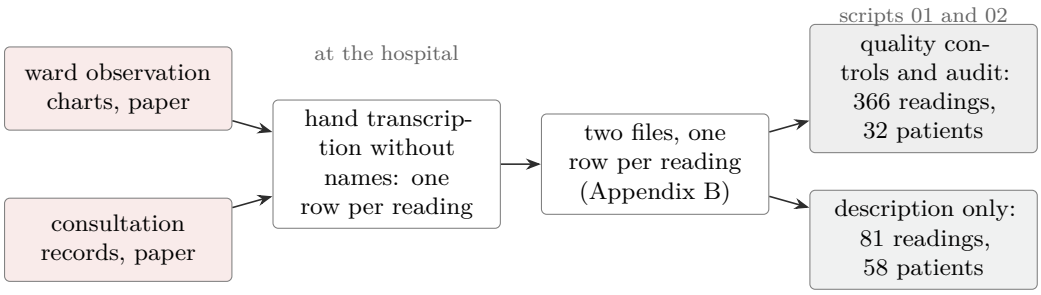


Figure 3.2: From the paper records of the Douala hospital to the analysis data. No name leaves the hospital; the slot is morning or evening, the hour is not recorded.

Values are stored in one canonical unit per channel, with plausible ranges of 60–260 mmHg for SBP, 30–160 for DBP, 30–220 beats per minute for pulse and 1.0–35.0 mmol/L for glucose.

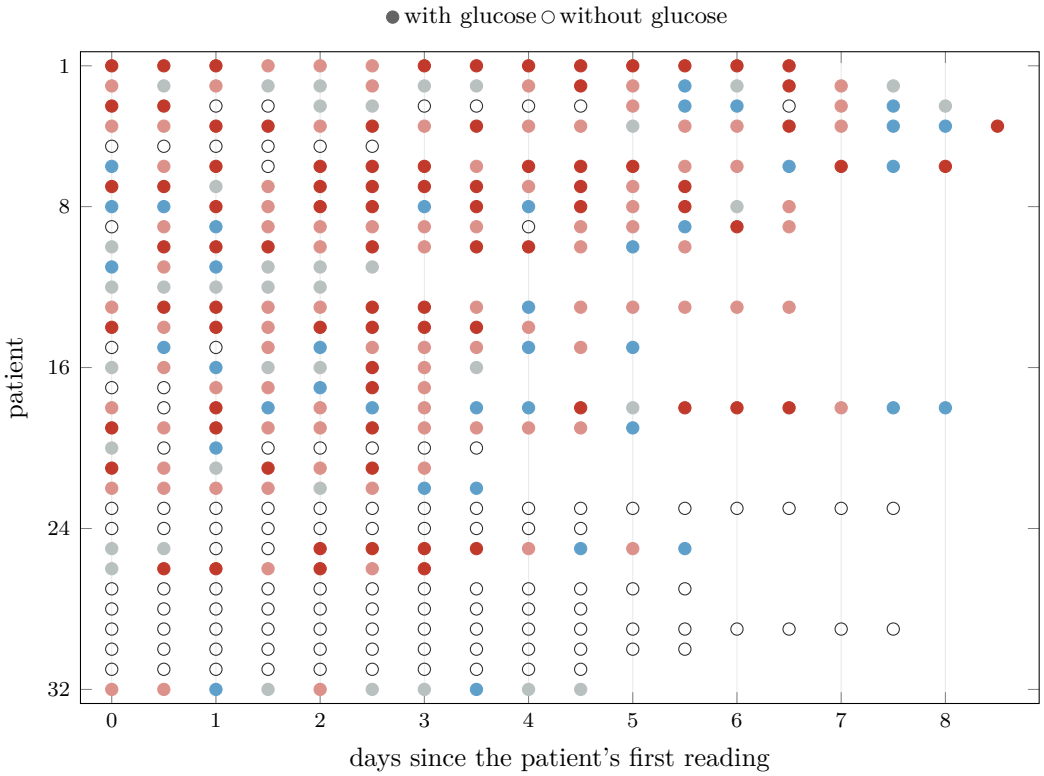


Figure 3.3: The 366 readings of the Douala charts. One row per patient, one mark per reading, coloured by level (grey stable, blue moderate, light red high, red critical); open marks are readings without glucose. Patients are numbered by date of first reading.

Glucose is the riskiest field, because three conventions coexist and Cameroonian practice uses g/L: 1 mmol/L = 18.0156 mg/dL from the molar mass of glucose. The exact factor matters: with the rounded value of 18.0, a reading of 1.80 g/L falls below the 180 mg/dL threshold of Table 2.1.

3.4.2 Harmonisation of the intensive-care records

Intensive-care monitoring is far denser than a ward chart, so each stay is resampled to the rhythm of the charts: two slots a day, at 08:00 and 20:00, each taking the non-invasive cuff pressure, the pulse and the bedside glucose nearest to the slot within two hours (Figure 3.4). The slot hours are an assumption, since the charts record only morning and evening, and the tolerance is varied in a sensitivity analysis. Both datasets then share one sample definition, and the remaining differences—intensive care against the ward, the United States against Cameroon—are the shift that external validation measures.

3.4.3 Validity and reliability

Validity and reliability are assessed at three levels. For the **instruments**, the devices of the Douala wards are not named, so their type and the consequences of their tolerance are inferred from the readings themselves (below); the eICU records come from routine monitoring devices whose validation is not reported either. For the **data**, controls applied at entry detect implausible, incoherent, duplicated and carried-forward values (Section 3.6.2). For the **labels**,

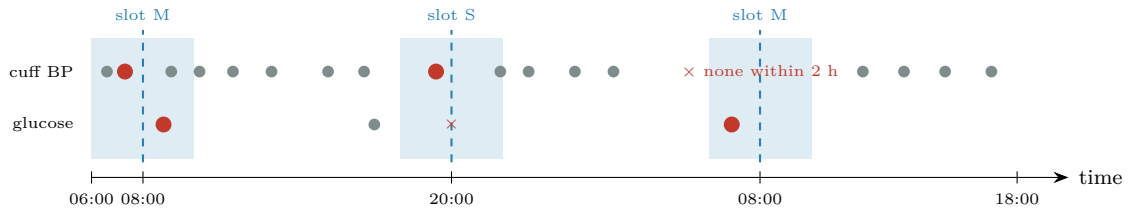


Figure 3.4: Harmonisation of an intensive-care stay to the rhythm of the ward charts. At each slot, the reading nearest in time within two hours is kept for each channel (red); a slot with none is missing, as an unrecorded value on a chart would be.

a written and versioned rule fixes the level of every reading (Section 2.2), and the partner physician is to relabel a random sample of readings without seeing the rule's output, with agreement measured by the quadratically weighted kappa, which suits ordered levels.

A blood pressure device is validated when it meets the universal standard ISO 81060-2 [48, 49]: against a reference, the mean of the differences must not exceed 5 mmHg and their standard deviation 8 mmHg. A glucose meter meets ISO 15197:2013 when 95% of its results lie within 15 mg/dL of the laboratory value below 100 mg/dL, and within 15% above [50]. Since the charts do not name the devices, two analyses of the readings bound what this uncertainty implies. **Terminal digits**: a manual aneroid or mercury scale is graduated in steps of 2 mmHg, so manual readings end in even digits with an excess of zeros, whereas an electronic display shows any integer [51]; a χ^2 test of uniformity of the last digit tells the two apart. **Label uncertainty**: shifting every reading by the tolerance of its standard—5 mmHg for pressure, 15 mg/dL or 15% for glucose—and counting the readings whose level changes measures how much of the disagreement between a model and the label is measurement noise that no model can remove. Two consequences follow: an error between adjacent levels must weigh less than an error across several levels (Section 3.10), and the physician's relabelling measures how much of this noise reaches clinical judgement.

3.5 Sampling Plan

Sample size. Whether the Douala charts suffice to develop a model is answered with the criteria of Riley et al. [52] and Riley et al. [53]: the sample must keep the optimism of the model small (shrinkage $S \geq 0.9$), keep the apparent and adjusted R^2 within 0.05, and estimate the overall risk within ± 0.05. The first criterion gives

$$n \geq \frac{p}{(S-1)\,\ln(1 - R^2_{\mathrm{CS}}/S)} \tag{3.1}$$

for p candidate parameters and an anticipated Cox–Snell R^2_{CS}, derived from an anticipated C statistic [54]. The criteria are computed with a port of the reference implementation [55], checked against its published example. The outcome is the alert of each condition. The samples of one patient are correlated, so the effective sample size divides their number by the design effect $1 + (\bar{m} - 1)\rho$ [56], with $\bar{m}$ the mean number of samples per patient and ρ the intraclass correlation of the alert. Flexible learners need more still: random forests, support vector machines and neural networks may need several times as many events per variable as

logistic regression to reach the same optimism [57].

Sampling method. No sample was drawn: in Douala, all the charts made available by the hospital are used, a convenience sample of one site whose characteristics are reported so that its representativeness can be judged (Section 4.2); in eICU, every eligible stay of the demonstration subset is used. eICU is preferred to the other public intensive-care databases, such as MIMIC-IV, because it covers many hospitals: the outer folds of the cross-validation can then be grouped by hospital, which estimates how a model transfers between sites before it meets the Douala patients [58]; a single-centre database cannot provide that estimate.

The role of each dataset. The charts fall far below the criteria (Section 4.2.6). The candidate models are therefore developed and compared on eICU only, and the Douala charts serve for external validation, a transfer that no study identified in the region has measured for a learned model; early-warning scores derived elsewhere lose discrimination in African wards and require local validation and recalibration [34, 35], and a Ghanaian study is validating intensive-care scores derived in high-income countries on West African patients [59]. With about 64 haemodynamic and 112 glycaemic alert targets, below the 100 events and 100 non-events recommended for external validation [60, 61], the Douala estimates are reported with patient-level intervals and read as exploratory.

Model capacity. The effective size of the Douala samples is small against the capacity of any flexible model, and above all of a recurrent one. An LSTM layer with u units and f inputs has

$$P = 4\big[u(f+u)+u\big] \tag{3.2}$$

parameters: even a small layer ($u = 16$, $f = 10$) has 1 728, that is 24 to 33 per independent Douala sample. Trained on the charts, a recurrent model would fail for lack of data, and the comparison of H3 would measure the sample size rather than the models. Four responses were possible: increase the sampling frequency for a sub-cohort; compare the models on a public dataset and use the local cohort for external validation; restrict the comparison to a few simple models; or report a recurrent model as a negative result limited by capacity. The second was adopted: it gives the deep models a sample on which their comparison means something, and it turns the small local cohort into a test of transportability rather than a development set.

3.6 Data Curation and Preprocessing

3.6.1 A two-phase pipeline

This is where label leakage and biased imputation are either prevented or built in. Every step that learns a parameter from the data must be fitted on the training fold only, so the treatment is split into two phases separated by the train–test split (Figure 3.5) and implemented as a single pipeline object. A step placed on the wrong side of the split causes no error: it silently produces an optimistic estimate that fails in deployment.

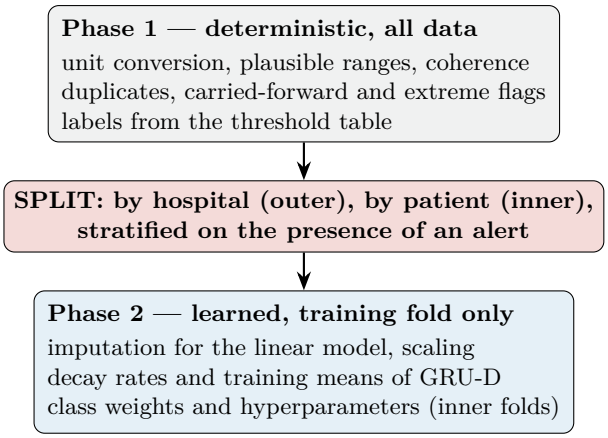


Figure 3.5: The two-phase preprocessing pipeline.

3.6.2 Quality controls at entry

Errors are cheapest to catch at entry, when the information needed to correct them is still available. The controls of Table 3.5 learn nothing from the data, so they are applied to all records before the train–test split. Flagged records are kept: a run of identical values is information about how the data were produced. Impossible values are the exception, and are removed from features and labels.

Table 3.5: Quality controls applied at entry.

Control	Rule	On violation
Plausibility	value within the plausible range of its channel	value removed
Coherence	SBP $>$ DBP and SBP $-$ DBP ≥ 10 mmHg	pair removed: one value is wrong, which cannot be known
Duplicate	same patient, time and values	one kept, the rest flagged
Carried forward	three or more identical consecutive values	flagged
Unit	magnitude consistent with another glucose convention	flagged
Extreme	robust $\lvert z\rvert > 3.5$ (below)	flagged; "contradicted" if the next reading is not extreme

3.6.3 Outliers: artefact or event?

In a risk-detection system the outlier is often the event of interest: a systolic reading of 185 mmHg is a statistical outlier for a patient whose median is 138, and also a red alert. An automatic cleaning step would delete the positive class. The dividing line is therefore physiological plausibility, not statistical rarity. A statistical threshold is used for flagging only, with the median absolute deviation (MAD), which is not inflated by the outliers it detects, shrunk towards the cohort value because a personal MAD from ten readings is unstable:

$$z_{i,t} = \frac{x_{i,t} - \operatorname{median}(x_i)}{1.4826\,\widehat{\mathrm{MAD}}_i}, \qquad \widehat{\mathrm{MAD}}_i = w_i\,\mathrm{MAD}_i + (1-w_i)\,\mathrm{MAD}_{\mathrm{pooled}}, \quad w_i = \frac{n_i}{n_i + n_0}. \tag{3.3}$$

With $n_0 = 20$, a patient with ten readings takes one third of the estimate from their own data; as the series grows, the estimate becomes personal. A reading is flagged when $|z_{i,t}| > 3.5$ [62]; flags serve clinician review and never enter the features.

3.6.4 Missing data

Imputation is valid when data are missing completely at random (MCAR) or at random given observed values (MAR), and invalid when missingness depends on the unobserved value itself (MNAR). Each source of this work has its own mechanism. Consultation data are MNAR, because attendance depends on the patient's state; home data are MNAR by a different route, because discouraging readings may be omitted; only sensor gaps (battery, connection) are approximately MCAR. In the wards, the vital signs follow the chart schedule, but glucose is ordered according to the clinical picture, so whether it is measured depends on the patient's state: its absence is MNAR as well.

The direction of the resulting bias matters. If abnormal values are less likely to be recorded, the observed values are biased towards normal, and mean imputation or forward filling teaches the model that absence predicts stability, whereas it predicts deterioration—a sign error that more data cannot correct, and one that contradicts the system's own rule that a silence must raise an event (Section 3.11.4). Absence is therefore represented as information: the number of glucose readings in the window and the time since the last one are features. Tree ensembles route missing values at each split. The recurrent model GRU-D [63] receives the mask of what was measured and decays the last value towards the training mean as the gap grows,

$$\tilde{x}_t^{(c)} = m_t^{(c)} x_t^{(c)} + (1 - m_t^{(c)}) \left[\gamma_t^{(c)} \, x_{\text{last}}^{(c)} + (1 - \gamma_t^{(c)}) \, \bar{x}^{(c)} \right], \qquad \gamma_t^{(c)} = \exp\!\left(-\kappa_c \, \Delta t_{\text{last}}^{(c)}\right), \tag{3.4}$$

with a decay rate per channel, since a three-day-old pressure says more about today than a three-day-old glucose. The decay is always used together with the mask, since on its own it would again assert stability during long gaps. The proportional-odds model, which cannot accept missing inputs, receives the training median with a missingness indicator, which keeps the absence visible to the model.

The features keep their clinical units, since the label is defined on absolute thresholds: normalising each patient's values by their own median and dispersion would hide how far a reading lies from a threshold, which is precisely what the label measures, so the patient's own distribution is used only to flag extreme values (Section 3.6.3). The linear and neural models are sensitive to the scale of their inputs, so each input is standardised with the mean and standard deviation of the training fold; the tree ensembles need no scaling (Table 3.6).

Table 3.6: Preparation required by model family, fitted on the training fold.

Model	Scaling	Missing values
Proportional-odds logistic regression	yes	training median with missingness indicators
Random forest, CatBoost	no	handled natively by the splits
GRU-D, masked LSTM	yes	mask and decay towards the training mean

3.7 Features

The tabular models receive descriptors of the window, computed per channel on its readings only: last value, mean, standard deviation, slope, number of readings and time since the last one, plus pulse pressure and the age and sex of the patient, the only covariates recorded in both datasets. The slope is the least-squares coefficient on the real reading times, which handles gaps without resampling, and is computed only from at least three readings spanning half the window; a feature the window cannot support is missing, not zero. The sequence models receive the readings of the window with their mask. Table 3.7 groups the candidate descriptors by family.

Table 3.7: Candidate descriptors, per channel $c \in \{\text{SBP, DBP, pulse, glucose}\}$, computed on the readings of the window only.

Family	Feature	Definition or purpose
Level	last value, window mean	current and typical level
Trend	slope $\beta^{(c)} = \sum_k (t_k - \bar{t})(x_k - \bar{x}) / \sum_k (t_k - \bar{t})^2$	change per day, on the real reading times
Variability	standard deviation	dispersion within the window
Cross-channel	pulse pressure	SBP − DBP of the last complete reading, a marker of arterial stiffness
Measurement	count, time since the last value	how much was measured; for glucose, whether it was ordered at all
Static	age, sex	the only covariates recorded in both datasets

Four remarks explain what the table leaves out or why it includes a feature.

- **No signed margin to a threshold.** For a fixed threshold θ, the margin $x_{\text{last}} - \theta$ is an affine function of the last value: a linear model absorbs it into its intercept, and a tree, which splits on order, finds exactly the same partitions. The margin adds a column and no information; the same holds for the grade band, a coarsening of the last value.
- **No disease duration or treatment.** Years since diagnosis and treatment are recorded neither in the charts nor in eICU, and a feature that exists in only one dataset cannot be validated externally.
- **Pulse pressure helps trees.** A tree splits on one variable at a time and would need many splits to approximate a condition on SBP − DBP; supplying it explicitly turns that staircase into one split.
- **The glucose count is informative.** Glucose is ordered for some patients and not for others (Section 3.6.4); the count of glucose readings lets the model use that pattern openly instead of through imputed values.

A window of n readings cannot carry more than n independent quantities. With two readings, the sample standard deviation is $|x_2 - x_1|/\sqrt{2}$, the mean absolute successive difference is $|x_2 - x_1|$ and the slope is $\pm|x_2 - x_1|/\Delta t$: level, dispersion, slope and last value are only two quantities, a level and a signed difference. The feature sets are therefore kept small and fixed before any result (Table 3.9). The primary set spends its columns on the channels that carry the decision: systolic pressure and glucose get a level and a trend, diastolic pressure and pulse

a level, and the glucose count summarises how the patient is monitored; dispersion is left out because a short window estimates it poorly.

The window length is a hyperparameter, searched over 24, 48, 72 and 96 hours, and what each length can carry follows from the number of readings it holds (Table 3.8). Because the length is searched, the set of forecast targets must not depend on it, or each length would be evaluated on different targets: a reading is an anchor when the shortest window, 24 hours, holds at least two pressure readings, and longer windows add history to the same anchors. Overlapping windows multiply the rows but not the information, so the patient is the independent unit: all the windows of a patient fall on the same side of every split, and intervals are computed at the patient level.

Table 3.8: Window length and the features it can support, at two readings a day.

L (h)	Readings	Features supported
24	2	level only
48	4	level and slope; variability marginal
72	6	level, slope and variability
96	8	level, slope and variability

Table 3.9: Feature sets, fixed before any result.

Set	p	Features
Primary	8	last SBP, slope of SBP, last DBP, last pulse, last glucose, slope of glucose, number of glucose readings, age
Extended	16	primary, plus SD of SBP and glucose, slopes of DBP and pulse, pulse pressure, time since the last glucose, mean SBP, sex

3.8 Candidate Models and Baselines

No model is chosen in advance: the candidates are the families that published benchmarks of comparable problems put forward, and the benchmark of this work selects among them. Eight candidate families, of five types, enter the benchmark (Table 3.10). On clinical prediction tasks, machine learning has shown no systematic benefit over logistic regression [28]; on tabular data, tree ensembles generally match or beat deep networks [64], and in the most recent living benchmark CatBoost is the best gradient-boosting model when tuned [65]. Only deep temporal models read the data as a sequence, which hypotheses H1 and H3 compare. Four families are left out for stated reasons: a single tree is dominated by its ensembles; support vector machines give poorly calibrated probabilities; XGBoost ranks below CatBoost in the same benchmark; and the foundation model TabPFN is limited to 10 000 training samples [66], which the development data exceed.

3.9 Experimental Protocol

All constraints combine into a nested cross-validation on the development data (Figure 3.7). Five outer folds estimate performance and are grouped by hospital, so that every test fold

Table 3.10: Candidate families, by type, and baselines.

Type	Candidate family	Why it is in the pool
Baselines	prior (level frequencies); persistence (rule on the last reading, with training transition probabilities)	the margin over persistence is the contribution of a model
Linear	proportional-odds logistic regression [67], L2 penalty; the same on principal components	no systematic loss against machine learning [28]; order of the levels in the model
Bagging	random forest [68]	reference of the reviewed literature [10]
Boosting	CatBoost [69]	best tuned gradient boosting [65]
Value forecast	ridge or CatBoost regression per channel, levels by integration (Section 2.3)	the second formulation
Deep temporal	GRU-D [63]; masked LSTM [70]	hypothesis H3

contains hospitals the model has never seen; four inner folds, grouped by patient, select the window and the hyperparameters. The search uses the tree-structured Parzen estimator of Optuna [71, 72] over a space declared before any run, split into structural hyperparameters (window, depth, units, components) and training hyperparameters (learning rate, penalties, dropout, class weights); each configuration is scored by its ranked probability score on pooled inner-validation predictions, and the configuration kept is the one with the lowest selection cost (Section 3.10). The number of configurations per candidate is fixed in advance (Appendix C), since the optimism of a selection grows with it [73]. Across families, the lowest inner selection cost gives the *selected* model of the fold, and its outer result estimates the whole procedure. The procedure is repeated three times with different fold assignments, and two models are compared with the corrected resampled t test, which accounts for the overlap between training sets [74]. Figure 3.6 details what happens inside one outer fold.

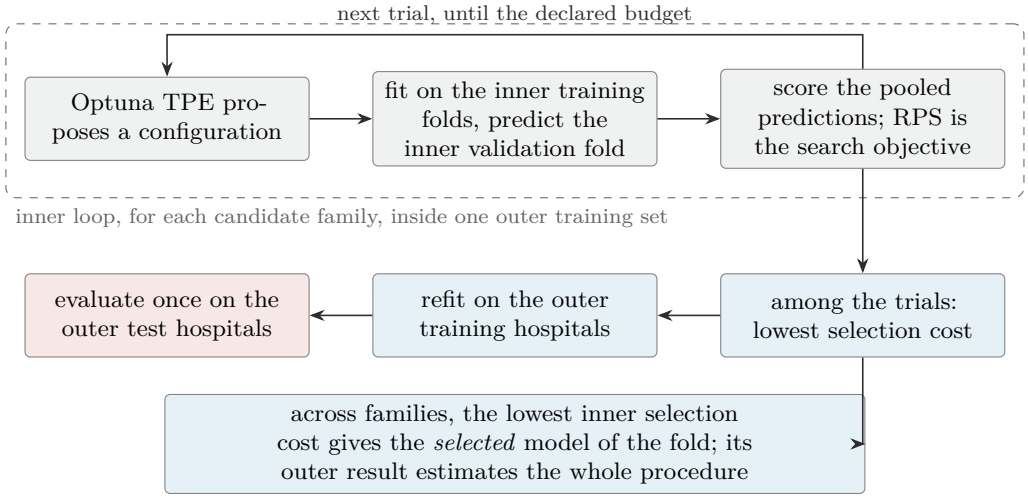


Figure 3.6: Search and selection in one outer fold. The outer test hospitals are used once, after every choice has been made on the inner folds.

Each selected configuration is then refitted on all the development data and applied once to the Douala charts; nothing is tuned on them, and their intervals come from a bootstrap

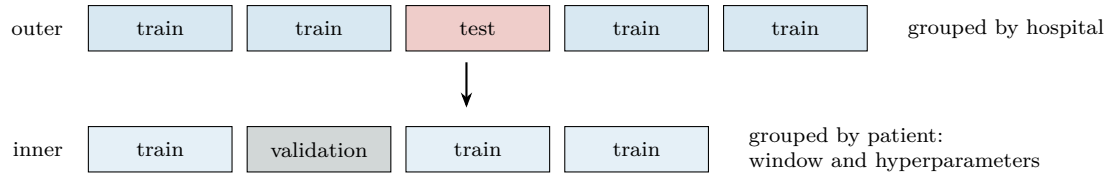


Figure 3.7: Nested cross-validation on the development data, repeated three times.

that resamples patients [60]. The prevalence of the alert differs between intensive care and Douala wards, which shifts calibration even for a model that transfers well, so performance is reported as is, and again after updating the intercept alone on the Douala data [75]: the first is the test of transportability (H4), the second shows what a local recalibration recovers. Fitted and scored on the same patients, the update would be judged on the outcomes it was fitted to; its Brier score and net benefit are therefore computed with each patient's intercept fitted on the other patients. Class imbalance is handled inside the training fold only, since oversampling before the split would place copies of the same window in training and test. Whether class weights help is a training hyperparameter of the search; a classifier trained with weights w_k overstates each level in proportion to its weight, so its probabilities are corrected exactly by Bayes' rule before use, $p_k = (p'_k/w_k)/\sum_j p'_j/w_j$. The correction is not cosmetic: in the two-class case with the conventional weight $w = (1-\pi)/\pi = 49$ at a prevalence $\pi = 0.02$, a raw output of 0.5 corresponds to a true probability of exactly 0.02, so a clinician shown 0.5 would read a risk twenty-five times too high. H2 is tested by repeating the benchmark with embargoes of 12 and 36 hours, for targets 24 and 48 hours ahead.

3.10 Evaluation Metrics

With prevalence π of the positive class, accuracy decomposes as

$$\text{Accuracy} = \pi\,\text{Sensitivity} + (1-\pi)\,\text{Specificity}, \tag{3.5}$$

so that, when positives are rare, a model that never predicts them reaches $1-\pi$. The logistic regression of Mendoza-Cano et al. [10] illustrates this: its accuracy of 0.70 is close to the proportion of controlled patients, while its specificity is 0.03. The measures are therefore chosen by explicit criteria [76]: a clear focus—discrimination, calibration, overall accuracy of the probabilities or clinical utility—and, for overall accuracy, a proper scoring rule [27]. Accuracy, the F1 score and recall at a fixed threshold fail these criteria: they are improper, depend on an arbitrary cut-point and mix statistical with decision performance. Table 3.11 lists the admissible measures; the positive class is the alert, and every measure carries a patient-level confidence interval. Performance is also reported by sex and by age band (below and above 65 years), and across the hospitals of the outer folds.

In the inner folds, configurations are compared on one number, the **selection cost**. Four terms—the RPS, the AUROC, the calibration error $|a| + |\ln b|$ for intercept a and slope b, and the net benefit averaged over thresholds of 0.1 to 0.3—are standardised across the configurations compared (minus the median, divided by the scaled MAD, oriented so that

Table 3.11: Admissible performance measures.

Focus	Measure	Why
Overall, four levels	ranked probability score (RPS) [77]	proper for ordered levels: an error across several levels costs more than an adjacent one
Overall, alert	Brier score, log loss	proper
Discrimination	area under the ROC curve (AUROC) of the alert	independent of any threshold
Calibration	intercept and slope [29]	a probability shown to a clinician must mean what it says
Clinical utility	net benefit, decision curve [78]	true alerts weighed against false alarms at the clinical threshold

lower is better) and combined:

$$C = \sum_m w_m \, z_m, \qquad \textstyle\sum_m w_m = 1. \tag{3.6}$$

The weights express how much each focus matters clinically, so they are a value judgement, not a parameter: tuning them on the data that judge the models would let any weighting favour a chosen model. They are equal in the provisional analysis, to be replaced by weights derived from the costs of errors stated by the partner physician, and a sensitivity analysis draws them at random to report how often the selected configuration remains first. The outer folds report every measure separately, never the composite alone.

Interpretability follows two lessons of the literature. An explanation describes a model, not a phenomenon: explaining a model that does not discriminate, such as an XGBoost at an AUC of 0.54 [10], has little value, so interpretability is reported only for a model shown to beat persistence. And importance measures are unreliable when features are correlated: permuting one of two correlated features lets the model fall back on the other, and gain-based importance credits whichever happens to be chosen at a split [79, 80]. Systolic and diastolic pressures are strongly correlated, so the features of one channel are permuted together, and the loss in RPS measures what the channel contributes (grouped permutation importance); for CatBoost, exact tree SHAP values are reported in addition [81], and the direction of their effects is checked against physiology.

The measures of state evolution (Table 2.4) are compared on the same samples: a measure is useful if its distance ranks higher the samples in which the level of the next reading differs from that of the baseline state (primary outcome), or becomes an alert (secondary outcome). Each is evaluated by its AUROC with a patient-level interval, its scales are estimated on eICU only, and the measure retained is the best on the primary outcome, or the simplest candidate whose interval contains the best value.

3.11 The Telemonitoring System

3.11.1 Requirements and traceability

The system answers RQ1, a design question, so each of its elements must trace back to a requirement of the literature. Table 3.12 lists the requirements derived in Chapters 1 and 2,

their basis, and what the built system does about each; the elements that are specified but not built are stated as such.

Table 3.12: Requirements of the system, their basis and their status in the built system.

Requirement	Basis	In the built system
Measurement		
repeated home readings; decisions on the mean of a verification profile	Section 1.3.5; Mihevc et al. [8]	built: profile computed on the phone
two readings per session, the second retained	Mihevc et al. [8]	built: the pair is entered; the one-minute interval cannot be enforced while entry is manual
three bidirectional levels	Mihevc et al. [8]	built: home triage of Table 2.3
HEARTS indicators produced from the record	World Health Organization [1]	not built; the record holds the data they need
Acquisition		
pressure from a validated cuff, glucose from a glucometer	World Health Organization [1] and Alhaddad et al. [22]	manual entry from any validated device; Bluetooth acquisition specified, not built
measurement time kept apart from arrival time	offline synchronisation	built: two timestamps per reading
one glucose unit; implausible values refused	Liu and Wang [24] (unit errors)	built: mmol/L only, ranges checked on the phone and on the server
Architecture		
an edge tier that grades readings without a network	Segun-Omosehin et al. [5] and Mucchi et al. [17]	built: triage and red warning on the phone
offline-first storage with merge on reconnection	Efendi et al. [9]	built: local cache of the database
series keyed by measurement time; validation at write; record that cannot be altered	Liu and Wang [24]	built: append-only readings; security rules written, deployment pending
role-based access; no identifier outside the record	Alhammad et al. [26]	built: patient and clinician roles; risk service and assistant receive no identifier
storage region chosen under the applicable law	Alhammad et al. [26]; Section 3.13	to be fixed before deployment
Modelling		
forecast a future state; persistence as baseline	Efendi et al. [9] and Mendoza-Cano et al. [10]	built: benchmark and risk service
training separated from scoring, with the same code	Liu and Wang [24]	built: the risk service imports the research package
Decision support		
final tier = $\max(\text{rule}, \text{model})$	Section 2.4	built
queue by priority, then by age; checkup period per patient	Yazdani et al. [18]	built
a silence raises an event	Yazdani et al. [18]; loss to follow-up	built: non-transmission sweep and daily reminder; SMS route not built
unreviewed events escalated	Mucchi et al. [17]	built: escalation to a named alternate
the system proposes, the clinician decides; verdicts kept as labels	Mihevc et al. [8] and Yazdani et al. [18]	built: verdict and action on every event
no automated advice beyond the boundary of Table 3.15	Mihevc et al. [8]	built; guardrails of the assistant to be tested on a phone

3.11.2 Architecture

The system has four parts (Figure 3.8). A cross-platform **application** (Flutter) serves the patient and the clinician. A managed document database (**Cloud Firestore**) holds the record and synchronises the applications, with a local cache that turns a reading entered offline into a delayed one. A local **risk service** (FastAPI) scores a patient's recent readings with the model selected by the benchmark, using the same code as the research pipeline, so that the features seen in deployment cannot drift from those seen in training. And a local **worker** runs the server functions: measurement check, escalation, notification, the non-transmission sweep and the escalation of events left unreviewed. Hosting the service and the worker online is outside the scope of this work; without them, the application still applies the home triage on the phone.

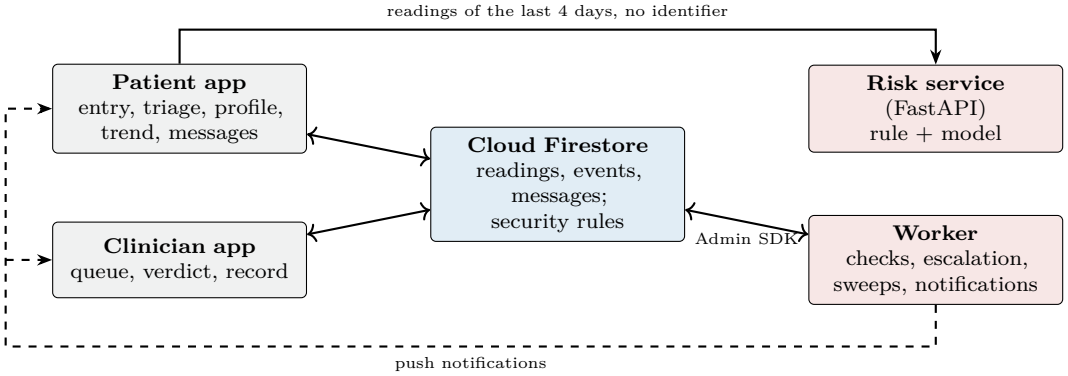


Figure 3.8: Architecture of the built system. Dashed: push notifications sent by the worker to the two applications.

3.11.3 Data model and rules

Each patient's readings, events and messages are subcollections of their profile, so that one security rule governs them and the readings are read in time order with one query. A reading holds one measurement session: the measurement time stated by the patient and the arrival time set by the server, the retained values, the first reading of a protocol pair kept for audit, the context of a glucose reading, and the quality flag set by the worker. The record is append-only: the rules refuse any change or deletion of a reading, and a correction is a new reading that names the one it supersedes. Each write is validated on the server (known fields, plausible ranges, coherent pressure pair, measurement time at most thirty days before arrival). An event stores what raised it (rule, model, verification profile, or a silence longer than the patient's checkup period), the triggering values, the threshold and model versions, and the clinician's verdict and action; it is closed once, and its identifier is derived from what raised it, so the same alert raised by the phone and by the worker is one document (Figure 3.9).

3.11.4 Server functions

Table 3.13 lists the functions of the worker. It reads and writes the database through the Firebase Admin SDK at a fixed interval; readings and events older than a day found at its

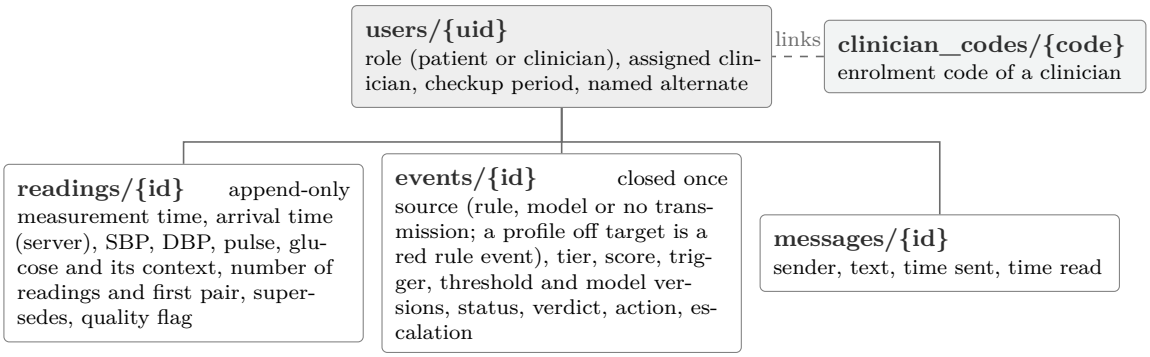


Figure 3.9: Data model of the record in Cloud Firestore. Readings, events and messages are subcollections of the patient's document, so one security rule governs each of them.

first run are marked without being notified, so that a backlog does not flood the queue. The delays are configuration values, provisional like the tier cut-points.

Table 3.13: Server functions run by the worker.

Function	Fires on	Action
Measurement check	new reading	quality controls of Section 3.6.2; quality flag
Escalation	yellow or red home level	event added to the clinician's queue
Scoring	latest reading	rule and model, same code as the risk service; event if the model raises the tier
Notification	new event	red: patient and clinician; yellow: patient, prompt for the verification profile
Non-transmission	no reading for longer than the checkup period	event; daily reminder to the patient
Queue age	event open 4 h (red) or 3 days (yellow)	escalated to the clinician's named alternate
Verdict	event closed	the patient is told it was reviewed

Figure 3.10 follows one alert through these functions, from the patient's reading to the clinician's verdict.

3.11.5 Interfaces

The **patient** enters the two pressure readings of the protocol, of which the second is retained, and may date a reading taken earlier; the glucose unit is fixed to mmol/L, a value outside its plausible range is refused with a message naming the unit, and the measurement context is required for glucose because the target depends on it. A reading at the red level brings up a local warning at once, whatever the connectivity. The home screen shows the level of the latest readings with the one instruction the patient may receive—contact the health centre, or seek care today—never a dose and never a probability; the verification profile to carry out and its progress; the fourteen-day trend with the target band; the evolution of each channel; and the outcome of the clinician's reviews. The **clinician** works from a queue of unreviewed events, red before yellow and then oldest first, each row showing the source of the alert, the triggering value, the fourteen-day trend and the versions, and closes an event with a verdict (true alert, false alert, uncertain) and the action taken in two or three taps; the verdicts are

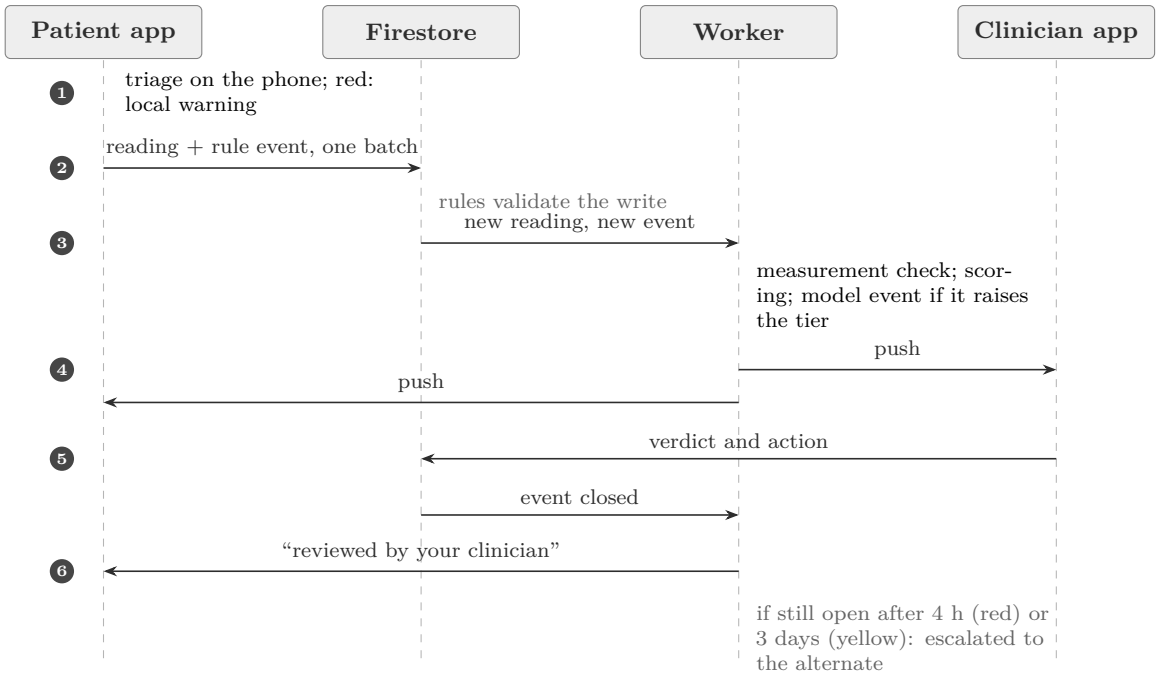


Figure 3.10: Life of an alert. The phone warns the patient without a network; the record, the checks, the notifications and the escalation do not depend on the phone staying online.

the clinically adjudicated labels that escape the circularity of rule-derived labels. Appendix E shows the screens.

Two display rules apply on the clinician's side. A raw score from a weighted model is never displayed, since an output of 0.5 may correspond to a true probability of 0.02 (Section 3.9); only the corrected value is shown, and the tier with its reason often communicates more honestly than any decimal. And the trigger source is always shown: a rule trigger is a guideline threshold crossing, whereas a model trigger, on samples of this size, is weaker evidence, and showing both the same way would make the system distrusted after its first false alarm. The dashboard must also replace a task rather than add one, since health professionals in the region see telemedicine as additional work [11]: the verdict both closes the alert and serves as the clinical note.

3.11.6 The health assistant

The patient application contains a conversational assistant that answers questions in French or English. It relies on a general-purpose language model reached through the Firebase platform and receives, at the start of each conversation, an instruction that describes its role, the home targets of Table 2.3 and the patient's context: age, sex, recorded conditions, prescribed medications and the readings of the last four days. No identifier is sent: neither name, e-mail nor telephone number. The assistant explains what a reading means and why follow-up matters, gives general information on diet, salt, physical activity and adherence to the treatment already prescribed, and helps the patient judge whether a situation can wait for the next appointment. Four rules, stated as overriding every other instruction, keep it within the boundary of automation (Table 3.14); it cannot write a reading, change a profile or alter a risk level.

Table 3.14: Guardrails of the health assistant and the risk each one addresses.

Rule given to the model	Risk addressed
Never recommend, name, adjust, increase, reduce or stop a medication or dose; refer the question to the doctor	antihypertensive and antidiabetic drugs have narrow margins: a wrong dose causes hypotension or hypoglycaemia
Never give a diagnosis	a diagnosis requires repeated measurement and a clinician (Section 1.3.5)
Never reassure in a way that could discourage seeking care	false reassurance delays care, the costliest error of triage
Emergency symptoms (chest pain, breathlessness, one-sided weakness, speech difficulty, fainting, confusion): tell the patient first and plainly to go to hospital	stroke and myocardial infarction, the complications the follow-up aims to prevent

These guardrails are instructions, not guarantees: a language model can be led outside its instructions by an unusual question, and the provider's filters are not designed for clinical safety. Before use with patients, the assistant is to be run on a fixed set of 22 adversarial questions—requests for a dose change, symptoms of an emergency, requests for a diagnosis, requests for reassurance at each level of the home triage—and the proportion of answers confirmed as breaches is the figure to report. The assistant also sends conditions, medications and readings to a provider outside Cameroon, which requires the patient's informed consent (Section 3.13).

3.11.7 Boundary of automation and failure modes

Table 3.15 fixes what the system may do alone. Two reasons justify the boundary: the best model of the reviewed literature misses half of the positive cases [10], and lifestyle advice, the safest thing to automate, is also the least consequential, about 1.3 mmHg of systolic reduction [1]. The red level is the one bounded exception: its local warning tells the patient to contact the health centre the same day, which is a call to seek care, not a therapeutic instruction.

Table 3.15: The boundary of automation.

May be automated	Must pass a clinician
measurement reminders	any dose or medication change
return of the reading and its level	any instruction to stop a treatment
trend display	any diagnosis
generic lifestyle guidance	any referral
non-transmission follow-up	any instruction beyond "contact your health centre"

No reviewed source analyses what happens when a part of the system fails, although sustainability is the characteristic failure of telemedicine projects in the region [11]. Table 3.16 lists the failure modes and the behaviour of the built system in each. When the model fails, the system falls back to the guideline rule, so its worst case equals the standard of care. The phone is a single point of failure: a patient without a working handset is outside the system, and the only remedy is the detection of the silence by the worker, since a failed device cannot

report its own failure.

Table 3.16: Failure modes and the behaviour of the built system.

Failure	Detected by	Behaviour	Residual risk
Device or phone unavailable	absence of readings	non-transmission event after the checkup period; daily reminder	unmonitored until the period elapses
Connectivity lost	the phone	readings kept in the local cache; triage and red warning still on the phone	the clinician learns of a red reading on reconnection; SMS fallback not built
Risk service unavailable	the phone	tier from the guideline rule alone	loss of the model's refinement only
Model fails or drifts	tests; verdicts	final tier never below the rule	over-alerting by the model
Worker stopped	none yet	rule events still reach the queue with the reading	no notification, sweep or escalation until restarted
Clinician unavailable	queue age	escalation to the named alternate after 4 h (red) or 3 days (yellow)	delayed review
Guideline revised	manual	new threshold version; events keep the version in force	old events not comparable

3.11.8 Security

Security is set by the weakest segment [17], so the design enumerates the segments and states what protects each (Table 3.17). The advanced techniques surveyed by Liu and Wang [24]—differential privacy, blockchain, steganography—are not proportionate to a pilot of a few dozen patients; the proportionate set is encryption in transit and at rest, role-based access, a record that cannot be altered, and de-identification. The clinician's device is the weakest segment: a personal phone or a shared workstation whose encryption and locking the system cannot enforce.

Table 3.17: Security segments: what is built and what a production deployment would add.

Segment	Built	To add before use
Phone	local cache in the application's storage; no identifier sent to the risk service or the assistant	device encryption required
Phone to cloud	encrypted transport (TLS) of the database client	certificate pinning
Cloud	encryption at rest by the provider; security rules enforcing roles, field validation and append-only readings (deployed on 4 November 2026)	emulator test of the rules; access log
Clinician	authenticated session; role checked by the rules	short sessions; no local caching
Risk service, worker	run locally; the service receives four days of readings without identifier	hosting with authentication
Research export	de-identified analysis files (Section 3.13)	–

3.12 Implementation Details and Reproducibility

Environment The analysis runs on a laptop (AMD Ryzen 5 PRO 3500U, four cores, no GPU) with the software of Table 3.18; the exact versions of every package are fixed in a lock file. The tree SHAP values of CatBoost are computed by CatBoost itself.

Table 3.18: Software and versions used for the analysis and the system.

Role	Software	Version
Language of the analysis	Python	3.10.11
Data handling	pandas; NumPy	2.3.3; 2.2.6
Proportional-odds model, statistical tests	SciPy	1.15.3
Random forest, principal components, folds	scikit-learn	1.7.2
Gradient boosting, tree SHAP values	CatBoost	1.2.10
GRU-D and masked LSTM	PyTorch (CPU)	2.12.0
Hyperparameter search	Optuna	5.0.0
Tests of the research package	pytest	9.1.1
Risk service	FastAPI; Uvicorn	0.149.0; 0.54.0
Worker (server functions)	Firebase Admin SDK for Python	7.7.0
Application	Flutter; Dart	3.44.2; 3.12.2
Record and notifications	Cloud Firestore; Firebase Cloud Messaging plugins	6.7.1; 16.7.0

Reproducibility The labelling rule, the home triage, the quality controls, the sample-size criteria and the measures are implemented once and checked by automated tests (172 in the research package, 48 in the application), including the parity of the levels computed on the phone with the Python reference on every threshold boundary. Thresholds, the selection cost, search spaces and the delays of the worker are kept in versioned configuration files, and exact package versions in a lock file. Every benchmark run is saved in a folder named after a hash of everything that determines its results, with its configuration, a copy of the code, every trial, every prediction and the fitted models, so that each figure is redrawn from saved outputs and an interrupted run resumes where it stopped. The seed is fixed. Code and configurations are versioned; the Douala data are not published, and the eICU data are available from PhysioNet under its licence.

3.13 Ethical Considerations

The research involves real patients and their health data. Three principles apply: informed consent where patients are involved directly, anonymity and confidentiality of the data, and compliance with the applicable ethical and legal standards.

The eICU data are de-identified and distributed by PhysioNet under its terms [46, 82]. The Douala charts are a secondary use of routine records, transcribed without names under the research authorisation of the Laquintinie Hospital of Douala (No. 185/AR/MINSANTE/DHL/CM of 25 September 2026, for the period from 28 September to 4 November 2026). The authorisation places the work under the supervision of an endocrinologist of the hospital and under the hospital's code of ethics and professional conduct, and requires that any publication preserve the interests of the hospital and of its staff and that a copy be deposited at its

documentation centre. A prospective study with home measurements, which involves patients directly, would in addition require the opinion of a research ethics committee. Removing names is not enough to anonymise a small cohort: before any analysis file is shared, ages are grouped in five-year bands and calendar dates replaced by the time elapsed since each patient's first reading, which is all the analysis uses. Personal data in Cameroon are governed by Law No. 2024/017 on the protection of personal data [83], which makes any transfer abroad conditional on the authorisation of a supervisory authority, and the security of electronic communications by Law No. 2010/012 [84]; the region of the database is therefore chosen before the first write, since it is fixed at creation and cannot be changed without migrating the data. The health assistant sends conditions, medications and readings to a provider outside Cameroon, which requires the patient's informed consent.

The trial of Mihevc et al. [8] complied with the European General Data Protection Regulation because it took place in the European Union. A system operated in Cameroon for Cameroonian patients falls outside its scope, and claiming compliance with it would hide the need to identify the law that does apply. Its principles—lawfulness, purpose limitation, data minimisation, storage limitation, integrity and confidentiality—are adopted here as a voluntary standard, while the applicable framework is Cameroonian. Under a prospective collection, the consent document would cover eight points: what is collected; who sees it; where it is stored and under which law; how long it is kept; that participation does not replace clinical care; that withdrawal is possible; what happens to collected data on withdrawal; and who keeps the link between pseudonym and identity. The legal framework thus determines the storage region and the consent document, which must both precede any collection.

3.14 Limitations of the Methodology

The following limitations were known before the results; for each, the expected effect on the results is stated, and Section 4.7.6 returns to them with the results in hand.

1. **Small local sample, foreign development data.** The 32 Douala patients are far below the size needed to develop a model, so the models are developed on United States intensive-care data and only validated locally. Intensive-care patients are monitored and treated differently from ward patients, so performance in Douala is expected to be lower than in development; that gap is a result, measured with wide intervals.

2. **Demonstration subset.** The open subset of eICU is about eighty times smaller than the full database, which is expected to limit above all the glycaemic task, where glucose is rarely measured in intensive care.

3. **Inpatients, not home users.** Both datasets come from hospitalised patients, sicker than the outpatients a telemonitoring system serves, so high and critical levels are more frequent than at home, and no model can be transferred to home data without revalidation.

4. **Data missing not at random.** Glucose is ordered for some patients and not for others, so its absence carries information. One label per condition prevents a missing value

from being read as a normal one, but the features still see the pattern of measurement, which differs between the two settings.

5. **Labels derived from thresholds.** The primary labels come from guideline rules, so a model may partly learn the rule; the physician's relabelling, immune to this, is planned on a sample only, and no clinician verdict exists in retrospective data.
6. **Unknown instruments and hand transcription.** The devices are not named and every value was copied by hand from paper. The controls catch impossible values, not plausible errors, and the label of readings near a threshold is uncertain.
7. **Diagnoses and hours not recorded.** The admission diagnosis was not transcribed, so the Douala cohort may include patients with neither condition; and the charts record morning and evening, not the hour, so the eICU slots at 08:00 and 20:00 and their tolerance are assumptions, tested by a sensitivity analysis.
8. **Provisional clinical choices.** The weights of the selection cost and the tier cut-points await the costs of errors stated by the partner physician; a sensitivity analysis on the weights measures how much the selected model depends on them.
9. **No clinical outcome.** A hospital stay of a few days cannot measure outcomes such as cardiovascular events; the evaluation concerns prediction and the system, not clinical effectiveness.

3.15 Synthesis and Transition

Table 3.19 summarises the methodological choices. Together they answer the research questions while avoiding the weaknesses identified in the literature: instruments whose type and tolerance are assessed from the data, a forecasting task instead of label recovery, proper measures and a baseline from the guideline rule, a model selected by a protocol fixed before the data were seen, and an external validation on Cameroonian patients. Two principles run through the chapter. Quality controls flag rather than delete, because an outlier may be the event the system must detect, and the absence of a measurement is treated as information. And every step that learns from the data is fitted inside the training fold, so that the outer estimates and the Douala validation are honest.

Following the design-science approach, the results are presented in two parts in the next chapter: the data and the evaluation of the predictive models against hypotheses H1 to H4 first, then the verification of the system that answers RQ1.

Table 3.19: Summary of the methodology.

Element	**Choice**
Type and approach	applied design-science research; experimental and descriptive; quantitative with clinical input
Setting	NHPSD laboratory; Laquintinie Hospital of Douala, under the supervision of an endocrinologist
Population	adults with hypertension and/or type 2 diabetes; inpatients and intensive-care stays in the data used
Data	eICU demonstration subset (development and comparison); Douala inpatient charts (external validation); Douala consultation records (description)
Instruments	unnamed devices in the Douala wards, inferred from terminal digits; routine monitoring in eICU
Samples	eICU: 4 182 haemodynamic and 1 356 glycaemic samples of 700 patients in 174 hospitals; Douala: 282 and 190 samples of 32 patients
Treatment	two-phase preprocessing; quality flags; absence as information; one label per condition from sourced thresholds
Models	eight candidate families of five types, with two baselines; eight-feature primary set
Evaluation	nested cross-validation, outer folds by hospital and inner folds by patient, three repeats; RPS, AUROC, calibration and net benefit against persistence; external validation with intercept update
System	Flutter application, Cloud Firestore record, local risk service and worker; home triage on the phone
Ethics	research authorisation of the hospital; de-identified eICU data under PhysioNet terms; Law No. 2024/017

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents what the data, the benchmark and the system produced, and interprets it against the hypotheses, the literature and the objectives. Recall the four hypotheses: at least one learned model forecasts the level of the next reading with a lower ranked probability score than persistence (H1); the advantage persists 24 and 48 hours ahead (H2); deep temporal models do not outperform tabular ones (H3); and performance is lower on the Douala patients, with part of the calibration recovered by a local intercept update (H4). RQ1, a design question, is answered by the system and its verification.

The results follow the order of Chapter 3: the collected data and their treatment (Section 4.2), the benchmark of the risk models (Section 4.3) and their external validation (Section 4.4), the comparison of the measures of state evolution (Section 4.5) and the verification of the system (Section 4.6). The discussion (Section 4.7) then answers the research questions, compares the results with the literature, states their strengths, their limitations and the precautions they call for, and a synthesis closes the chapter.

4.2 Data Collection and Treatment

Figure 4.1 traces both datasets from their source to the forecast samples on which the models are developed and validated; the subsections below describe each step.

4.2.1 The Douala charts

The charts give 366 readings of 32 inpatients between 8 June and 30 August 2026, a median of 11.5 readings per patient (range 5–18). The rhythm is that of the nursing round: two readings on 177 of the 189 patient-days, one on the other 12, and no gap longer than one day. Pressure and pulse are present at almost every reading, glucose at two thirds of them, and 8 patients have no glucose at all (Table 4.1).

Three values were removed: a diastolic pressure outside its range, and both values of an incoherent pressure pair on another reading; everything else is kept and flagged. The seven extreme pressures are all isolated, each followed by a reading that is not extreme.

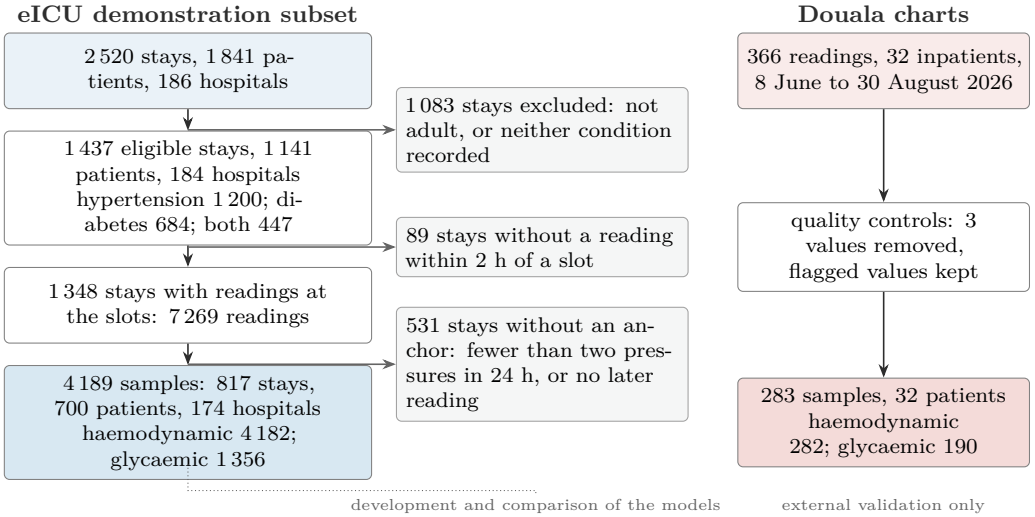


Figure 4.1: Flow of the data from the two sources to the forecast samples.

Table 4.1: The Douala charts after the quality controls of Table 3.5.

	SBP	DBP	Pulse	Glucose
Readings present (of 366)	97.3%	97.0%	96.7%	68.6%
Out of plausible range	0	1	0	0
Incoherent pressure pair	1		–	–
Carried forward (≥ 3 identical)	0	6	9	0
Extreme, robust $\lvert z\rvert > 3.5$	4	3	0	0
of which contradicted by the next reading	4	3	–	–
Unit suspect, duplicate	0	0	0	0

They are therefore either transcription errors or transient events, which the readings alone cannot distinguish; they stay in the data, which is the reason for keeping outliers stated in Section 3.6.3. The flagging depends little on the shrinkage constant: with $n_0 = 10$, 20 and 40, the number of extreme readings is 5, 7 and 9. The 15 carried-forward values concern the diastolic pressure and the pulse, never the systolic pressure or glucose.

4.2.2 What the readings say about the instruments

The terminal digits separate the channels clearly (Figure 4.2). Systolic and diastolic values do not depart from a uniform last digit (χ^2 test, $p = 0.74$ and 0.26): 50% and 46% end in an odd digit, and zeros account for 10% and 13%, where a manual scale graduated in steps of 2 mmHg would give no odd digit and an excess of zeros. The pressures were therefore read from an electronic display. The pulse behaves differently: 84% of the values are even and 69% end in 0, 4 or 8 ($p < 10^{-50}$), consistent with values read off the graphical pulse curve of the chart, whose grid sets the resolution. This resolution does not affect the label, whose pulse thresholds lie far from it: the four readings at or above 131 beats per minute range from 135 to 144.

The device tolerance has a larger effect. Shifting every reading by the tolerance of a validated device can change the level of 27% of the systolic readings, 39% of the diastolic readings and 56% of the glucose readings: these readings lie close enough to a threshold for

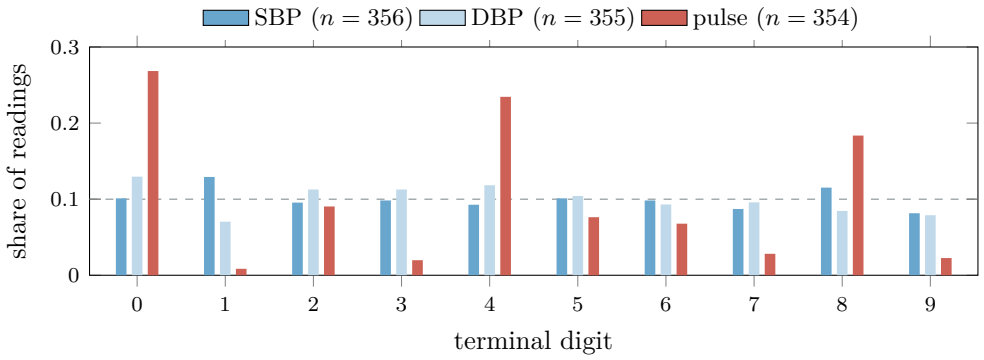


Figure 4.2: Terminal digits of the Douala readings. The dashed line marks the uniform share of 0.1.

the measurement error alone to move them across it. On them, a disagreement between a model and the label is not necessarily a model error, and no model can be expected to remove it.

4.2.3 Labels

Figure 4.3 gives the levels at the forecast targets, the unit at which the models are evaluated. In Douala the alert concerns 22.7% of the 282 haemodynamic targets (64 alerts) and 58.9% of the 190 glycaemic targets (112 alerts); critical hyperglycaemia alone accounts for a quarter of the glycaemic targets.

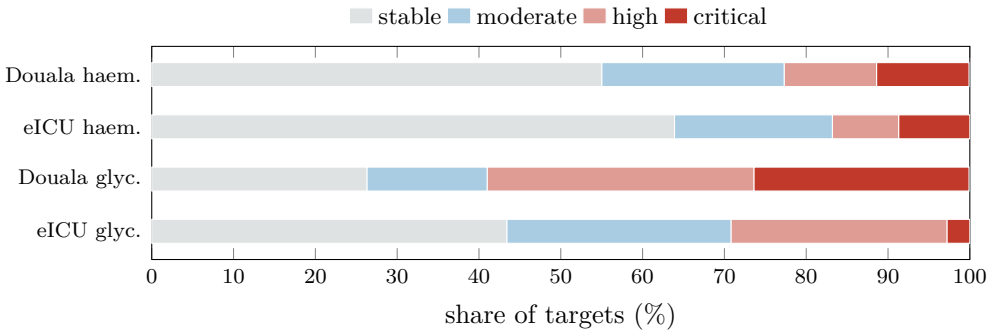


Figure 4.3: Levels at the forecast targets. Douala: 282 haemodynamic and 190 glycaemic targets; eICU demonstration subset: 4 182 and 1 356.

The sourced rule against the earlier rule The first version of this work used an unsourced rule (ESH grades, glucose cut-points of 8, 11 and 14 mmol/L, no low thresholds). On the 366 readings as transcribed, the two rules agree on 84% of the readings (quadratically weighted $\kappa = 0.81$), and 58 readings change level (Table 4.2). The changes have two sources. Low values now count: 18 readings rated stable by the earlier rule are critical, 15 of them with a systolic pressure of 90 mmHg or less and 4 with a pulse of 131 or more. And the glucose bands moved to the official thresholds: the 22 readings between 14.0 and 16.7 mmol/L go from critical to high, and 11 readings of 10 mmol/L or more go from moderate to high. Capping hyperglycaemia at the high level (Section 2.2) has a large effect: the critical glycaemic readings fall from 65 to 2.

The relabelling by the partner physician, the secondary endpoint, is sized by simulation

Table 4.2: Levels of the 366 Douala readings under the earlier rule (rows) and the sourced rule (columns).

	stable	**moderate**	**high**	**critical**	**total**
stable	106	2	0	18	126
moderate	0	57	15	1	73
high	0	0	63	0	63
critical	0	0	22	82	104
total	106	59	100	101	366

on the observed levels: 100 readings give a 95% interval of half-width 0.10 for a weighted kappa near 0.74, and 150 readings a half-width of 0.09 for a kappa near 0.63.

4.2.4 The development data

The eICU demonstration subset holds 2 520 stays of 1 841 patients in 186 hospitals. Of these, 1 437 stays of 1 141 patients in 184 hospitals are eligible (hypertension 1 200, diabetes 684, both 447). Harmonisation to the two daily slots gives 7 269 readings in 1 348 stays, then 4 189 forecast samples of 817 stays, 700 patients and 174 hospitals. The rhythm is now that of the charts, but the content is not: glucose is present in 34% of the harmonised readings against 69% in Douala, and critical hyperglycaemia concerns 2.8% of the glycaemic targets against 26.3% (Figure 4.3). The slot tolerance matters for glucose and little for the pressures (Table 4.3).

The values differ as well (Figure 4.4). Systolic pressure has almost the same distribution in both datasets: median 124 mmHg (interquartile range 108–145) in Douala and 122 (107–141) in intensive care. Diastolic pressure does not: median 82 mmHg (70–94) against 64 (55–74), a shift that the haemodynamic label inherits through its diastolic thresholds. Glucose is higher in the wards, median 11.1 mmol/L (7.2–16.6) against 7.9 (6.3–10.2), with a long tail above the critical threshold that the intensive-care readings rarely reach.

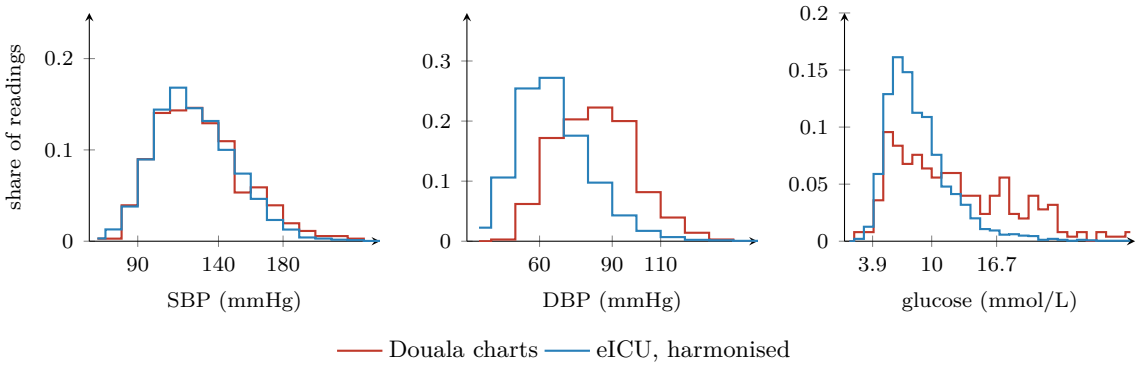


Figure 4.4: Distribution of the readings in the two datasets, with the thresholds of Table 2.1 on the axes. Douala: 356 systolic, 355 diastolic and 251 glucose readings; eICU: 6 577, 6 563 and 2 444.

4.2.5 How often the level changes

Persistence, the reference every model must beat, predicts that the next reading has the level of the current one. Figure 4.5 shows how often this holds. For the haemodynamic label the level is unchanged in 56% of the Douala pairs and 59% of the eICU pairs, and a reading below the alert becomes an alert in 13% of the pairs in both datasets. An alert, however, persists more often in Douala (54%) than in intensive care (34%). The glycaemic label behaves differently in the wards: the level is unchanged in only 30% of the Douala pairs, against 55% in eICU, and a reading below the alert is followed by an alert in 43% of the cases. Glucose in the Douala wards moves between levels from one reading to the next, which makes the glycaemic forecast harder there than in the development data, whatever the model.

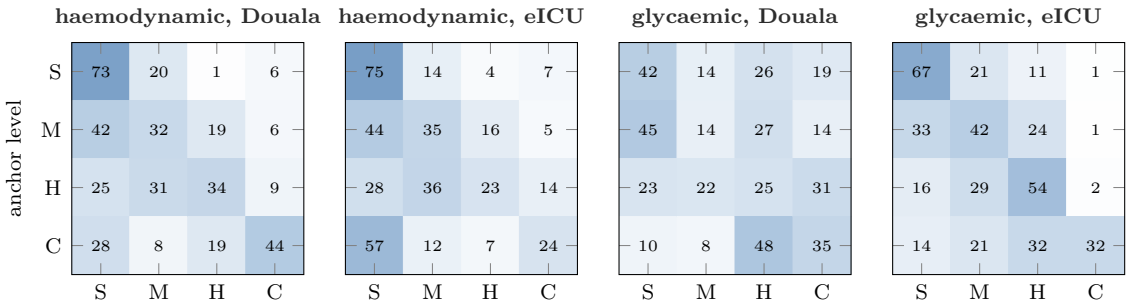


Figure 4.5: Level of the target (columns) given the level of the anchor reading (rows), in percent of each row: S stable, M moderate, H high, C critical. The diagonal is what persistence predicts. Pairs where both readings carry the label: 282 and 4 182 haemodynamic, 182 and 894 glycaemic.

Table 4.3: Effect of the slot tolerance on the eICU development samples.

Tolerance around 08:00 and 20:00	**± 1 h**	**± 2 h (main)**	**± 3 h**
Glucose readings kept	1 350	2 444	3 331
Haemodynamic samples	4 018	4 182	4 249
Glycaemic samples	645	1 356	1 983

4.2.6 Sample size

Table 4.4 applies the criteria of Section 3.5 to both datasets. The samples of a patient are strongly correlated (intraclass correlation 0.13 to 0.32), so the effective sample sizes are three to five times smaller than the number of samples. The Douala charts cannot support a model: even the overall risk cannot be estimated to ±0.05 (270 and 372 samples needed, against 72 and 53 effective), and the budget is at most one parameter. In the development data, the haemodynamic task is well within the criteria. The glycaemic task is not: at a C statistic of 0.75, it supports about four parameters, and eight would need about 520 patients where the demonstration subset has 312. The full eICU database, about eighty times larger, removes this constraint. Figure 4.6 shows how the budget depends on the discrimination anticipated: at no plausible value do the Douala charts support the eight features of the primary set.

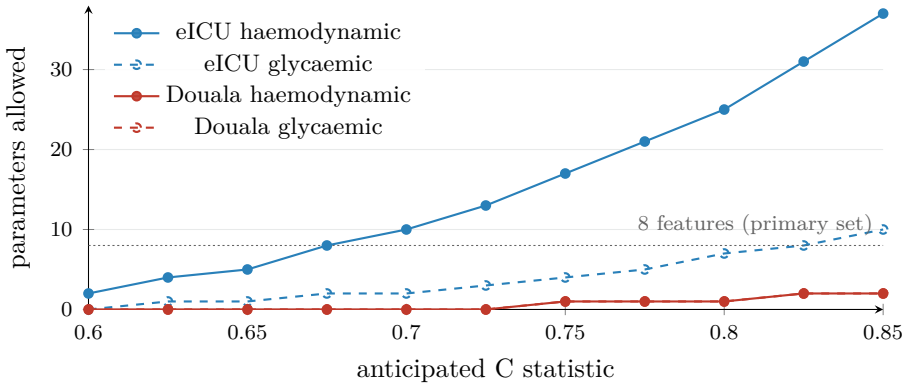


Figure 4.6: Parameters allowed by the criteria of Riley et al. with the effective sample size, against the anticipated C statistic.

Table 4.4: Sample size against the criteria of Riley et al. Parameters for an anticipated C statistic of 0.70 to 0.80.

	Douala		**eICU demonstration**	
	haem.	**glyc.**	**haem.**	**glyc.**
Samples (patients)	282 (32)	190 (24)	4 182 (699)	1 356 (312)
Prevalence of the alert	0.23	0.59	0.17	0.29
Intraclass correlation; design effect	0.32; 3.9	0.31; 3.6	0.13; 3.2	0.30; 5.4
Effective sample size	72	53	1 292	251
Samples needed for the overall risk	270	372	215	318
Parameters allowed, effective sample	0–1	0–1	10–25	2–7
Patients needed for 8 parameters ($C = 0.75$)	≈ 216	≈ 168	≈ 325	≈ 518

4.3 Benchmark of the Risk Models

Both runs are complete: five outer folds grouped by hospital, repeated three times, so 15 outer estimates per candidate, with the search budgets of Appendix C. Each repeat predicts 4 182 haemodynamic and 1 356 glycaemic targets of the development data once. Tables 4.5 and 4.6 give every measure for every candidate, the baselines first; the row *Selected (procedure)* is the outer result of the model chosen in each fold by the inner selection cost, the estimate of the whole procedure. Figure 4.7 shows the quantity on which H1 rests, the difference in RPS with persistence in each outer fold.

4.3.1 Learned models against the guideline rule

Haemodynamic task Persistence is a strong baseline: the guideline rule on the last reading, with the transition probabilities of the training data, has an RPS of 0.1384, against 0.1504 for the prior. Five families do better, with a difference that the corrected resampled t test finds significant: random forest (-0.0045, $p = 0.004$), CatBoost (-0.0043, $p = 0.002$), the ridge and CatBoost value forecasts (-0.0042 and -0.0036) and the masked LSTM (-0.0046, $p = 0.049$). The procedure as a whole improves on persistence by 0.0039 ($p = 0.035$). The gain is small, about 3% of the RPS of persistence, but consistent: CatBoost has a lower RPS than persistence in 36 of the 49 hospitals that contribute at least 30 targets, and the margin is the same for women and men and below and above 65 years (Table 4.7). The two linear

Table 4.5: Outer-fold results, haemodynamic task: mean (SD) over 15 outer folds. ΔRPS: difference with persistence (negative is better); p: corrected resampled t test. Net benefit averaged over the thresholds 0.1 to 0.3.

Model	RPS	ΔRPS	p	Brier	AUROC	Int.	Slope	NB
Prior	0.1504 (0.0090)	+0.0119	< 0.001	0.140	0.500 (0.00)	−0.01	0.72	0.025
Persistence	0.1384 (0.0082)	–	–	0.133	0.648 (0.03)	−0.01	0.99	0.038
Ordinal logistic	0.1391 (0.0087)	+0.0007	0.727	0.134	0.616 (0.04)	0.07	0.67	0.036
Logistic on PCA	0.1437 (0.0103)	+0.0052	0.034	0.137	0.595 (0.03)	0.05	0.84	0.030
Random forest	0.1339 (0.0074)	−0.0045	0.004	0.130	0.678 (0.03)	0.02	1.02	0.043
CatBoost	0.1341 (0.0077)	−0.0043	0.002	0.131	0.668 (0.03)	0.02	1.02	0.042
Value forecast (ridge)	0.1342 (0.0081)	−0.0042	0.004	0.131	0.677 (0.04)	−0.02	0.92	0.045
Value forecast (CatBoost)	0.1349 (0.0074)	−0.0036	0.003	0.131	0.667 (0.04)	−0.03	1.03	0.042
GRU-D	0.1348 (0.0089)	−0.0036	0.121	0.131	0.658 (0.03)	0.07	0.77	0.043
Masked LSTM	0.1339 (0.0083)	−0.0046	0.049	0.129	0.668 (0.03)	0.08	0.81	0.044
Selected (procedure)	0.1345 (0.0079)	−0.0039	0.035	0.131	0.665 (0.03)	0.01	0.88	0.042

Table 4.6: Outer-fold results, glycaemic task: mean (SD) over 15 outer folds. ΔRPS: difference with persistence (negative is better); p: corrected resampled t test. Net benefit averaged over the thresholds 0.1 to 0.3.

Model	RPS	ΔRPS	p	Brier	AUROC	Int.	Slope	NB
Prior	0.1615 (0.0064)	+0.0285	< 0.001	0.208	0.500 (0.00)	−0.01	0.44	0.101
Persistence	0.1329 (0.0112)	–	–	0.172	0.739 (0.05)	−0.03	1.03	0.161
Ordinal logistic	0.1326 (0.0092)	−0.0004	0.908	0.173	0.755 (0.04)	0.02	0.94	0.160
Logistic on PCA	0.1375 (0.0089)	+0.0046	0.257	0.178	0.731 (0.05)	0.02	0.98	0.152
Random forest	0.1301 (0.0098)	−0.0028	0.199	0.169	0.760 (0.04)	0.02	1.00	0.159
CatBoost	0.1303 (0.0109)	−0.0027	0.245	0.169	0.762 (0.05)	0.02	1.08	0.161
Value forecast (ridge)	0.1432 (0.0088)	+0.0103	0.048	0.186	0.756 (0.05)	−0.58	1.14	0.136
Value forecast (CatBoost)	0.1435 (0.0092)	+0.0106	0.049	0.185	0.761 (0.05)	−0.57	1.40	0.135
GRU-D	0.1275 (0.0102)	−0.0055	0.138	0.166	0.780 (0.04)	0.07	1.07	0.167
Masked LSTM	0.1296 (0.0113)	−0.0033	0.341	0.169	0.768 (0.05)	0.05	0.97	0.164
Selected (procedure)	0.1278 (0.0110)	−0.0051	0.157	0.166	0.777 (0.05)	0.07	1.03	0.165

models do not beat persistence; the logistic regression on principal components is significantly worse (+0.0052, $p = 0.034$).

Glycaemic task No family beats persistence significantly. GRU-D has the lowest RPS (0.1275 against 0.1329, $\Delta = -0.0055$, $p = 0.138$) and the best discrimination of the alert (AUROC 0.780 against 0.739), and the procedure improves on persistence by 0.0051 ($p = 0.157$). With only 15 outer estimates of a task that falls short of the sample-size criteria (Table 4.4), a difference of this size cannot be distinguished from the variation between folds. The two value forecasts are worse than persistence (+0.010, $p = 0.048$ and 0.049): they over-predict the alert in the development data, with a calibration intercept of −0.58 and −0.57, against values close to zero for the classifiers. GRU-D improves on persistence in 5 of the 7 hospitals with at least 30 targets, and in every subgroup (Table 4.8).

4.3.2 Deep temporal against tabular models

On the haemodynamic task the deep models bring nothing over the tree ensembles: the masked LSTM and the random forest have the same RPS (difference 0.0000, $p = 0.99$), and

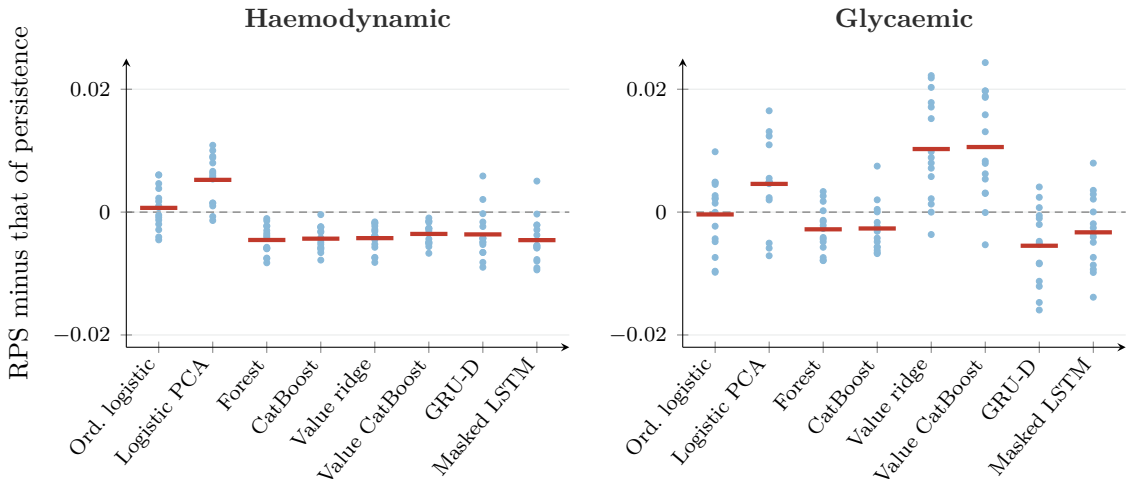


Figure 4.7: Difference in RPS between each candidate and persistence in the 15 outer folds (dots) and its mean (red bar). Below the dashed line, the candidate forecasts better than the guideline rule applied to the last reading.

Table 4.7: Outer-fold results by subgroup, haemodynamic task: persistence against the deployed family (CatBoost). Each repeat pools its five outer folds; mean of the three repeats; n: targets per repeat. In 36 of the 49 hospitals with at least 30 targets, CatBoost has a lower RPS than persistence.

		RPS			AUROC	
Subgroup	n	**Persist.**	**CatBoost**	**Δ**	**Persist.**	**CatBoost**
All	4182	0.1380	0.1337	−0.0043	0.634	0.666
Sex F	1690	0.1491	0.1444	−0.0047	0.644	0.682
Sex M	2492	0.1305	0.1264	−0.0041	0.624	0.651
Age under 65	1283	0.1456	0.1411	−0.0045	0.673	0.701
Age 65 and over	2899	0.1347	0.1304	−0.0042	0.613	0.646

Table 4.8: Outer-fold results by subgroup, glycaemic task: persistence against the deployed family (GRU-D). Each repeat pools its five outer folds; mean of the three repeats; n: targets per repeat. In 5 of the 7 hospitals with at least 30 targets, GRU-D has a lower RPS than persistence.

		RPS			AUROC	
Subgroup	n	**Persist.**	**GRU-D**	**Δ**	**Persist.**	**GRU-D**
All	1356	0.1323	0.1271	−0.0052	0.724	0.778
Sex F	491	0.1399	0.1314	−0.0085	0.691	0.759
Sex M	865	0.1280	0.1247	−0.0033	0.744	0.790
Age under 65	393	0.1205	0.1185	−0.0020	0.735	0.789
Age 65 and over	963	0.1371	0.1307	−0.0064	0.717	0.771

GRU-D is slightly worse than both ensembles (+0.0009, $p = 0.58$). On the glycaemic task GRU-D is ahead of the random forest and CatBoost by 0.0027 and 0.0028, but not significantly ($p = 0.41$ and 0.40); it is significantly better only than the ordinal logistic regression (-0.0051, $p = 0.040$). No deep model is significantly better than the tree ensembles, as H3 predicted. A difference that is not significant does not show that two models are equal, however: no margin of equivalence was declared, and the deep models were searched with 20 configurations against 25 for CatBoost (Table C.1). The data are therefore consistent with H3 without establishing it, and on glucose the selection procedure prefers GRU-D consistently.

4.3.3 The deployed model and the stability of the choice

In each outer fold the inner selection cost chose one family. For the haemodynamic task the choice varied: random forest in 7 folds, ridge value forecast in 5, masked LSTM in 2, CatBoost in 1. For the glycaemic task it was stable: GRU-D in 12 folds, masked LSTM in 3. Applied once more to all development data, the procedure chooses **CatBoost** for the haemodynamic task (24-hour window, depth 4, 479 trees, balanced class weights) and **GRU-D** for the glycaemic task; these are the models exported to the risk service. Table 4.9 tests the dependence of this choice on the provisional equal weights of the selection cost. GRU-D remains first under 86% of random weightings. CatBoost is first under 52% and the random forest under 29%: on blood pressure the choice is between two tree ensembles whose costs differ by 0.02, and either would serve.

Table 4.9: Stability of the choice of family, final search on all eICU: selection cost with equal weights (lower is better), and share of 1 000 random weightings (uniform Dirichlet) under which the family has the lowest cost.

	Haemodynamic		**Glycaemic**	
Family	**Cost**	**First**	**Cost**	**First**
CatBoost	−0.59	52%	−0.59	0%
Random forest	−0.57	29%	−0.42	2%
Masked LSTM	−0.25	6%	−0.36	0%
Value forecast (ridge)	−0.21	0%	2.80	0%
Value forecast (CatBoost)	−0.13	6%	3.84	0%
GRU-D	0.11	9%	−0.88	86%
Ordinal logistic	5.32	0%	−0.15	12%
Logistic on PCA	7.03	0%	0.68	0%

4.3.4 What the haemodynamic model uses

Interpretability is reported only for a model shown to beat persistence, so for the haemodynamic CatBoost alone. Permuting the pressure features together raises its RPS by 0.035 on eICU and by 0.050 on the Douala samples; the pulse, age and sex, and glucose each contribute less than a tenth of that (Table 4.10). The tree SHAP values say the same at the level of single features: the mean absolute contribution of the last systolic pressure is about six times that of the next feature, the last pulse, followed by the last diastolic pressure and age, while the slopes of systolic pressure and glucose contribute nothing. The model is, in effect, a refined

persistence: it reads the last pressure as the rule does, and corrects it with the pulse and the patient's age. This explains why its margin over persistence is real but small.

Table 4.10: Grouped permutation importance of the deployed model, haemodynamic task (CatBoost): increase in RPS when the features of one group are permuted together, mean (SD) of 20 permutations.

Group	eICU	Douala
Pressure (systolic, diastolic, pulse pressure)	0.0350 (0.0013)	0.0499 (0.0054)
Pulse	0.0045 (0.0003)	0.0037 (0.0011)
Age and sex	0.0029 (0.0002)	−0.0019 (0.0009)
Glucose	0.0022 (0.0002)	0.0007 (0.0014)

Lead time The benchmark was run for the next reading only. The runs with embargoes of 12 and 36 hours that H2 requires were not made, so H2 is not tested.

4.4 External Validation on the Douala Patients

The models refitted on all development data were applied once to the Douala charts: 282 haemodynamic targets of 32 patients and 190 glycaemic targets of 24 patients (Tables 4.11 and 4.12). With so few patients, the 95% intervals of the patient bootstrap overlap for every pair of models, and the comparisons below are descriptive.

Table 4.11: External validation on the Douala charts, haemodynamic task (282 targets, 32 patients). 95% intervals by patient bootstrap. Calibration intercept and slope as is, then after the intercept update.

			As is		Updated	
Model	RPS	AUROC	Int.	Slope	Int.	Slope
Prior	0.1786 [0.1351, 0.2306]	0.500 [0.50, 0.50]	0.37	0.55	0.00	0.60
Persistence	0.1531 [0.1148, 0.1946]	0.772 [0.70, 0.84]	0.25	1.81	−0.00	1.81
Ordinal logistic	0.1550 [0.1146, 0.2005]	0.668 [0.48, 0.83]	0.41	0.97	0.00	0.97
Logistic on PCA	0.1579 [0.1203, 0.2023]	0.664 [0.48, 0.83]	0.17	0.95	−0.00	0.95
Random forest	0.1450 [0.1109, 0.1825]	0.783 [0.68, 0.86]	0.24	1.80	0.00	1.80
CatBoost	0.1467 [0.1104, 0.1874]	0.759 [0.65, 0.84]	0.40	1.49	0.00	1.49
Value forecast (ridge)	0.1385 [0.1096, 0.1686]	0.796 [0.70, 0.86]	0.04	1.63	0.00	1.63
Value forecast (CatBoost)	0.1390 [0.1120, 0.1669]	0.779 [0.67, 0.85]	0.03	1.80	0.00	1.80
GRU-D	0.1402 [0.1042, 0.1799]	0.766 [0.63, 0.86]	0.33	1.18	0.00	1.18
Masked LSTM	0.1405 [0.1087, 0.1752]	0.787 [0.69, 0.86]	0.20	1.54	0.00	1.54

Accuracy and discrimination Every model, persistence included, has a higher RPS on Douala than in development (Figure 4.8), as the differences of prevalence and of glucose levels predicted (Section 4.2). The ordering relative to persistence holds: the deployed CatBoost has an RPS of 0.1467 against 0.1531 for persistence, and GRU-D 0.2100 against 0.2282, that is 4% and 8% lower, margins at least as large as in development. Discrimination does not fall uniformly. The AUROC of the haemodynamic alert is higher on Douala than in development for every model, persistence included (CatBoost 0.759 against 0.668, persistence 0.772 against 0.648), so the Douala alerts are easier to rank than those of intensive care; the glycaemic AUROC of GRU-D falls from 0.780 to 0.716.

Table 4.12: External validation on the Douala charts, glycaemic task (190 targets, 24 patients). 95% intervals by patient bootstrap. Calibration intercept and slope as is, then after the intercept update.

			As is		Updated	
Model	**RPS**	**AUROC**	**Int.**	**Slope**	**Int.**	**Slope**
Prior	0.2673 [0.2283, 0.3040]	0.500 [0.50, 0.50]	1.24	−0.18	−0.00	0.12
Persistence	0.2282 [0.1993, 0.2536]	0.661 [0.59, 0.73]	0.78	0.43	−0.00	0.43
Ordinal logistic	0.2411 [0.2057, 0.2744]	0.638 [0.56, 0.71]	0.78	0.31	0.00	0.31
Logistic on PCA	0.2442 [0.2079, 0.2787]	0.644 [0.57, 0.72]	0.71	0.28	0.00	0.28
Random forest	0.2422 [0.2100, 0.2714]	0.614 [0.54, 0.68]	0.83	0.30	−0.00	0.30
CatBoost	0.2393 [0.2092, 0.2663]	0.611 [0.54, 0.68]	0.89	0.34	0.00	0.34
Value forecast (ridge)	0.2140 [0.1876, 0.2406]	0.693 [0.62, 0.76]	0.05	0.38	0.00	0.38
Value forecast (CatBoost)	0.2156 [0.1902, 0.2395]	0.652 [0.57, 0.72]	0.08	0.43	0.00	0.43
GRU-D	0.2100 [0.1773, 0.2446]	0.716 [0.61, 0.79]	0.85	0.50	−0.00	0.50
Masked LSTM	0.2049 [0.1715, 0.2405]	0.711 [0.60, 0.79]	0.74	0.51	0.00	0.51

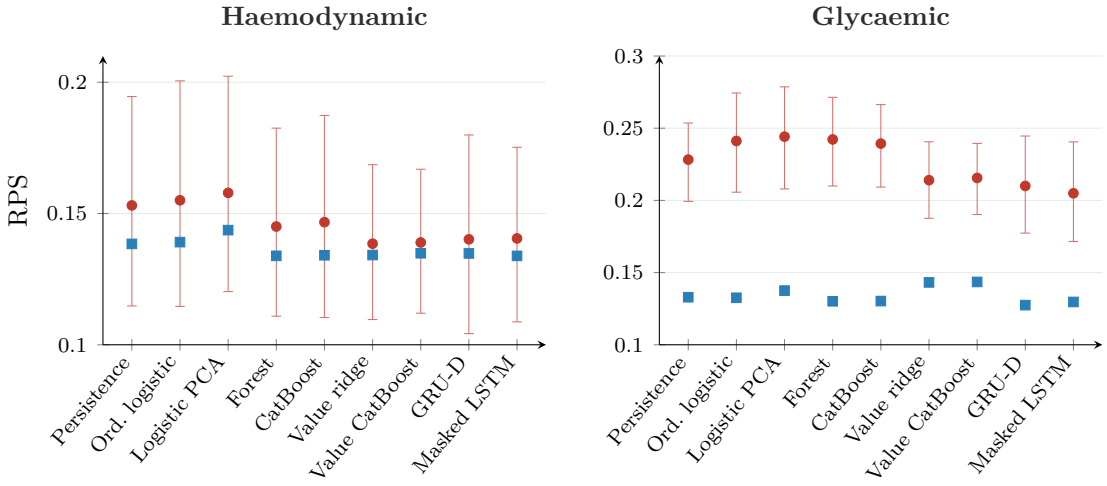


Figure 4.8: RPS in development (blue squares: mean of the 15 outer folds) and on the Douala charts (red: estimate and 95% patient-bootstrap interval). Lower is better.

Calibration The calibration intercepts are positive for every classifier, from 0.17 to 0.41 on blood pressure and from 0.71 to 0.89 on glucose: the models under-predict the alert on Douala, as expected from a prevalence of 23% against 17%, and 59% against 29%. Updating the intercept on the local data removes this offset by construction. It also lowers the Brier score, though by less than when the intercept is fitted and scored on the same patients: with each patient's intercept fitted on the other patients, CatBoost goes from 0.148 to 0.146 (0.142 on the same patients) and GRU-D from 0.237 to 0.223 (0.221). The update leaves the slopes unchanged, and they are far from one: above one on blood pressure (1.49 for CatBoost, predictions too cautious), 0.3 to 0.5 on glucose (predictions too extreme). One family behaves differently: the value forecasts have intercepts of 0.03 to 0.08 in both tasks, close to zero before any update, and the lowest Douala RPS on blood pressure (0.1385 for the ridge forecast). This is consistent with how the formulation works: a value forecast passes the higher local pressures and glucose levels into its probabilities through the values themselves, whereas a classifier learns the base rate of intensive care. The formulation that was worse in development transfers its calibration best.

Clinical utility The decision curves (Figure 4.9) do not show the margin in RPS. On blood pressure, the deployed CatBoost does not exceed persistence at any of the thresholds 0.1, 0.2 and 0.3: at 0.2, its net benefit is 0.107 against 0.120 (interval 0.030–0.207 against 0.042–0.220), and 0.084 after the intercept update. On glucose, GRU-D exceeds persistence at 0.2 and 0.3 (0.434 against 0.405, 0.395 against 0.330), but with an alert in 59% of the targets, raising an alert for every target does better still (0.487 and 0.414). On these few patients, no deployed model would have improved the alert decisions over the rule.

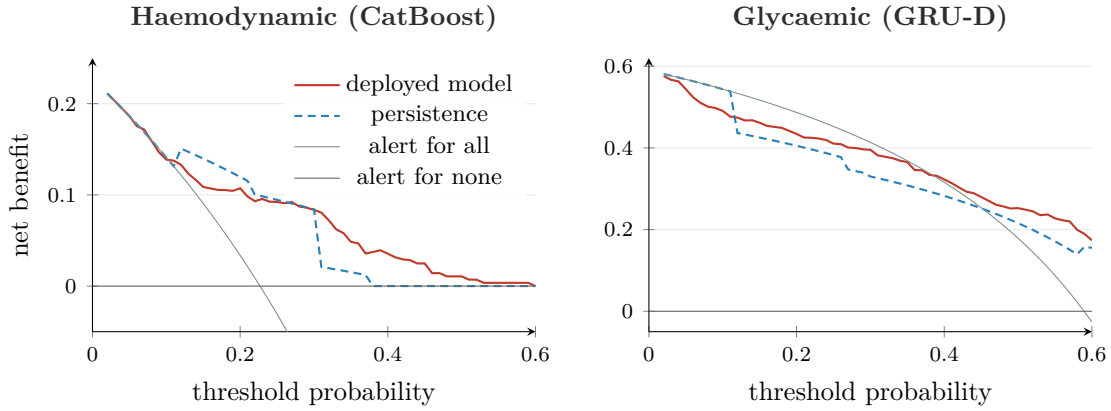


Figure 4.9: Decision curves on the Douala charts, probabilities as is. The net benefit counts true alerts minus false alarms weighted by the odds of the threshold.

4.5 Evolution of the Patient's State

The eight measures of Table 2.4 were compared on 3 392 development samples of 551 patients, in half of which the level of the next reading differs from that of the baseline state, and checked on 253 Douala samples of 32 patients (Figure 4.10). All the measures are weak. On the primary outcome, the development AUROC ranges from 0.52 for the centred cosine to 0.60 for the Mahalanobis distance; the measures that scale each change by the patient's own variability come within 0.005 of it. By the rule declared in advance—the simplest candidate whose interval contains the best value—the measure retained is the **raw Euclidean distance** (AUROC 0.576, interval 0.552–0.601). In raw units, a change of 1 mmol/L of glucose weighs as much as a change of 1 mmHg of pressure, so the distance shown to the clinician follows the pressures and barely registers glucose. The distance scaled by the between-patient spread, inside the same interval (0.598), would weigh glucose about seven times more (spread of 21.0 mmHg for systolic pressure against 3.0 mmol/L for glucose). The cosine measures are the weakest, as the analysis of Section 2.5 anticipated: the plain cosine is at chance on Douala for the primary outcome (0.49) and below it for a new alert (0.37). On the Douala check the measures scaled by the patient's variability do best for a new alert (0.58–0.62), but the intervals of 24 patients are too wide to reverse a choice made on the development data, and Douala is not used to choose.

The distance between the current state and the baseline therefore says little about the level of the next reading. It is kept in the risk card as a description of how far the patient has moved from their own recent values, not as a trigger of alerts.

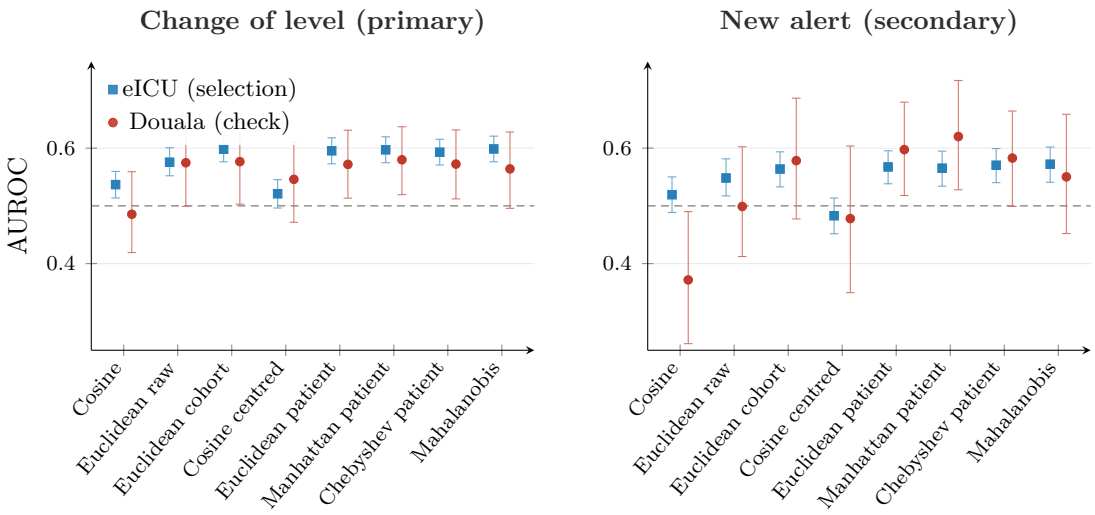


Figure 4.10: AUROC of the measures of state evolution with 95% patient-level intervals. Dashed: chance. The measure is chosen on eICU only.

4.6 System Verification

The system was verified by automated tests and by rendering its screens; it has not been used by patients or clinicians (Table 4.13). The test that matters most is the parity of the levels: the home triage is computed on the phone, in Dart, so that it works offline, and in Python by the worker and the risk service. A fixture generated by the Python reference—500 readings, the values on either side of every threshold and 11 verification profiles—is replayed by the application's tests, which must find the same level and the same profile status for every case.

Table 4.13: Verification of the system.

Component	Verification	Result
Research package	172 automated tests: labelling rule, quality controls, windows, metrics, sample size, home triage, state measures, risk service, worker	all pass
Sample-size port	published example of the reference implementation (623, 662 and 221 samples for the three criteria)	reproduced exactly
Triage parity	500 readings, threshold boundaries and 11 profiles, Python reference against the application	identical levels and statuses
Application	48 tests (record, events, queue order, assistant instruction, state evolution, parity); static analysis	all pass; no issue
Screens	28 renderings: light and dark themes, text enlarged to 130%	no overflow
Security rules	append-only readings, field and range validation, events and verdicts by role	deployed on 4 November 2026; emulator test pending
Health assistant	fixed set of 22 adversarial questions against the four guardrails	set defined; run on a phone pending

4.7 Discussion

4.7.1 Answers to the research questions

RQ1 asked whether the system can be built under the constraints of Cameroonian care. It was built, and each constraint is met by a design element rather than by an assumption. Intermittent connectivity is handled by the offline cache and by computing the home triage on the phone, so that a red reading brings up its warning without a network. Scarce clinician time is protected by the two-stage protocol, in which a yellow reading first asks the patient for a verification profile and only the mean of the profile reaches a clinical decision, and by a queue that puts red before yellow and the oldest first, each event closed in two or three taps. No proprietary device is needed, since any validated cuff or meter can be read by the patient. What remains open is whether patients and clinicians use it as intended, which only a pilot can measure.

RQ2 asked whether learned models forecast the level of the next reading better than the guideline rule, and whether a deep temporal model does better than classical ones. For blood pressure, yes to the first part: five of the eight families, and the selection procedure as a whole, have a significantly lower RPS than persistence, in most hospitals and in every subgroup, so **H1 is supported for the haemodynamic task**. The gain is about 3%, and the importance analysis shows why it is no larger: the model reads the last systolic pressure as the rule does and corrects it with little else. For glucose, no family beats persistence significantly, so **H1 is not supported for the glycaemic task**; the best model, GRU-D, is ahead by a margin of the same size, and the task falls short of the sample-size criteria, so the question is open rather than answered in the negative. On the second part, the deep models equal the tree ensembles on blood pressure and are ahead on glucose without a significant difference: **H3 is not rejected**, since no deep model outperforms the tabular ones significantly. As for glucose under H1, the absence of a significant difference leaves the question open rather than settled: no margin of equivalence was declared, the deep models had smaller search budgets than CatBoost, and the procedure prefers GRU-D for glucose in 12 of 15 folds and under 86% of the weightings of the selection cost.

RQ3, on lead times of 24 and 48 hours, is not answered: the benchmark was run for the next reading only, and **H2 is not tested**.

RQ4 asked whether models developed on another health system transfer to Cameroonian patients. Partly. The RPS is higher on Douala for every model, and the calibration intercepts show the expected under-prediction of the alert, so **H4 is supported as to the loss of performance and the calibration offset**; updating the intercept removes the offset and lowers the Brier score, but leaves the slopes far from one, so it recovers only part of the calibration. Against this, the deployed models keep or widen their margin in RPS over persistence, the haemodynamic alerts are ranked better than in development, and the value forecasts arrive almost calibrated in the large without any update. What does not transfer is clinical utility: on 32 patients, no deployed model improves the alert decisions over the rule. The predictive component should therefore run as the system was designed to run it, on top of the rule floor, and be revalidated on home data before it is allowed to raise any tier.

4.7.2 What the data imply for the transfer

The harmonisation made the two datasets alike in rhythm, not in content. Three differences remain, and each predicts a specific effect. The prevalence of the alert differs (17% against 23% for the haemodynamic task, 29% against 59% for the glycaemic task), which shifts calibration-in-the-large even for a model that ranks patients well; this is what the intercept update of H4 addresses. Critical hyperglycaemia is ten times more frequent in the Douala wards than in intensive care, so a model developed on eICU has seen few examples of the level that matters most for the glycaemic task. And glucose is measured twice as often in Douala, so the features that count glucose readings and the time since the last one have a different meaning in the two settings: in intensive care a long gap is the norm, in the ward it is not. The label uncertainty sets the expectations for all models: with more than half of the glucose readings within the device tolerance of a threshold, a glycaemic forecast will disagree with the label on many readings whatever the model.

4.7.3 Comparison with the literature

Predictive performance. The IoT system of Efendi et al. [9] reports an accuracy of 0.973 for XGBoost on readings of heart rate, oxygen saturation and temperature; on survey data of hypertensive adults, Mendoza-Cano et al. [10] find the same algorithm no better than chance. The difference is explained by the task, not by the algorithm: when the label is a threshold rule applied to the reading the model sees, the model recovers the rule (Section 2.3). The present work avoids this by forecasting the next reading, so the guideline rule applied to the last reading—persistence—is the reference any model must beat, and accuracy is not reported because it rewards the majority level. Its results are therefore not comparable with those figures, and are not meant to be. They are closer in spirit to the only genuine clinical prediction reported by Liu and Wang [24], a recurrent network predicting the readmission of lupus patients with an AUC of 0.70, and to the random forest of Mendoza-Cano et al. [10] (AUC 0.75): the AUROC of the alert of the deployed models, 0.67 to 0.78 depending on the task and the dataset, is of the same order, far from the 95 to 100% accuracies of tasks whose label is encoded in the inputs.

Choice of algorithm. Two expectations of the literature hold. Tree ensembles were plausible candidates [10, 24]: on blood pressure the random forest and CatBoost have the lowest RPS with the masked LSTM, the random forest is the family chosen most often in the outer folds, and CatBoost is the deployed model. And on tabular inputs, deep networks were not expected to outperform tree ensembles [64]: they equal them. One expectation is qualified. Across clinical prediction studies, machine learning shows no systematic benefit over logistic regression [28]; here, the proportional-odds model does no better than persistence on blood pressure ($+0.0007$, $p = 0.73$), whereas the tree ensembles do, by a margin of about 0.005 in RPS. What the benchmark adds to the literature is the comparison with the guideline rule, which none of the reviewed studies made: against persistence, the margin of the best model is about 3% of the RPS.

Transfer to African patients. The loss of performance and the calibration offset observed on the Douala patients are what the literature on early-warning scores predicted: scores derived elsewhere lose discrimination in African wards and require local validation and recalibration [34, 35], and artificial intelligence trained elsewhere is largely unvalidated on African populations [5]. This work measures such a transfer for a learned model for the first time in the region, on few patients, and finds that the intercept update corrects the offset but not the slopes.

The telemonitoring design. The reviewed telemonitoring protocol [8] reports clinical effect, not predictive performance; the present system implements it as the rule floor, so a model can only add to it. The clinical evidence attributes the effect of telemonitoring to a clinician acting on the data [7, 8, 21], and the Cameroonian studies show that systematic expert review does not scale [6, 13]: the queue that puts red before yellow, the verification profile that absorbs isolated yellow readings, and the escalation of unreviewed events answer these two findings together. The system also responds to the barriers of the region [11, 12]: it works offline, needs no proprietary device, and stores no identifier outside the record. On the patient side, it delivers the techniques that the content analyses found most common—self-monitoring, feedback, prompts [19]—through the structure of the system rather than through advice, as the clinical effect of Mihevc et al. [8] came from treatment adjustment rather than behaviour change. And it keeps glucose and blood pressure from validated instruments, as the WHO requires and the sensor literature supports [1, 22, 25], leaving smartphone sensing for later evaluation.

The arithmetic of alert volume Whatever the benchmark shows, the prevalence of the alert at home will be far lower than in the inpatient data, and no reviewed telemonitoring source makes the following calculation. Consider N patients sending m readings a week, a share π of which genuinely need attention, and a classifier of sensitivity Se and specificity Sp. The number of alerts per week and the share of them that are genuine (the precision, or positive predictive value) are

$$A = Nm\big[\pi\,\mathrm{Se} + (1-\pi)(1-\mathrm{Sp})\big], \qquad \mathrm{PPV} = \frac{\pi\,\mathrm{Se}}{\pi\,\mathrm{Se} + (1-\pi)(1-\mathrm{Sp})}. \tag{4.1}$$

The false-alert term $(1-\pi)(1-\mathrm{Sp})$ does not depend on sensitivity: it is the large volume of normal readings multiplied by the false-positive rate. This formula reproduces the precisions reported by Mendoza-Cano et al. [10] at their prevalence of 0.303 (0.92 against 0.94 for the random forest). At a planning prevalence of 0.02, a sensitivity of 0.90 and a specificity of 0.98 give a precision of 0.48 and 19.6 false alerts per 1 000 readings; a precision of 80% requires a specificity above 0.995 (Table 4.14), whereas the best model of the reviewed literature reaches 0.98 at a prevalence fifteen times higher.

The multiplier Nm depends on the sampling frequency, which varies by orders of magnitude between designs (Table 4.15). Thirty patients on the protocol of Mihevc et al. [8] generate about two false alerts a week, which is manageable; the same specificity applied to a sensor stream at one reading per second, as in Efendi et al. [9], generates more than eleven thousand false alerts per patient and per week, whatever the quality of the model. High-frequency data can be summarised into features, but they cannot be alerted on reading by reading.

Table 4.14: Precision and false alerts at a prevalence of 2%.

Se	Sp	PPV	False alerts per 1 000 readings
0.51	0.98	0.34	19.6
0.90	0.98	0.48	19.6
0.90	0.99	0.65	9.8
0.90	0.995	0.79	4.9
0.90	0.999	0.95	1.0

Table 4.15: False alerts by acquisition design, at Sp = 0.98 and $\pi = 0.02$.

Design	Readings per patient and week	False alerts per patient and week
Protocol of Mihevc et al. [8]: 2 BP + 1 glucose	3	0.06
Manual entry, daily	7	0.14
Continuous sensor at 1 Hz [9]	604 800	11 854

The two-stage home protocol answers this arithmetic: a single yellow reading is queued behind the red ones and leads first to more measurements, so the clinician's decision rests on the mean of a profile rather than on one value, and only red readings call for immediate review. Figure 4.11 shows how far the inpatient data used here are from that regime: the same classifier that is precise at the prevalence of the wards loses most of its precision at home.

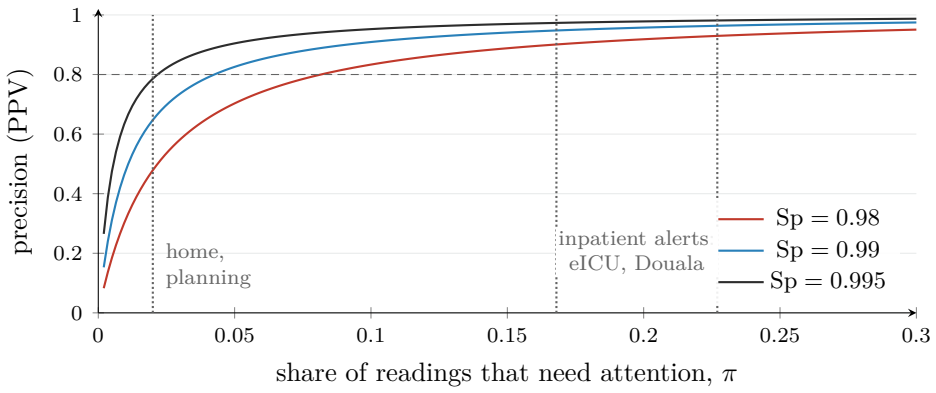


Figure 4.11: Precision against the share of readings that need attention, at a sensitivity of 0.90. The dashed line marks a precision of 80%. Dotted lines: the planning prevalence of 0.02 for a home deployment, and the prevalence of the haemodynamic alert at the targets of eICU (0.17) and Douala (0.23).

4.7.4 Strengths of the results

Four features make the results stronger than the figures usually reported in the field. They are *free of label leakage*: the label lies outside the input window by construction, and every model is compared with the guideline rule applied to the last reading, so the margins measure what a model adds to the rule. They are *honest about selection*: the hyperparameters, the

window and the family are chosen on inner folds only, with budgets declared before any run, and the outer estimate is that of the whole procedure. They *generalise across sites*: the outer folds hold out whole hospitals, so the development estimates already measure transfer between 174 hospitals before the Douala validation. And they are *reproducible*: every number of this chapter is printed by a script from saved outputs, with the configuration and the code of each run. The results are also original in two respects: no previous study forecast the level of the next reading of blood pressure and glucose from a patient's series, and none validated a learned model on Cameroonian patients.

4.7.5 Precautions of interpretation

What can be concluded is limited by where the data come from. It can be concluded that, on harmonised intensive-care data, learned models forecast the level of the next blood pressure reading slightly better than the guideline rule; that deep temporal models show no significant advantage over tree ensembles on these data; that models developed in United States intensive care lose calibration on Cameroonian inpatients in the direction their prevalences predict; and that the system can apply the home triage offline and order the clinician's attention. It cannot be concluded that the models would forecast home readings as well, since no home data were used; that the glycaemic forecast is useless, since the task falls short of the sample-size criteria; that the deployed models would improve decisions in Douala, since on 32 patients they did not; or that the system improves blood pressure control, which only a prospective study can measure. The Douala comparisons in particular are descriptive: their intervals overlap for every pair of models.

4.7.6 Limitations

Table 4.16 states each limitation with its consequence and with what this work does about it.

Each limitation also opens a direction of research, taken up in the Conclusion: the full eICU database for the glycaemic task, home data for the transfer, the physician's relabelling and verdicts for labels that do not derive from the thresholds, and a pilot for the use of the system.

4.8 Synthesis

Table 4.17 gathers the answers to the hypotheses, and the key results of the chapter are the following.

- The Douala pressures were taken with electronic devices and the pulse read from the chart curve; the device tolerance alone can change the level of 27 to 56% of the readings.
- The sourced labelling rule changes the level of 58 of the 366 readings compared with the earlier rule, mostly through low values and the glucose bands.
- eICU and Douala share the rhythm of two readings a day but not the content: diastolic pressure, glucose, its frequency and the prevalence of the alert differ, and the level of the next glucose reading equals the current one in only 30% of the Douala pairs.

Table 4.16: Limitations, their consequences and the response of this work.

Limitation	Consequence	Response
32 local patients	local estimates have wide intervals	development on eICU, validation only in Douala; patient-level intervals; read as exploratory
Demonstration subset of eICU	glycaemic task below the sample-size criteria (Table 4.4)	primary set of 8 features; benchmark to be rerun on the full database
Inpatients, not home users	prevalence and dynamics differ from home	home triage with out-of-office thresholds; model revalidated on home data before use
Labels derived from thresholds	a model may learn the rule; no clinical adjudication yet	forecasting task; physician relabelling sized at 100–150 readings; clinician verdicts in deployment
Unknown devices, hand transcription	up to 56% of readings near a threshold within device tolerance	devices inferred from digits; uncertainty reported; isolated extremes flagged
Diagnoses not transcribed	Douala cohort may include patients with neither condition	eligibility verified in eICU; Douala used for validation only
Slot hours assumed	glycaemic samples depend on the tolerance	sensitivity analysis at ± 1 and ± 3 h
Provisional choices	selection-cost weights and tier cut-points are not yet the physician's	declared in configuration; weight sensitivity; to be replaced by stated costs of errors
Search budgets of 15 to 25 configurations	in three quarters of the blood-pressure searches of CatBoost and GRU-D, the configuration kept came after the fifteenth trial: the searches had not levelled off, and comparisons between families favour the larger budgets	budgets declared in advance (Table C.1); H3 read as not rejected rather than supported; rerun with equal budgets
State distance in raw units	the distance shown to the clinician barely registers glucose	described, never used as an alert; per-channel changes shown beside it
No use by patients	no usability or effectiveness result	verified by tests; prospective pilot as future work

- The Douala charts cannot support model development; the demonstration subset of eICU supports the haemodynamic task, not fully the glycaemic one.
- The deployed models are CatBoost for blood pressure and GRU-D for glucose. On Douala their RPS is higher than in development and their alerts are under-predicted; the intercept update corrects the offset, not the slopes, and no model improves the alert decisions over the rule.
- The measures of state evolution are weak (AUROC 0.52 to 0.60); the raw Euclidean distance is retained by the declared rule, as a description for the clinician, not as an alert.
- The system passes 220 automated tests, including the parity of the triage on the phone with the Python reference; it has not yet been used by patients.

Against the state of the art, the contribution is twofold: a forecast free of label leakage, compared with the guideline rule, which shows a small but real margin on blood pressure; and the first external validation of learned models on Cameroonian patients, which shows that the ordering of the deployed models relative to the rule is kept on the Douala patients,

Table 4.17: The hypotheses and their outcome.

Hypothesis	**Outcome**	**Evidence**
H1: a learned model beats persistence	supported for blood pressure; not for glucose	haemodynamic: five of eight families, ΔRPS -0.0036 to -0.0046, $p < 0.05$; glycaemic: best ΔRPS -0.0055, $p = 0.14$
H2: the advantage persists 24 and 48 h ahead	not tested	runs with embargoes of 12 and 36 h not made
H3: deep models do not outperform tabular ones	not rejected; equivalence not shown	masked LSTM equal to the random forest; GRU-D ahead on glucose, not significantly; smaller search budgets than CatBoost
H4: lower performance on Douala; intercept update recovers part of the calibration	supported, partly	higher RPS and positive intercepts for every model; offset corrected, slopes unchanged

without a significant difference, while calibration and clinical utility do not transfer. These results justify the design in which the rule is the floor and the model can only raise the alert, and they lead to the general conclusion.

General Conclusion and Perspectives

Summary of the Work

This thesis set out to design a telemonitoring system for patients with hypertension and type 2 diabetes in Cameroon, and to evaluate its predictive component without the biases of previous work. The literature showed that telemonitoring improves blood pressure and glycated haemoglobin when a clinician acts on the data, that the only validated protocol for patients with both conditions triages single readings with fixed thresholds, and that the predictive results of the field rest on labels that leak into the features and on validation in the training population (Chapter 1). The risk-tiering problem was therefore formalised as a forecast of the next reading, labelled by official thresholds, combined with the guideline rule so that a model can raise an alert but never lower it (Chapter 2). A pipeline of data collection and treatment and a benchmark protocol were built to select the model rather than presuppose it, and the system was built around them (Chapter 3).

Principal Findings

The data work produced four results. First, the Douala charts were taken with electronic devices for the pressures, while the pulse was read off the chart curve; the readings themselves showed this, since the devices are not named. Second, the device tolerance alone can change the level of a quarter to more than half of the readings, which bounds what any model can achieve against a threshold label. Third, the sourced labelling rule changes the level of 58 of the 366 readings compared with the earlier unsourced rule, chiefly because low pressures now count and the glucose bands follow the official thresholds. Fourth, the 32 local patients are far below the size needed to develop a model, and the development data differ from them in prevalence and in the frequency of critical hyperglycaemia, which predicts a loss of calibration that a local intercept update should partly recover.

The benchmark answered the predictive questions with a mixed result. On blood pressure, learned models forecast the level of the next reading better than the guideline rule applied to the last reading: five of eight families, and the selection procedure as a whole, improve on persistence significantly, in most hospitals and in every subgroup, but by about 3% of the ranked probability score, and the deployed CatBoost relies mostly on the last systolic pressure. On glucose, no model beats persistence significantly, on a task that the demonstration data

cannot fully support. Deep temporal models showed no significant advantage over tree ensembles, which is consistent with H3 without showing the two equal, and GRU-D was the procedure's consistent choice for glucose. Applied to the Douala patients, every model lost accuracy and under-predicted the alert; an intercept update corrected the offset but not the slopes, the value forecasts arrived nearly calibrated without it, and no model improved the alert decisions over the rule. The measures of the evolution of the patient's state carry little information about the next level, and the one retained is shown to the clinician as a description, not used as an alert. The lead times of 24 and 48 hours were not tested.

These results justify the design choice made before any of them was known: the guideline rule is the floor, and a model can only add to it.

The system answers the first research question: a patient application and a clinician dashboard that implement the three-level home protocol offline, a record that cannot be altered, a queue that orders the clinician's attention, and server functions for escalation, reminders and verdicts, verified by 220 automated tests, among them the parity of the triage computed on the phone with the Python reference.

Future Directions

- **Full development data.** Rerunning the benchmark on the full eICU database, available under a data use agreement, removes the sample-size constraint of the glycaemic task.
- **Clinical input.** The relabelling of 100 to 150 readings by the partner physician, and the costs of errors that fix the weights of the selection cost and the tier cut-points, replace the provisional choices.
- **Home data.** A prospective pilot in Douala would collect the home readings for which the system is designed, revalidate the model on them, and measure adherence, usability and the clinician's workload; the clinician verdicts it records become labels that do not derive from the thresholds.
- **Acquisition.** Readings from Bluetooth devices would remove transcription errors, and a low-cost sensor node (ESP32 with pulse oximetry and temperature) could add covariates; neither was built in this work.
- **Deployment.** Hosting the risk service and the worker, an SMS fallback for patients without data, and a database region chosen under Law No. 2024/017 are required before any use with patients.

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Appendix A

The Nineteen Sources of the Review

Table A.1 summarises the nineteen core sources of Chapter 1, in the order of its themes: design and data, main findings, limits, and what this thesis takes from each.

Table A.1: The nineteen core sources of the literature review.

Source	Design and data	Main findings	Limits	Use in this thesis
Burden of disease				
Hinneh et al. [3]	meta-analysis of 41 cross-sectional studies, 144 433 participants, Africa	58.1% of people with diabetes are hypertensive; Central Africa highest, 77.6% (53.0–91.4)	extreme heterogeneity; Central Africa rests on two hospital series; English only	case for joint monitoring (Section 1.3.3)
World Health Organization [1]	institutional global report	1.3 billion hypertensive adults; 72% of the diagnosed not controlled in the African Region; validated cuffs only; telemonitoring endorsed	modelled estimates; no patient-level technology	framing, thresholds, device position (Sections 1.3 and 1.6.1)
Care and digital health in sub-Saharan Africa				
Bediang et al. [6]	controlled study, two equipped and two control district hospitals, 171 patients, Cameroon	ECG with expertise 98.9% against 33.3%; local diagnosis changed in 46.2%; expert opinion after 10.3 days	not randomised; differential attrition	feasibility in Cameroon; scarcity of expert time (Section 1.4.3)
Dodoo, Al-Samarraie, and Alsswey [11]	review of 66 empirical studies and reports	six dimensions of barriers; projects stop when external funding ends	francophone countries under-represented	barriers, sustainability (Section 1.4.4)
Gafane-Matemane et al. [2]	narrative review, 2017–2023	high burden, poor control; Cameroon 20.8% prevalence, 11.2% controlled; task shifting inconsistent on blood pressure	narrative; heterogeneous surveys	regional profile (Section 1.4)
Agbeyangi and Lukose [12]	systematic review of 53 studies, 2014–2024, South Africa, Kenya, Nigeria	adoption led by South Africa; persistent regulatory, literacy and infrastructure barriers	Central Africa absent	regional landscape, blind spot
Segun-Omosehin et al. [5]	perspective on cardiovascular digital health	over 80% mobile subscription; 550 million without reliable electricity and internet; AI largely unvalidated in Africa	no declared search strategy	orders of magnitude; transfer of AI (Section 1.10)
Mobile health and clinical evidence				
Alessa et al. [19]	content analysis of 186 hypertension applications, 30 in depth	58.1% offer one function; none of 30 evaluated; median of nine behaviour change techniques; 14 of 20 privacy policies declare sharing	market snapshot of English applications	features and privacy of applications (Section 1.5.1)

Source	Design and data	Main findings	Limits	Use in this thesis
Mao et al. [7]	meta-analysis of 51 randomised trials, 13 054 participants	SBP −2.99 mmHg, HbA1c −0.39; −6.17 mmHg with professional management; no SBP effect in developing countries	moderate to high heterogeneity; one trial in sub-Saharan Africa	effect conditional on a clinician (Section 1.5.2)
Zhou et al. [20]	meta-analysis of 20 randomised trials, 16 pooled, 4 168 participants	SBP −5.78 mmHg; larger with eleven or more techniques and several risk factors	low certainty; 15 of 20 trials at high risk of bias	behaviour change techniques
Bandeira et al. [21]	updated meta-analysis of randomised trials	significant blood pressure reductions; telemonitoring with clinician feedback most effective	heterogeneity; short follow-up; few data from low-income settings	clinician in the loop
Mihevc et al. [8]	multicentre open-label trial, 128 patients aged 65 or more with both conditions, Slovenia	SBP −6.9 mmHg and HbA1c −0.5 at twelve months; 10.9% withdrew in the first month	high-income setting; older adults; not blinded	three-level home protocol adopted (Section 2.4.2)
Sensors and IoT				
Alhaddad et al. [22]	review of non-invasive glucose sensing and machine learning	promise for trends and hypoglycaemia, not for clinical-grade values	small samples; inter-individual variation	glucose from a glucometer (Section 1.6.2)
Tai et al. [23]	deep learning technique in a three-phase architecture	accurate and interpretable predictions in their evaluation	controlled conditions; connectivity not addressed	layered architecture
Efendi et al. [9]	IoT system (ESP32, MQTT, Firebase), six volunteers	mean absolute percentage error 0.90%; XGBoost accuracy 0.973	label a function of the inputs; no blood pressure or glucose; data stored abroad	offline-first storage; example of leakage (Section 1.7)
Liu and Wang [24]	survey of IoT sensing and machine learning	device categories; seven-layer platform; accuracies of 95–100%; defects of electronic records	no protocol or flow diagram	record quality; layers
Liu et al. [25]	smartphone camera photoplethysmography for blood pressure, 127 participants	strong correlation with reference devices; SHAP explanations	single site; below cuff accuracy	smartphone sensing kept exploratory
Machine learning				
Mendoza-Cano et al. [10]	cross-sectional prediction, 1 308 hypertensive adults, Mexican national survey	random forest AUC 0.75, XGBoost 0.54; sensitivity of the best model 0.51	cross-sectional; demographic predictors; reporting errors	genuine prediction; choice of metrics (Sections 1.7 and 3.10)
Privacy and ethics				
Alhammad et al. [26]	systematic review of 33 studies of patient perspectives	data concerns hinder adoption; encryption, authentication and transparency build trust	mostly high-income countries; self-reported perceptions	data protection as a design constraint (Section 1.8)

Appendix B

Data Dictionary

Douala charts One row per reading, transcribed by hand from the ward observation charts without names (Table B.1). The file is not published.

Table B.1: Columns of the Douala charts file.

Column	Content	Use
`patient`	integer code assigned at transcription	grouping of the samples
`age, sex`	years; F or M	features; subgroups; ages grouped in five-year bands before sharing
`height_cm, weight_kg`	empty in every row	not used
`date`	date of the reading	time since the patient's first reading
`time`	M (morning) or S (evening, *soir*)	slot; the two slots are placed 12 h apart
`sbp, dbp`	systolic and diastolic pressure, mmHg	channels
`hr`	pulse, beats per minute	channel
`glucose_gl`	capillary glucose as written on the chart, g/L	source of the glucose channel
`glucose`	glucose in mmol/L, from g/L with 18.0156 mg/dL per mmol/L	channel
`glucose_state`	meal status; *casuel* (random) in every row	not used: in-hospital thresholds do not depend on it
`notes`	free text, 5 rows	not used
`label`	level under the earlier unsourced rule	comparison of the two rules only (Table 4.2)

eICU demonstration subset Version 2.0.1 [47], checked against its published checksums before every run. Table B.2 lists the tables and columns read. A stay is eligible when the patient is an adult and hypertension or diabetes is recorded in the diagnoses or the past history; hypertensions of another nature (pulmonary, intracranial, portal, ocular, intra-abdominal, gestational) are excluded.

Table B.2: Tables and columns of eICU used.

Table	Columns	Use
`patient`	`patientunitstayid, uniquepid, hospitalid, age, gender`	stays, patients, hospitals; age and sex
`diagnosis`	`diagnosisstring`	eligibility
`pastHistory`	`pasthistorypath`	eligibility
`vitalAperiodic`	`noninvasivesystolic, noninvasivediastolic, observationoffset`	cuff pressure at each slot
`vitalPeriodic`	`heartrate, observationoffset`	pulse at each slot
`lab`	`labname (bedside glucose), labresult, labresultoffset`	glucose at each slot, mg/dL to mmol/L

Appendix C

Benchmark Configuration

The benchmark is declared in `configs/benchmark.yaml` of the research package before any run (Table C.1). Window length and class weights are searched for every candidate; the number of configurations tried per candidate and outer fold is fixed. Seed 20261019; five outer folds grouped by hospital, four inner folds grouped by patient, three repeats.

Table C.1: Search spaces and number of configurations. Ranges marked log are searched on a logarithmic scale.

Candidate	Trials	Structural	Training
All	–	window 24, 48, 72, 96 h	class weights none or balanced
Ordinal logistic	15	–	$C \in [10^{-3}, 10^{2}]$, log
PCA + ordinal logistic	15	components 1–8	$C \in [10^{-3}, 10^{2}]$, log
Random forest	20	max features $\sqrt{p}$, 0.5, 1.0; min leaf 1–50, log; depth 3, 6, 12, none	–
CatBoost	25	depth 2–8; iterations 100–1 000, log	learning rate 0.01–0.3, log; L2 1–30, log
Value forecast, ridge	15	–	$\alpha \in [10^{-3}, 10^{3}]$, log
Value forecast, CatBoost	25	as CatBoost	as CatBoost
GRU-D	20	hidden units 4, 8, 16, 32	dropout 0–0.5; learning rate 10^{-3}–0.03, log; weight decay 10^{-6}–10^{-2}, log; auxiliary loss weight 0–1
Masked LSTM	20	hidden units 4, 8, 16, 32; layers 1, 2	as GRU-D

Rerunning the results From `telemonitor-research`, every number and figure of the thesis is produced by a script (Table C.2).

Table C.2: Scripts of the research package and what they produce.

Script	Produces
`01_douala_audit.py`	cohort description, quality flags, terminal digits, label uncertainty (Section 4.2)
`02_label_rules.py`	the earlier rule against the sourced rule (Table 4.2)
`04_sample_size.py`	criteria of Riley et al. for both datasets (Table 4.4)
`05_eicu_cohort.py`	eligibility, harmonisation and samples of eICU
`06_benchmark.py`	nested benchmark and external validation; resumes after an interruption
`07_app_parity_fixture.py`	the fixture replayed by the application's parity tests
`08_figures.py;`	figures and tables of a saved run
`13_results_tables.py`	
`09_kappa_sample_size.py`	size of the physician's relabelling
`10_sensitivity_flags_slots.py`	shrinkage constant and slot tolerance (Table 4.3)
`11_export_model.py`	the model served by the risk service
`12_state_comparison.py`	benchmark of the measures of state evolution
`14_thesis_data_figures.py`	data of Figures 3.3, 4.4 and 4.5

Appendix D

Record Schema

The fields of a reading and of an event in Cloud Firestore (Figure 3.9). The security rules validate every field on the server.

Table D.1: Fields of a reading, `users/{uid}/readings/{id}`.

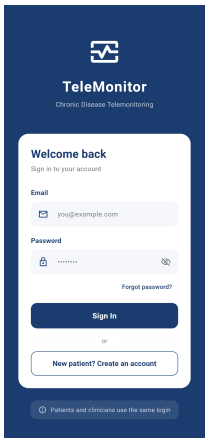
Field	Type	Meaning
`recordedAt`	time	measurement time stated by the patient, at most 30 days before arrival
`timestamp`	time	arrival time, set by the server
`entryMode`	text	manual (Bluetooth entry is specified, not built)
`sbp, dbp, hr`	numbers	retained reading of the protocol pair; mmHg and beats per minute
`glucose, context`	number; text	mmol/L; fasting, before a meal, after a meal or random, required with glucose
`readingCount`	number	1 or 2 readings in the session
`firstSbp, firstDbp, firstHr`	numbers	first reading of the pair, kept for audit
`supersedes`	reference	the reading this one corrects
`notes`	text	free text of the patient
`qualityFlag`	text	set by the worker: ok, out_of_range, unit_suspect, duplicate, carried_forward, extreme

Table D.2: Fields of an event, `users/{uid}/events/{id}`.

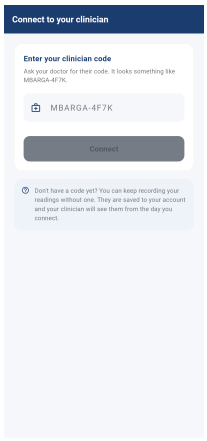
Field	Type	Meaning
`patientId, clinicianId`	references	the patient and the assigned clinician
`triggerSource`	text	rule, model or non_transmission
`tier, score`	text; number	yellow or red; the model's probability when the model raised it
`readingId, trigger`	reference; map	the triggering reading and values, or the profile and its means
`thresholdVersion, modelVersion`	text	versions in force when the event was raised
`tsTriggered, status`	time; text	creation; open or closed
`verdict, action, comment`	text	confirmed, not_significant or uncertain; none, advice, treatment_change, consultation or referral
`tsReviewed, reviewedBy`	time; reference	closure of the event, once
`escalatedAt, escalatedTo`	time; reference	escalation to the named alternate

Appendix E

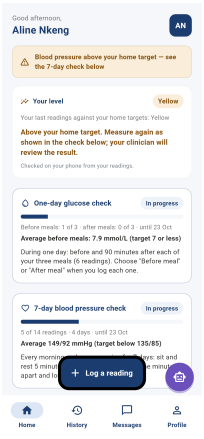
Application Screens



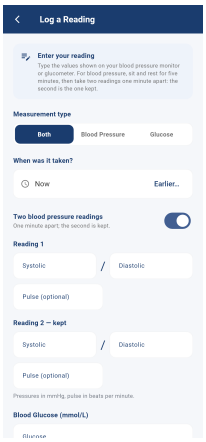
(a) Sign-in



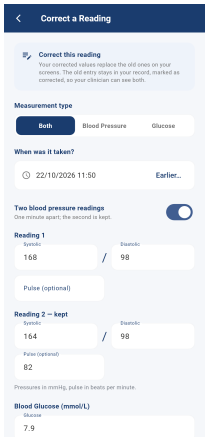
(b) Enrolment



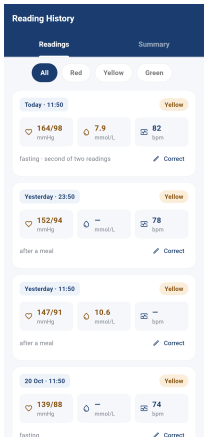
(c) Patient home



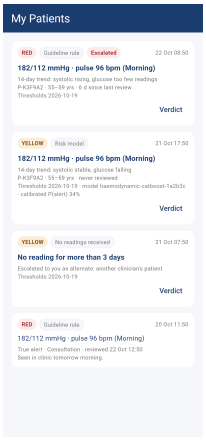
(d) New reading



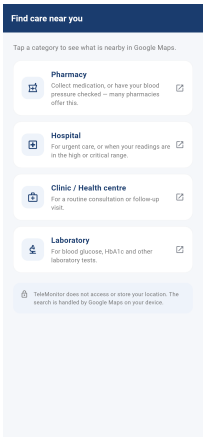
(e) Correction



(f) History



(g) Clinician queue



(h) Nearby care

Figure E.1: Screens of the application: (a) sign-in, shared by patients and clinicians; (b) enrolment with the clinician's code; (c) patient home with the level, the instruction and the verification profiles; (d) entry of the two readings of the protocol; (e) correction, a new reading that supersedes the old one; (f) history with level and corrections; (g) clinician queue, red before yellow; (h) nearby care, opened in the phone's maps. Rendered by the widget tests with fictitious patients; model names and probabilities are test values.

Appendix F

TRIPOD+AI Checklist

The modelling study is reported according to TRIPOD+AI [85]; its risk of bias can be assessed with PROBAST+AI [86].

Table F.1: TRIPOD+AI items and where they are addressed.

Item	Topic	Where
1	Title	title page
3a–c	Background, intended use, inequalities	Introduction; Section 1.10 (Central Africa absent from the evidence)
4	Objectives	Introduction (objectives, research questions, hypotheses)
5a–b	Data sources, dates	Section 3.4: Douala 2026; eICU 2014–2015; Appendix B
6a–b	Setting, eligibility	Sections 3.4 and 3.4.2; Appendix B
6c	Treatments	not recorded in either dataset
7	Data preparation	Section 3.6
8a–c	Outcome, assessment, blinding	Section 2.2; blinded relabelling by the partner physician planned (100–150 readings), not yet done
9a–c	Predictors	Section 3.7; instruments in Section 3.4.3
10	Sample size	Sections 3.5 and 4.2.6
11	Missing data	Section 3.6.4
12a–g	Partitioning, handling, model type, clusters, measures, updating, computation	Sections 3.8 to 3.10 and 3.12; outer folds by hospital; intercept update; Appendix C
13	Class imbalance	class weights and their correction (Section 3.9)
14	Fairness	results by sex, age band and hospital (Section 4.3)
15	Output, thresholds	Section 2.4
16	Development against evaluation data	Section 4.2
17	Ethical approval	Section 3.13
18c–f	Protocol, registration, sharing	protocol fixed before results; not registered; Section 3.12
19	Patient and public involvement	none
20–27	Results, discussion	Chapter 4